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Title 33 — Navigation and Navigable Waters

Chapter II — Corps of Engineers, Department of the Army, Department of Defense

Part 222 Engineering and Design

§ 222.2 Acquisition of lands downstream from spillways for hydrologic safety purposes.

§ 222.3 Clearances for power and communication lines over reservoirs.

§ 222.4 Reporting earthquake effects.

§ 222.5 Water control management (ER 1110-2-240).

§ 222.6 National Program for Inspection of Non-Federal Dams.

PART 222—ENGINEERING AND DESIGN

Authority: 23 U.S.C. 116(d); delegation in 49 CFR 1.45(b); 33 U.S.C. 467 *et seq.*; 33 U.S.C. 701, 701b, and 701c-1 and specific legislative authorization Acts and Public Laws listed in appendix E of § 222.7.

§ 222.2 Acquisition of lands downstream from spillways for hydrologic safety purposes.

- (a) **Purpose.** This regulation provides guidance on the acquisition of lands downstream from spillways for the purpose of protecting the public from hazards imposed by spillway discharges. Guidance contained herein is in addition to ER 405-2-150.
- (b) **Applicability.** This regulation is applicable to all OCE elements and all field operating agencies having civil works responsibilities.
- (c) **Reference.** ER 405-2-150.
- (d) **Discussion.** A policy of public safety awareness will be adhered to in all phases of design and operation of dam and lake projects to assure adequate security for the general public in areas downstream from spillways. A real estate interest will be required in those areas downstream of a spillway where spillway discharge could create or significantly increase a hazardous condition. The real estate interest will extend downstream to where the spillway discharge would not significantly increase hazards. A real estate interest is not required in areas where flood conditions would clearly be nonhazardous.
- (e) **Hydrologic criteria.** The construction and operation of a dam and spillway may create or aggravate a potential hazard in the spillway discharge area. Therefore, an appropriate solution should be developed in a systematic manner. All pertinent facts need to be considered to assure that the risk to non-Federal interests does not exceed conditions that would prevail without the project. General hydrologic engineering considerations are as follows:
 - (1) **Probability of spillway use.** Pool elevation versus probability of filling relationships can change materially after initial construction. Spillway use may be more frequent than anticipated. The infrequent use of a spillway is not a basis for the lack of adequate downstream real estate interest.

- (2) **Changes in project functions.** Water resource needs within river basins change and pool levels may be adjusted to provide more conservation storage, particularly when high-level limited-service spillways are provided. Such changes normally increase spillway use and are to be considered.
- (3) **Volume and velocity of spillway flow.** The amount of flow and destructive force of the flow from a spillway during floods up to the spillway design flood can vary from insignificant to extremely hazardous magnitudes. The severity and area of hazard associated with spillway discharge will vary depending on specific project site conditions. Therefore, the hazard is to be analyzed on a project-by-project basis.
- (4) **Development within floodway.** If development within the floodway downstream from a spillway is not present at the time of project construction, the existence of the reservoir may encourage development. Adverse terrain conditions do not preclude development. Sparse present development is not a basis for lack of real estate acquisition.
- (5) **Debris movement within floodway.** The availability of erodible material in a spillway flow area intensifies the hazards of spillway flow. In fact, debris may be transported to downstream areas that otherwise would not be adversely affected. Extreme erosion may result from high velocities and turbulence. Both debris and erosion must be evaluated and considered.
- (6) **Flood warning and response potential.** Small projects generally have short time periods available to warn downstream inhabitants and may be unattended prior to spillway use. The ability to convince individuals to leave most of their worldly possessions to the ravages of spillway flow may be severely limited. In some cases flood warning systems may be necessary; however, this subject is beyond the scope of this regulation. Warning systems are not an adequate substitute for a real estate interest in lands downstream of spillways.
- (7) **Location of spillway.** Spillways should be located to minimize the hazards associated with their discharge and the total project cost (cost of spillway structure and downstream lands). Spillways, outlet works, stilling basins, and outlet channels should be designed to minimize hazards to downstream interest insofar as is engineeringly and economically reasonable.
- (f) **Real estate.** The real estate interest required downstream of spillways will be adequate to assure carrying out project purposes and to protect non-Federal interest from hazards created by spillway flows. The interest may be either fee or permanent easement. A permanent easement must exclude all overnight and/or permanent habitation, structures subject to damage by spillway flows and activities that would increase the potential hazards. No real estate interest is required for:
 - (1) Areas where the imposed or aggravated flood condition is non-hazardous. Affected interest should be informed of the nature of the imposed non-hazardous flood condition.
 - (2) Areas where the construction and operation of a dam and spillway do not increase or create a hazardous condition.
- (g) **Alternative land uses.** In some cases land downstream from spillways can be effectively used for purposes other than hydrologic safety. Therefore, the entire cost of these lands may not be an additional project cost. For example, the lands downstream of a spillway may be used for wildlife management essential to project purposes in lieu of other lands suitable for similar purposes at another location.
- (h) **Procedural guidance.** Procedures regarding the application of the principles outlined in the above paragraphs are as follows:

- (1) For various flood magnitudes up to the probable maximum flood determine the “with” and “without project” flood conditions downstream of a dam spillway for the following:
 - (i) Flooded area.
 - (ii) Flood depth.
 - (iii) Flood duration.
 - (iv) Velocities.
 - (v) Debris and erosion.
- (2) Determine the combinations of flood magnitudes and the above flood conditions that could be the most hazardous and/or result in the greatest increase in hazard from “without” to “with project” flood conditions. Designate these combinations of flood magnitude and flood conditions as the critical conditions.
- (3) For the critical conditions selected above outline the areas where the project could increase and/or create (impose) one or more of the critical conditions. Areas where spillway flows do not create or increase flood conditions are excluded from further analysis.
- (4) Determine where the imposed critical conditions as outlined above would be hazardous and non-hazardous. Non-hazardous areas are defined as those areas where:
 - (i) Flood depths are maximum of 2 feet in urban and rural areas.
 - (ii) Flood depths are essentially non-damaging to urban property.
 - (iii) Flood durations are a maximum of 3 hours in urban areas and 24 hours in agricultural areas.
 - (iv) Velocities do not exceed 4 feet per second.
 - (v) Debris and erosion potentials are minimal.
 - (vi) Imposed flood conditions would be infrequent. That is, the exceedence frequency should be less than 1 percent. Hazardous areas are those where any of the above criteria are exceeded.
- (5) Based upon the information developed above and the principles outlined in paragraphs (c) through (f) of this section, decide on the extent of area and estate required for hydrologic safety purposes.
 - (i) **Reporting.** Lands to be acquired downstream from spillways and intended purposes will be identified and the cost included in feasibility reports and real estate design memoranda. Additional specific information in support of land acquisition should be provided in Phase I or Phase II general design memoranda (GDM) and dam modernization reports. This information should include topographic maps, area flooded maps, velocities, erosion and debris areas “with” and “without” the project. Real estate boundaries and discussions of items in paragraph (h)(4) are also essential in the GDM's and dam modernization reports.

[43 FR 35481, Aug. 10, 1978. Redesignated at 60 FR 19851, Apr. 21, 1995]

§ 222.3 Clearances for power and communication lines over reservoirs.

- (a) **Purpose.** This regulation prescribes the minimum vertical clearances to be provided when relocating existing or constructing new power and communication lines over waters of reservoir projects.
- (b) **Applicability.** This regulation applies to all field operating agencies having Civil Works responsibilities.

(c) **References.**

- (1) ER 1180-1-1 (Section 73).
- (2) National Electrical Safety Code (ANSI C2), available from IEEE Service Center, 445 Hoes Lane, Piscataway, N.J. 08854.

(d) **Definitions —**

- (1) **Design high water level.** The design high water level above which clearances are to be provided shall be either:
 - (i) The elevation of the envelope profile of the 50 year flood, or flood series, routed through the reservoir with a full conservation pool after 50 years of sedimentation, or
 - (ii) the elevation of the top of the flood control pool, whichever is higher.
 - (2) **Low point of line.** The low point of the line shall be the elevation of the lowest point of the line taking into consideration all factors including temperature, loading and length of spans as outlined in the National Electrical Safety Code.
 - (3) **Minimum vertical clearance.** The minimum vertical clearance shall be the distance from the design high water level (paragraph (d)(1) of this section) to the low point of the line (paragraph (d)(2) of this section).
- (e) **Required clearances.** Minimum vertical clearances for power and communication lines over reservoirs shall not be less than required by section 23, rule 232 of the latest revision of the National Electrical Safety Code (ANSI C2).
- (1) In general, minimum vertical clearances shall not be less than shown in Table 232-1, Item 7, of ANSI C2, even for reservoirs or areas not suitable for sailboating or where sailboating is prohibited.
 - (2) If clearances not in accordance with Table 232-1 of ANSI C2 are proposed, justification for the clearances should be provided.
- (f) **Navigable waters.** For parts of reservoirs that are designated as navigable waters of the United States, greater clearances will be provided if so required. The clearances required over navigable waters are covered by 33 CFR 322.5(i)(2) and are not affected by this regulation.

[43 FR 14013, Apr. 4, 1978. Redesignated at 60 FR 19851, Apr. 21, 1995]

§ 222.4 Reporting earthquake effects.

- (a) **Purpose.** This regulation states policy, defines objectives, assigns functions, and establishes procedures for assuring the structural integrity and operational adequacy of major Civil Works structures following the occurrence of significant earthquakes. It primarily concerns damage surveys following the occurrences of earthquakes.
- (b) **Applicability.** This regulation is applicable to all field operating agencies having Civil Works responsibilities.
- (c) **References.**
 - (1) ER 1110-2-100 (§ 222.2).
 - (2) ER 1110-2-1806.

(3) ER 1110-2-8150.

(4) ER 1130-2-419.

(5) State-of-the-Art for Assessing Earthquake Hazards in the United States—WES Miscellaneous Papers S-73-1—Reports 1 thru 14. Available from U.S. Army Engineer Waterways Experiment Station, P.O. Box 631, Vicksburg, Mississippi 39180.

(d) **Policy.** Civil Works structures which could be caused to fail or partially fail by an earthquake and whose failure or partial failure would endanger the lives of the public and/or cause substantial property damage, will be evaluated following potentially damaging earthquakes to insure their continued structural stability, safety and operational adequacy. These structures include dams, navigation locks, powerhouses, and appurtenant structures, (intakes, outlet works, buildings, tunnels, paved spillways) which are operated by the Corps of Engineers and for which the Corps is fully responsible. Also included are major levees, floodwalls, and similar facilities designed and constructed by the Corps of Engineers and for whose structural safety and stability the Corps has a public obligation to be aware of although not responsible for their maintenance and operation. The evaluation of these structures will be based upon post-earthquake inspections which will be conducted to detect conditions of significant structural distress and to provide a basis for timely initiation of restorative and remedial measures.

(e) **Post-earthquake inspections and evaluation surveys —**

(1) **Limitations of present knowledge.** The design of structures for earthquake loading is limited by the infrequent opportunity to compare actual performance with the design. Damage which would affect the function of the project is unlikely if peak accelerations are below 0.1g.; but it cannot be assumed that a structure will not be damaged from earthquake loadings below that for which it was designed. Furthermore, earthquakes have occurred in several parts of the country where significant seismic activity had not been predicted by some seismic zoning maps. This indicates the possibility that earthquake induced loads may not have been adequately considered in the design of older structures.

(2) **Types of reportable damage.** Many types of structural damage can be induced by ground motion from earthquakes or from large nuclear blasts (which also tend to induce ground vibrations in the more damaging lower frequency ranges). Any post-earthquake change in appearance or functional capability of a major Civil Works structure should be evaluated and reported. Examples are symptoms of induced stresses in buildings made evident by cracked plaster, windows or tile, or in binding of doors or windows; cracked or shifted bridge pier footings or other concrete structures; turbidity or changed static level of water wells; cracks in concrete dams or earth embankments; and misalignment of hydraulic control structures or gates. Induced dynamic loading on earth dams may result in loss of freeboard by settlement, or cause localized quick conditions within the embankment sections or earth foundations. Also, new seepage paths may be opened up within the foundation or through the embankment section. Ground motion induced landslides may occur in susceptible areas of the reservoir rim, causing embankment overtopping by waves and serious damage. All such unusual conditions should be evaluated and reported.

(f) **Inspection and evaluation programs.**

(1) If the project is located in an area where the earthquake causes significant damage (Modified Mercalli Intensity VII or greater) to structures in the vicinity, the Chief, Engineering Division, should be immediately notified and an engineering evaluation and inspection team should be sent to the project.

- (2) If the project is located in an area where the earthquake is felt but causes no or insignificant damage (Modified Mercalli Intensity VI or less) to structures in the vicinity of the project, project operations personnel should make an immediate inspection. This inspection should determine:
 - (i) Whether there is evidence of earthquake damage or disturbance, and
 - (ii) whether seismic instrumentation, where present, has been triggered. The Chief, Engineering Division should be notified by phone of the results of the inspection. If damage is observed, which is considered to threaten the immediate safety or operational capability of the project, immediate action should be taken as covered in paragraph (f)(1) of this section. For other situations, the Chief of Engineering Division will determine the need for and urgency for an engineering inspection.
- (3) When an engineering inspection of structures is deemed necessary following a significant earthquake, HQDA (DAEN-CWE) WASH DC 20314 will be notified of the inspection program as soon as it is established.
- (4) As a general rule, the structures which would be of concern following an earthquake are also the structures which are involved in the inspection program under ER 1110-2-100. Whenever feasible, instrumentation and prototype testing programs undertaken under ER 1110-2-100 to monitor structural performance and under ER 1110-2-8150 to develop design criteria will be utilized in the post-earthquake safety evaluation programs. Additional special types of instrumentation will be incorporated in selected structures in which it may be desirable to measure forces, pressures, loads, stresses, strains, displacements, deflections, or other conditions relating to damage and structural safety and stability in case of an earthquake.
- (5) Where determined necessary, a detailed, systematic engineering inspection will be made of the post-earthquake condition of each structure, taking into account its distinctive features. For structures which have incurred earthquake damage a formal technical report will be prepared in a format similar to inspection reports required under ER 1110-2-100. (Exempt from requirements control under paragraph 7-2b, AR 335-15.) The report will include summaries of the instrumentation and other observation data for each inspection, for permanent record and reference purposes. This report will be used to form a basis for major remedial work when required. Where accelerometers or other types of strong motion instruments have been installed, readings and interpretations from these instruments should also be included in the report. The report will contain recommendations for remedial work when appropriate, and will be transmitted through the Division Engineer for review and to HQDA (DAEN-CWE) WASH DC 20314 for review and approval. For structures incurring no damage a simple statement to this effect will be all that is required in the report, unless seismic instrumentation at the project is activated. (See paragraph (h)(4) of this section.)
- (g) **Training.** The dam safety training program covered by paragraph 6 of ER 1130-2-419 should include post-earthquake inspections and the types of damage operations personnel should look for.
- (h) **Responsibilities.**
 - (1) The Engineering Divisions of the District offices will formulate the inspection program, conduct the post-earthquake inspections, process and analyze the data of instrumental and other observations, evaluate the resulting condition of the structures, and prepare the inspection reports. The Engineering division is also responsible for planning special instrumentation felt necessary in selected structures under this program. Engineering Division is responsible for providing the training discussed in paragraph (g) of this section.

- (2) The Construction Divisions of the District offices will be responsible for the installation of the earthquake instrumentation devices and for data collection if an earthquake occurs during the construction period.
- (3) The Operations Division of the District offices will be responsible for the immediate assessment of earthquake damage and notifying the Chief, Engineering Division as discussed in paragraphs (f)(1) and (2). The Operations Division will also be responsible for earthquake data collection after the construction period in accordance with the instrumental observation programs, and will assist and participate in the post-earthquake inspections.
- (4) The U.S. Geological Survey has the responsibility for servicing and collecting all data from strong motion instrumentation at Corps of Engineers dam projects following an earthquake occurrence. However, the U.S. Army Waterways Experiment Station (WES) is assigned the responsibility for analyzing and interpreting these earthquake data. Whenever a recordable earthquake record is obtained from seismic instrumentation at a Corps project, the Division will send a report of all pertinent instrumentation data to the Waterways Experiment Station, ATTN: WESGH, P.O. Box 631, Vicksburg, Mississippi 39180. The report on each project should include a complete description of the locations and types of instruments and a copy of the instrumental records from each of the strong motion machines activated. (Exempt from requirements control under paragraph 7-2v, AR 335-15).
- (5) The Engineering Divisions of the Division offices will select structures for special instrumentation for earthquake effects, and will review and monitor the data collection, processing, evaluating, and inspecting activities. They will also be specifically responsible for promptly informing HQDA (DAEN-CWE) WASH DC 20314, when evaluation of the condition of the structure or analyses of the instrumentation data indicate the stability of a structure is questionable. (Exempt for requirements control under paragraph 7-2o, AR 335-15.)
- (6) Division Engineers are responsible for issuing any supplementary regulations necessary to adapt the policies and instructions herein to the specific conditions within their Division.
- (i) **Funding.** Funding for the evaluation and inspection program will be under the Appropriation 96X3123, Operations and Maintenance, General. Funds required for the inspections, including Travel and Per Diem costs incurred by personnel of the Division office or the Office, Chief of Engineers, will be from allocations made to the various projects for the fiscal year in which the inspection occurs.

[44 FR 43469, July 25, 1979. Redesignated at 60 FR 19851, Apr. 21, 1995]

§ 222.5 Water control management (ER 1110-2-240).

- (a) **Purpose.** This regulation prescribes policies and procedures to be followed by the U.S. Army Corps of Engineers in carrying out water control management activities, including establishment of water control plans for Corps and non-Corps projects, as required by Federal laws and directives.
- (b) **Applicability.** This regulation is applicable to all field operating activities having civil works responsibilities.
- (c) **References.** Appendix A lists U.S. Army Corps of Engineers publications and sections of Federal statutes and regulations that are referenced herein.
- (d) **Authorities —**

(1) ***U.S. Army Corps of Engineers projects.*** Authorities for allocation of storage and regulation of projects owned and operated by the Corps of Engineers are contained in legislative authorization acts and referenced project documents. These public laws and project documents usually contain provisions for development of water control plans, and appropriate revisions thereto, under the discretionary authority of the Chief of Engineers. Some modifications in project operation are permitted under congressional enactments subsequent to original project authorization. Questions that require interpretations of authorizations affecting regulation of specific reservoirs will be referred to CDR USACE (DAEN-CWE-HW), WASH DC 20314, with appropriate background information and analysis, for resolution.

(2) ***Non-Corps projects.*** The Corps of Engineers is responsible for prescribing flood control and navigation regulations for certain reservoir projects constructed or operated by other Federal, non-Federal or private agencies. There are several classes of such projects: Those authorized by special acts of Congress; those for which licenses issued by the Federal Energy Regulatory Commission (formerly Federal Power Commission) provide that operation shall be in accordance with instructions of the Secretary of the Army; those covered by agreements between the operating agency and the Corps of Engineers; and those that fall under the terms of general legislative and administrative provisions. These authorities, of illustrative examples, are described briefly in Appendix B.

(e) ***Terminology: Water control plans and reservoir regulation schedules.***

(1) Water control plans include coordinated regulation schedules for project/system regulation and such additional provisions as may be required to collect, analyze and disseminate basic data, prepare detailed operating instructions, assure project safety and carry out regulation of projects in an appropriate manner.

(2) The term "reservoir regulation schedule" refers to a compilation of operating criteria, guidelines, rule curves and specifications that govern basically the storage and release functions of a reservoir. In general, schedules indicate limiting rates of reservoir releases required during various seasons of the year to meet all functional objectives of the particular project, acting separately or in combination with other projects in a system. Schedules are usually expressed in the form of graphs and tabulations, supplemented by concise specifications.

(f) ***General policies.***

(1) Water control plans will be developed for reservoirs, locks and dams, reregulation and major control structures and interrelated systems to conform with objectives and specific provisions of authorizing legislation and applicable Corps of Engineers reports. They will include any applicable authorities established after project construction. The water control plans will be prepared giving appropriate consideration to all applicable Congressional Acts relating to operation of Federal facilities, *i.e.*, Fish and Wildlife Coordination Act (Pub. L. 85-624), Federal Water Project Recreation Act-Uniform Policies (Pub. L. 89-72), National Environmental Policy Act of 1969 (Pub. L. 91-190), and Clean Water Act of 1977 (Pub. L. 95-217). Thorough analysis and testing studies will be made as necessary to establish the optimum water control plans possible within prevailing constraints.

(2) Necessary actions will be taken to keep approved water control plans up-to-date. For this purpose, plans will be subject to continuing and progressive study by personnel in field offices of the Corps of Engineers. These personnel will be professionally qualified in technical areas involved and familiar with comprehensive project objectives and other factors affecting water control. Organizational requirements for water control management are further discussed in ER 1110-2-1400.

- (3) Water control plans developed for specific projects and reservoir systems will be clearly documented in appropriate water control manuals. These manuals will be prepared to meet initial requirements when storage in the reservoir begins. They will be revised as necessary to conform with changing requirements resulting from developments in the project area and downstream, improvements in technology, new legislation and other relevant factors, provided such revisions comply with existing Federal regulations and established Corps of Engineers policy.
- (4) Development and execution of water control plans will include appropriate consideration for efficient water management in conformance with the emphasis on water conservation as a national priority. The objectives of efficient water control management are to produce beneficial water savings and improvements in the availability and quality of water resulting from project regulation/operation. Balanced resource use through improved regulation should be developed to conserve as much water as possible and maximize all project functions consistent with project/system management. Continuous examination should be made of regulation schedules, possible need for storage reallocation (within existing authority and constraints) and to identify needed changes in normal regulation. Emphasis should be placed on evaluating conditions that could require deviation from normal release schedules as part of drought contingency plans (ER 1110-2-1941).
- (5) Adequate provisions for collection, analysis and dissemination of basic data, the formulation of specific project regulation directives, and the performance of project regulation will be established at field level.
- (6) Appropriate provisions will be made for monitoring project operations, formulating advisories to higher authorities, and disseminating information to others concerned. These actions are required to facilitate proper regulation of systems and to keep the public fully informed regarding all pertinent water control matters.
- (7) In development and execution of water control plans, appropriate attention will be given to project safety in accordance with ER 1130-2-417 and ER 1130-2-419 so as to insure that all water impounding structures are operated for the safety of users of the facilities and the general public. Care will be exercised in the development of reservoir regulation schedules to assure that controlled releases minimize project impacts and do not jeopardize the safety of persons engaged in activities downstream of the facility. Water control plans will include provisions for issuing adequate warnings or otherwise alerting all affected interests to possible hazards from project regulation activities.
- (8) In carrying out water control activities, Corps of Engineers personnel must recognize and observe the legal responsibility of the National Weather Service (NWS), National Oceanic and Atmospheric Administration (NOAA), for issuing weather forecasts and flood warnings, including river discharges and stages. River forecasts prepared by the Corps of Engineers in the execution of its responsibilities should not be released to the general public, unless the NWS is willing to make the release or agrees to such dissemination. However, release to interested parties of factual information on current storms or river conditions and properly quoted NWS forecasts is permissible. District offices are encouraged to provide assistance to communities and individuals regarding the impact of forecasted floods. Typical advice would be to provide approximate water surface elevations at locations upstream and downstream of the NWS forecasting stream gages. Announcement of anticipated changes in reservoir release rates as far in advance as possible to the general public is the responsibility of Corps of Engineers water control managers for projects under their jurisdiction.

- (9) Water control plans will be developed in concert with all basin interests which are or could be impacted by or have an influence on project regulation. Close coordination will be maintained with all appropriate international, Federal, State, regional and local agencies in the development and execution of water control plans. Effective public information programs will be developed and maintained so as to inform and educate the public regarding Corps of Engineers water control management activities.
- (10) Fiscal year budget requests for water control management activities will be prepared and submitted to the Office of the Chief of Engineers in accordance with requirements established in Engineer Circular on Annual Budget Requests for Civil Works Activities. The total annual costs of all activities and facilities that support the water control functions, (excluding physical operation of projects, but including flood control and navigation regulation of projects subject to 33 CFR 208.11) are to be reported. Information on the Water Control Data Systems and associated Communications Category of the Plant Replacement and Improvement Program will be submitted with the annual budget. Reporting will be in accordance with the annual Engineer Circular on Civil Works Operations and Maintenance, General Program.

(g) **Responsibilities: US Army Corps of Engineers projects –**

- (1) **Preparation of water control plans and manuals.** Normally, district commanders are primarily responsible for background studies and for developing plans and manuals required for reservoirs, locks and dams, reregulation and major control structures and interrelated systems in their respective district areas. Policies and general guidelines are prescribed by OCE engineer regulations while specific requirements to implement OCE guidance are established by the division commanders concerned. Master Water Control Manuals for river basins that include more than one district are usually prepared by or under direct supervision of division representatives. Division commanders are responsible for providing such management and technical assistance as may be required to assure that plans and manuals are prepared on a timely and adequate basis to meet water control requirements in the division area, and for pertinent coordination among districts, divisions, and other appropriate entities.
- (2) **Public involvement and information –**
 - (i) **Public meeting and public involvement.** The Corps of Engineers will sponsor public involvement activities, as appropriate, to appraise the general public of the water control plan. In developing or modifying water control manuals, the following criteria is applicable.
 - (A) Conditions that require public involvement and public meetings include: Development of a new water control manual that includes a water control plan; or revision or update of a water control manual that changes the water control plan.
 - (B) Revisions to water control manuals that are administratively or informational in nature and that do not change the water control plan do not require public meetings.
 - (C) For those conditions described in paragraph (g)(2)(i)(A) of this section, the Corps will provide information to the public concerning proposed water control management decisions at least 30 days in advance of a public meeting. In so doing, a separate document(s) should be prepared that explains the recommended water control plan or change, and provides technical information explaining the basis for the recommendation. It should include a description of its impacts (both monetary and nonmonetary) for

various purposes, and the comparisons with alternative plans or changes and their effects. The plan or manual will be prepared only after the public involvement process associated with its development or change is complete.

(D) For those conditions described in paragraph (g)(2)(i)(A) of this section, the responsible division office will send each proposed water control manual to the Army Corps of Engineers Headquarters, Attn: CECW-EH-W for review and comments prior to approval by the responsible division office.

(ii) **Information availability.** The water control manual will be made available for examination by the general public upon request at the appropriate office of the Corps of Engineers. Public notice shall be given in the event of occurring or anticipated significant changes in reservoir storage or flow releases. The method of conveying this information shall be commensurate with the urgency of the situation and the lead time available.

(3) **Authority for approval of plans and manuals.** Division commanders are delegated authority for approval of water control plans and manuals, and associated activities.

(4) **OCE role in water control activities.** OCE will establish policies and guidelines applicable to all field offices and for such actions as are necessary to assure a reasonable degree of consistency in basic policies and practices in all Division areas. Assistance will be provided to field offices during emergencies and upon special request.

(5) **Methods improvement and staff training.** Division and district commanders are responsible for conducting appropriate programs for improving technical methods applicable to water control activities in their respective areas. Suitable training programs should be maintained to assure a satisfactory performance capability in water control activities. Appropriate coordination of such programs with similar activities in other areas will be accomplished to avoid duplication of effort, and to foster desirable exchange of ideas and developments. Initiative in re-evaluating methods and guidelines previously established in official documents referred to in paragraph (e) of this section is encouraged where needs are evident. However, proposals for major deviations from basic concepts, policies and general practices reflected in official publications will be submitted to CDR USACE (DAEN-CWE) WASH DC 20314 for concurrence or comment before being adopted for substantial application in actual project regulation at field level.

(h) **Directives and technical instruction manuals.**

(1) Directives issued through OCE Engineer Regulations will be used to foster consistency in policies and basic practices. They will be supplemented as needed by other forms of communication.

(2) Engineering Manuals (EM) and Engineer Technical Letters (ETL) are issued by OCE to serve as general guidelines and technical aids in developing water control plans and manuals for individual projects or systems.

(3) EM 1110-2-3600 discusses principles and concepts involved in developing water control plans. Instructions relating to preparation of "Water Control Manuals for specific projects" are included. EM 1110-2-3600 should be used as a general guide to water control activities. The instructions are sufficiently flexible to permit adaptation to specific regions. Supplemental information regarding technical methods is provided in numerous documents distributed to field offices as "hydrologic references."

- (4) Special assistance in technical studies is available from the Hydrologic Engineering Center, Corps of Engineers, 609 Second Street, Davis, California 95616 and DAEN-CWE-HW.

(i) ***Water control manuals for US Army Corps of Engineers projects.***

- (1) As used herein, the term “water control manual” refers to manuals that relate primarily to the functional regulation of an individual project or system of projects. Although such manuals normally include background information concerning physical features of projects, they do not prescribe rules or methods for physical maintenance or care of facilities, which are covered in other documents. (References 15 and 23, appendix A.)
- (2) Water control manuals prepared in substantially the detail and format specified in instructions referred to in paragraph 8 are required for all reservoirs under the supervision of the Corps of Engineers, regardless of the purpose or size of the project. Water Control manuals are also required for lock and dam, reregulation and major control structure projects that are physically regulated by the Corps of Engineers. Where there are several projects in a drainage basin with interrelated purposes, a “Master Manual” shall be prepared. The effects of non-Corps projects will be considered in appropriate detail, including an indication of provisions for interagency coordination.
- (3) “Preliminary water control manuals,” for projects regulated by the Corps of Engineers should contain regulation schedules in sufficient detail to establish the basic plan of initial project regulation.
- (4) As a general rule, preliminary manuals should be superseded by more detailed interim or “final” manuals within approximately one year after the project is placed in operation.
- (5) Each water control manual will contain a section on special regulations to be conducted during emergency situations, including droughts. Preplanned operations and coordination are essential to effective relief or assistance.
- (6) One copy of all water control manuals and subsequent revisions shall be forwarded to DAEN-CWE-HW for file purposes as soon as practicable after completion, preferably within 30 days from date of approval at the division level.

(j) ***Policies and requirements for preparing regulations for non-Corps projects.***

- (1) Division and district commanders will develop water control plans as required by section 7 of the 1944 Flood Control Act, the Federal Power Act and section 9 of Pub. L. 436-83 for all projects located within their areas, in conformance with ER 1110-2-241, 33 CFR part 208. That regulation prescribes the policy and general procedures for regulating reservoir projects capable of regulation for flood control or navigation, except projects owned and operated by the Corps of Engineers; the International Boundary and Water Commission, United States and Mexico; those under the jurisdiction of the International Joint Commission, United States and Canada, and the Columbia River Treaty. ER 1110-2-241, 33 CFR part 208 permits the promulgation of specific regulations for a project in compliance with the authorizing acts, when agreement on acceptable regulations cannot be reached between the Corps Engineers and the owners. Appendix B provides a summary of the Corps of Engineers responsibilities for prescribing regulations for non-Corps reservoir projects.
- (2) Water control plans will be developed and processed as soon as possible for applicable projects already completed and being operated by other entities, including projects built by the Corps of Engineers and turned over to others for operation.

- (3) In so far as practicable, water control plans for non-Corps projects should be developed in cooperation with owning/operating agencies involved during project planning stages. Thus, tentative agreements on contents, including pertinent regulation schedules and diagrams, can be accomplished prior to completion of the project.
- (4) The magnitude and nature of storage allocations for flood control or navigation purposes in non-Corps projects are governed basically by conditions of project authorizations or other legislative provisions and may include any or all of the following types of storage assignments:
 - (i) Year-round allocations: Storage remains the same all year.
 - (ii) Seasonal allocations: Storage varies on a fixed seasonal basis.
 - (iii) Variable allocations of flood control from year to year, depending on hydrologic parameters, such as snow cover.
- (5) Water control plans should be developed to attain maximum flood control or navigation benefits, consistent with other project requirements, from the storage space provided for these purposes. When reservoir storage capacity of the category referred to in paragraph (j)(4)(iii) is utilized for flood control or navigation, jointly with other objectives, the hydrologic parameters and related rules developed under provisions of ER 1110-2-241, 33 CFR part 208 should conform as equitably as possible with the multiple-purpose objectives established in project authorizations and other pertinent legislation.
- (6) Storage allocations made for flood control or navigation purposes in non-Corps projects are not subject to modifications by the Corps of Engineers as a prerequisite for prescribing 33 CFR 208.11 regulations. However, regulations developed for use of such storage should be predicated on a mutual understanding between representatives of the Corps and the operating agency concerning the conditions of the allocations in order to assure reasonable achievement of basic objectives intended. In the event field representatives of the Corps of Engineers, and the operating agency are unable to reach necessary agreements after all reasonable possibilities have been explored, appropriate background explanations and recommendations should be submitted to DAEN-CWE-HW for consideration.
- (7) The Chief of Engineers is responsible for prescribing regulations for use of flood control or navigation storage and/or project operation under the provisions of the referenced legislative acts. Accordingly, any regulations established should designate the division/district commander who is responsible to the Chief of Engineers as the representative to issue any special instructions required under the regulation. However, to the extent practicable, project regulations should be written to permit operation of the project by the owner without interpretations of the regulations by the designated representative of the Commander during operating periods.
- (8) Responsibility for compliance with 33 CFR 208.11 regulations rests with the operating agency. The division or district commander of the area in which the project is located will be kept informed regarding project operations to verify reasonable conformance with the regulations. The Chief of Engineers or his designated representative may authorize or direct deviation from the established water control plan when conditions warrant such deviation. In the event unapproved deviations from the prescribed regulations seem evident, the division or district commander concerned will bring the matter to the attention of the operating agency by appropriate means.

If corrective actions are not taken promptly, the operating agency should be notified of the apparent deviation in writing as a matter of record. Should an impasse arise, in that the project owner or the designated operating entity persists in noncompliance with regulations prescribed by the Corps of Engineers, the Office of Chief Counsel should be advised through normal channels and requested to take necessary measures to assure compliance.

(9) Regulations should contain information regarding the required exchange of basic data between the representative of the operating agency and the U.S. Army Corps of Engineers, that are pertinent to regulation and coordination of interrelated projects in the region.

(10) All 33 CFR 208.11 regulations shall contain provisions authorizing the operating agency to temporarily deviate from the regulations in the event that it is necessary for emergency reasons to protect the safety of the dam, to avoid health hazards, and to alleviate other critical situations.

(k) ***Developing and processing regulations for non-Corps projects.*** Guidelines concerning technical studies and development of regulations are contained in ER 1110-2-241, 33 CFR part 208 and EM 1110-2-3600. Appendix C of this regulation summarizes steps normally followed in developing and processing regulations for non-Corps projects.

(l) ***Water control during project construction stage.*** Water control plans discussed in preceding paragraphs are intended primarily for application after the dam, spillway and outlet structures; major relocations; land acquisitions, administrative arrangements and other project requirements have reached stages that permit relatively normal project regulation. With respect to non-Corps projects, regulations normally become applicable when water control agreements have been signed by the designated signatories, subject to special provisions in specific cases. In some instances, implementation of regulations has been delayed by legal provisions, contract limitations, or other considerations. These delays can result in loss of potential project benefits and possible hazards. Accordingly, it is essential that appropriate water control and contingency plans be established for use from the date any storage may accumulate behind a partially completed dam until the project is formally accepted for normal operations. Division commanders shall make certain that construction-stage regulation plans are established and maintained in a timely and adequate manner for projects under the supervision of the Corps of Engineers. In addition, the problems referred to should be discussed with authorities who are responsible for non-Corps projects, with the objective of assuring that such projects operate as safely and effectively as possible during the critical construction stage and any period that may elapse before regular operating arrangements have been established. These special regulation plans should include consideration for protection of construction operations; safety of downstream interests that might be jeopardized by failure of partially completed embankments; requirements for minimizing adverse effects on partially completed relocations or incomplete land acquisition; and the need for obtaining benefits from project storage that can be safely achieved during the construction and early operation period.

(m) ***Advisories to OCE regarding water control activities —***

(1) ***General.*** Division commanders will keep the Chief of Engineers currently informed of any unusual problems or activities associated with water control that impact on his responsibilities.

(2) ***Annual division water control management report (RCS DAEN-CWE-16(R1)).*** Division commanders will submit an annual report on water control management activities within their division. The annual report will be submitted to (DAEN-CWE-HW) by 1 February each year and cover significant activities of the previous water year and a description of activities to be accomplished for the current year. Funding information for water control activities will be provided in the letter of transmittal for in-

house use only. The primary objective of this summary is to keep the Chief of Engineers informed regarding overall water management activities Corps-wide, thus providing a basis to carry out OCE responsibilities set forth in paragraph (g)(4) of this section.

- (3) **Status of water control manuals.** A brief discussion shall be prepared annually by each division commander, as a separate section of the annual report on water control management activities discussed in paragraph (m)(2) of this section listing all projects currently in operation in his area, or expected to begin operation within one-year, with a designation of the status of water control manuals. The report should also list projects for which the Corps of Engineers is responsible for prescribing regulations, as defined in ER 1110-2-241, 33 CFR part 208.
- (4) **Monthly water control charts (RCS DAEN-CWE-6 (R1)).** A monthly record of reservoirs/lakes operated by the Corps of Engineers and other agencies, in accordance with 33 CFR 208.11, will be promptly prepared and maintained by district/division commanders in a form readily available for transmittal to the Chief of Engineers, or others, upon request. Record data may be prepared in either graphical form as shown in EM 1110-2-3600, or tabular form as shown in the sample tabulation in appendix D.
- (5) **Annual division water quality reports (RCS DAEN-CWE-15).** By Executive Order 12088, the President ordered the head of each Executive Agency to be responsible for ensuring that all necessary actions are taken for prevention, control, and abatement of environmental pollution with respect to Federal facilities and activities under control of the agency. General guidance is provided in references 24 and 25, appendix A, for carrying out this agency's responsibility. Annual division water quality reports are required by reference 24, appendix A. The report is submitted in two parts. The first part addresses the division Water quality management plan while the second part presents specific project information. A major objective of this report is to summarize information pertinent to water quality aspects of overall water management responsibilities. The annual division water quality report may be submitted along with the annual report on water control management activities discussed in paragraph 13b above.
- (6) **Master plans for water control data systems (RCS DAEN-CWE-21).**
 - (i) A water control data system is all of the equipment within a division which is used to acquire, process, display and distribute information for real-time project regulation and associated interagency coordination. A subsystem is all equipment as defined previously within a district. A network is all equipment as defined previously which is used to regulate a single project or a group of projects which must be regulated interdependently.
 - (ii) Master plans for water control data systems and significant revisions thereto will be prepared by division water control managers and submitted to DAEN-CWE-HW by 1 February each year for review and approval of engineering aspects. Engineering approval does not constitute funding approval. After engineering approval is obtained, equipment in the master plan is eligible for consideration in the funding processes described in ER 1125-2-301 and engineering circulars on the annual budget request for civil works activities. Master plans will be maintained current and will:
 - (A) Outline the system performance requirements, including those resulting from any expected expansions of Corps missions.
 - (B) Describe the extent to which existing facilities fulfill performance requirements.
 - (C) Describe alternative approaches which will upgrade the system to meet the requirements not fulfilled by existing facilities, or are more cost effective than the existing system.

- (D) Justify and recommend a system considering timeliness, reliability, economics and other factors deemed important.
 - (E) Delineate system scope, implementation schedules, proposed annual capital expenditures by district, total costs, and sources of funding.
 - (iii) Modified master plans should be submitted to DAEN-CWE-HW by 1 February, whenever revisions are required, to include equipment not previously approved or changes in scope or approach. Submittal by the February date will allow adequate time for OCE review and approval prior to annual budget submittals.
 - (iv) Division commanders are delegated authority to approve detailed plans for subsystems and networks of approved master plans. Plans approved by the division commander should meet the following conditions:
 - (A) The plan conforms to an approved master plan.
 - (B) The equipment is capable of functioning independently.
 - (C) An evaluation of alternatives has been completed considering reliability, cost and other important factors.
 - (D) The plan is economically justified, except in special cases where legal requirements dictate performance standards which cannot be economically justified.
 - (v) Copies of plans approved by the division commander shall be forwarded to appropriate elements in OCE in support of funding requests and to obtain approval of Automatic Data Processing Equipment (ADPE), when applicable.
 - (vi) Water control data systems may be funded from Plant Revolving Fund; O&M General; Flood Control, MR&T, and Construction, General. Funding for water control equipment that serves two or more projects will be from Plant Revolving Fund in accordance with ER 1125-2-301. District and division water control managers will coordinate plant revolving fund requests with their respective Plant Replacement and Improvement Program (PRIP) representatives following guidance provided in ER 1125-2-301. Budget funding requests under the proper appropriation title should be submitted only if the equipment is identified in an approved master plan.
 - (vii) Justification for the Automatic Data Processing Equipment (ADPE) aspects of water control data systems must conform to AR 18-1, Appendix I or J as required. The "Funding for ADPE" paragraph in Appendixes I and J must cite the source of funds and reference relevant information in the approved master plan and detailed plan.
 - (viii) Division water control managers will submit annual letter summaries of the status of their respective water control systems and five-year plan for improvements. These summaries will be submitted to DAEN-CWE by 1 June for coordination with DAEN-CWO, CWB and DSZ-A, prior to the annual budget request. Summaries should not be used to obtain approval of significant changes in master plans. Sources of funding for all items for each district and for the division should be delineated so that total system expenditures and funding requests are identified. Changes in the master plan submitted 1 February should be documented in this letter summary if the changes were approved.
- (7) *Summary of runoff potentials in current season (RCS DAEN-CWO-2).*

- (i) The Chief of Engineers and staff require information to respond to inquiries from members of Congress and others regarding runoff potentials. Therefore, the division commander will submit a snowmelt runoff and flood potential letter report covering the snow accumulation and runoff period, beginning generally in February and continuing monthly, until the potential no longer exist. Dispatch of supplemental reports will be determined by the urgencies of situations as they occur. The reports will be forwarded as soon as hydrologic data are available, but not later than the 10th of the month. For further information on reporting refer to ER 500-1-1, 33 CFR part 203.
- (ii) During major drought situations or low-flow conditions, narrative summaries of the situation should be furnished to alert the Chief of Engineers regarding the possibility of serious runoff deficiencies that are likely to call for actions associated with Corps of Engineers reservoirs.
- (iii) The reports referred to in paragraphs (m)(7) (i) and (ii) of this section will include general summaries regarding the status of reservoir storage, existing and forecasted at the time of the reports.

(8) **Reports on project operations during flood emergencies.** Information on project regulations to be included in reports submitted to the Chief of Engineers during flood emergencies in accordance with ER 500-1-1 include rate of inflow and outflow in CFS, reservoir levels, predicted maximum level and anticipated date, and percent of flood control storage utilized to date. Maximum use should be made of computerized communication facilities in reporting project status to DAEN-CWO-E/CWE-HW in accordance with the requirements of ER 500-1-1, 33 CFR part 203.

(9) **Post-flood summaries of project regulation.** Project regulation effects including evaluation of the stage reductions at key stations and estimates of damages prevented by projects will be included in the post flood reports required by ER 500-1-1, 33 CFR part 203.

(n) **Water Control Management Boards.**

(1) The Columbia River Treaty Permanent Engineering Board was formed in accordance with the Columbia River Treaty with Canada. This board, composed of U.S. and Canadian members, oversees the implementation of the Treaty as carried out by the U.S. and Canadian Entities.

(2) The Mississippi River Water Control Management Board was established by ER 15-2-13. It consists of the Division Commanders from LMVD, MRD, NCD, ORD, and SWD with the Director of Civil Works serving as chairman. The purposes of the Board are:

- (i) To provide oversight and guidance during the development of basin-wide management plans for Mississippi River Basin projects for which the US Army Corps of Engineers has operation/ regulation responsibilities.
- (ii) To serve as a forum for resolution of water control problems among US Army Corps of Engineers Divisions within the Mississippi River Basin when agreement is otherwise unobtainable.

(o) **List of projects.** Projects owned and operated by the Corps of Engineers subject to this regulation are listed with pertinent data in Appendix E. This list will be updated periodically to include Corps projects completed in the future. Federal legislation, Federal regulations and local agreements have given the Corps of Engineers wide responsibilities for operating projects which it does not own. Non-Corps projects subject to this regulation are included in Appendix A of ER 1110-2-241.

Appendix A to § 222.5—References

1. The Federal Power Act, Pub. L. 436-83, approved 10 June 1920, as amended (41 Stat. 1063; 16 U.S.C. 791(a))
2. Section 3 of the Flood Control Act approved 22 June 1936, as amended (49 Stat. 1571; 33 U.S.C. 701(c))
3. Section 9(b) of Reclamation Project Act of 1939, approved 4 August 1939 (53 Stat. 1187; 43 U.S.C. 485)
4. Section 7 of the Flood Control Act approved 22 December 1944 (58 Stat. 890; 33 U.S.C. 709)
5. Section 5 of Small Reclamation Projects Act of 6 August 1956, as amended (70 Stat. 1046; 43 U.S.C. 422(e))
6. Section 9 of Pub. L. 436-83d Congress (68 Stat. 303)
7. The Fish and Wildlife Coordination Act of 1958, Pub. L. 85-624
8. The Federal Water Project Recreation Act Uniform Policies, Pub. L. 89-72
9. The National Environmental Policy Act of 1969, Pub. L. 91-190
10. The Clean Water Act of 1977, Pub. L. 95-217
11. Executive Order 12088, Federal Compliance with Pollution Control Standards, 13 October 1978
12. 33 CFR 208.10, Local flood protection works; maintenance and operation of structures and facilities (9 FR 9999; 9 FR 10203)
13. 33 CFR 208.11, Regulations for use of Storage Allocated for Flood Control or Navigation and/or Project Operation at Reservoirs subject to Prescription of Rules and Regulations by the Secretary of the Army in the Interest of Flood Control and Navigation (43 FR 47184)
14. AR 18-1
15. ER 11-2-101
16. ER 15-2-13
17. ER 500-1-1, 33 CFR part 203
18. ER 1110-2-241, 33 CFR part 208
19. ER 1110-2-1400
20. ER 1110-2-1402
21. ER 1110-2-1941

- 22. ER 1125-2-301
- 23. ER 1130-2-303
- 24. ER 1130-2-334
- 25. ER 1130-2-415
- 26. ER 1130-2-417
- 27. ER 1130-2-419
- 28. EM 1110-2-3600

Appendix B to § 222.5—Summary of Corps of Engineers Responsibilities for Prescribing Regulations for Non-Corps Reservoir Projects

Summary

1. (a) "Regulations for Use of Storage Allocated for Flood Control or Navigation and/or Project Operation at Reservoirs subject to Prescription of Rules and Regulations by the Secretary of the Army in the Interest of Flood Control and Navigation" (33 CFR 208.11) prescribe the responsibilities and general procedures for regulating reservoir projects capable of regulation for flood control or navigation and the use of storage allocated for such purposes and provided on the basis of flood control and navigation, except projects owned and operated by the Corps of Engineers; the International Boundary and Water Commission, United States and Mexico; and those under the jurisdiction of the International Joint Commission, United States and Canada, and the Columbia River Treaty.

(b) Pertinent information on projects for which regulations are prescribed under Section 7 of the 1944 Flood Control Act, (Pub. L. 78-58 Stat. 890 (33 U.S.C. 709)) the Federal Power Act (41 Stat. 1063 (16 U.S.C. 791(A))) and Section 9 of Pub. L. 436-83d Congress (68 Stat. 303) is published in the FEDERAL REGISTER in accordance with 33 CFR 208.11.

Publication in the FEDERAL REGISTER establishes the fact and the date of a project's regulation plan promulgation.

2. Section 7 of Act of Congress approved 22 December 1944 (58 Stat. 890; 33 U.S.C. 709), reads as follows:

"Hereafter, it shall be the duty of the Secretary of War to prescribe regulations for the use of storage allocated for flood control or navigation at all reservoirs constructed wholly or in part with Federal funds provided on the basis of such purposes, and the operation of any such project shall be in accordance with such regulations: *Provided*, That this section shall not apply to the Tennessee Valley Authority, except that in case of danger from floods on the Lower Ohio and Mississippi Rivers the Tennessee Valley Authority is directed to regulate the release of water from the Tennessee River into the Ohio River in accordance with such instructions as may be issued by the War Department."

3. Section 9(b) of the Reclamation Project Act of 1939, approved 4 August 1939 (53 Stat. 1189, 43 U.S.C. 485), provides that the Secretary of the Interior may allocate to flood control or navigation as part of the cost of new projects or supplemental works; and that in connection therewith he shall consult with the Chief of Engineers and may perform any necessary investigations under a cooperative agreement with the Secretary of the Army. These projects are subject to 33 CFR 208.11 regulations.
4. Several dams have been constructed by State agencies under provisions of legislative acts wherein the Secretary of the Army is directed to prescribe rules and regulations for project operation in the interest of flood control and navigation. These projects are subject to 33 CFR 208.11 regulations.
5. There are few dams constructed under Emergency Conservation work authority or similar programs, where the Corps of Engineers has performed major repairs or rehabilitation, that are operated and maintained by local agencies which are subject to 33 CFR 208.11 regulations.
6. The Federal Power Act, approved 10 June 1920, as amended (41 Stat. 1063, 16 U.S.C. 791 (A)), established the Federal Power Commission, now Federal Energy Regulatory Commission (FERC), with authority to issue licenses for constructing, operating, and maintaining dams or other project works for the development of navigation, for utilization of water power and for other beneficial public uses in any streams over which Congress has jurisdiction. The Chief of Engineers is called upon for advice and assistance as needed in formulating reservoir regulation requirements somewhat as follows:
 - a. In response to requests from the FERC, opinions and technical appraisals are furnished by the Corps of Engineers for consideration prior to issuance of licenses by the FERC. Such assistance may be limited to general presentations, or may include relatively detailed proposals for water control plans, depending upon the nature and scope of projects under consideration. The information furnished is subject to such consideration and use as the Chairman, FERC, deems appropriate. This may result in inclusion of simple provisions in licenses without elaboration, or relatively detailed requirements for reservoir regulation schedules and plans.
 - b. Some special acts of Congress provide for construction of dams and reservoirs by non-Federal agencies or private firms under licenses issued by the FERC, subject to stipulation that the operation and maintenance of the dams shall be subject to reasonable rules and regulations of the Secretary of the Army in the interest of flood control and navigation. Ordinarily no Federal funds are involved, thus Section 7 of the 1944 Flood Control Act does not apply. However, if issuance of regulations by the Secretary of the Army is required by the authority under which flood control or navigation provisions are included as functions of the specific project or otherwise specified in the FERC license, regulation plans will be prescribed in accordance with 33 CFR 208.11 regulations.
7. Projects constructed by the Corps of Engineers for local flood protection purposes are subject to conditions of local cooperation as provided in Section 3 of the Flood Control Act approved 22 June 1936, as amended. One of those conditions is that a responsible local agency will maintain and operate all works after completion in accordance with regulations prescribed by the Secretary of the Army. Most such projects consist mainly of levees and flood walls with appurtenant drainage structures. Regulations for operation and maintenance of these projects has been prescribed by the Secretary of the Army in 33 CFR 208.10. When a reservoir is included in such a project, it may be appropriate to apply 33 CFR 208.10 in establishing regulations for operation, without requiring their publication in the FEDERAL REGISTER. For example, if the reservoir controls a small drainage area, has an uncontrolled flood control outlet with automatic operation or contains less than 12,500

acre-feet of flood control or navigation storage, 33 CFR 208.10 may be suitable. However, 33 CFR 208.11 regulations normally would be applicable in prescribing flood control regulations for the individual reservoir, if the project has a gated flood control outlet by which the local agency can regulate floods.

8. Regulation plans for projects owned by the Corps of Engineers are not prescribed in accordance with 33 CFR 208.11. However, regulation plans for projects constructed by the Corps of Engineers and turned over to other agencies or local interests for operation may be prescribed in accordance with 33 CFR 208.11.

9. The Small Reclamation Projects Act of 6 August 1956 provides that the Secretary of the Interior may make loans or grants to local agencies for the construction of reclamation projects. Section 5 of the Act provides in part that the contract covering any such grant shall set forth that operation be in accordance with regulations prescribed by the head of the Federal department or agency primarily concerned. Normally, 33 CFR 208.11 is not applicable to these projects.

Appendix C to § 222.5—Procedures for Developing and Processing Regulations for Non-Corps Projects in Conformance with 33 CFR 208.11

1. *Sequence of actions.* a. Discussions leading to a clarification of conditions governing allocations of storage capacity to flood control or navigation purposes and project regulation are initiated by District/Division Engineers through contacts with owners and/or operating agencies concerned at regional level.

b. Background information on the project and conditions requiring flood control or navigation services, and other relevant factors, are assembled by the District Engineer and incorporated in a "Preliminary Information Report". The Preliminary Information Report will be submitted to the Division Engineer for review and approval. Normally, the agency having jurisdiction over the particular project is expected to furnish information on project features, the basis for storage allocations and any other available data pertinent to the studies. The Corps of Engineers supplements this information as required.

c. Studies required to develop reservoir regulation schedules and plans usually will be conducted by Corps of Engineers personnel at District level, except where the project regulation affects flows in more than one district, in which case the studies will be conducted by or under supervision of Division personnel. Assistance as may be available from the project operating agency or others concerned will be solicited.

d. When necessary agreements are reached at district level, and regulations developed in accordance with 33 CFR 208.11 and EM 1110-2-3600, they will be submitted to the Division Commander for review and approval, with information copies for DAEN-CWE-HW. Usually the regulations include diagrams of operating parameters.

e. For projects owned by the Bureau of Reclamation, the respective Regional Directors are designated as duly authorized representatives of the Commissioner of Reclamation. By letter of 20 October 1976, the Commissioner delegated responsibilities to the Regional Directors as follows: "Regarding the designated authorization of representatives of the Commissioner of Reclamation in matters relating to the development and processing of Section 7 flood control regulations, we are designating each Regional Director as our duly authorized representative to sign all letters of understanding, water control agreements, water control diagrams, water control release schedules and other documents which may become part of the prescribed regulations. The Regional Director also will be responsible for obtaining the signature of the designated operating agency on these documents where such is required. Regarding internal coordination within the

Bureau of Reclamation, the Regional Directors will obtain the review and approval of this office and at appropriate offices with our Engineering and Research Center, Denver, Colorado, prior to signing water control documents.”

f. In accordance with the delegation cited in paragraph e, 33 CFR 208.11 regulations pertaining to Bureau of Reclamation projects will be processed as follows:

(1) After regulation documents submitted by District Commanders are reviewed and approved by the Division Commander they are transmitted to the respective Regional Director of the Bureau of Reclamation for concurrence of comment, with a request that tracings of regulation diagrams be signed and returned to the Division Commander.

(2) If any questions arise at this stage appropriate actions are taken to resolve differences. Otherwise, the duplicate tracings of the regulation diagram are signed by the Division Commander and transmitted to the office of the project owner for filing.

(3) After full agreement has been reached in steps (1) and (2), the text of proposed regulations is prepared in final form. Copies of any diagrams involved are included for information only.

(4) A letter announcing completion of action on processing the regulations, with pertinent project data as specified in paragraph 208.11(d)(11) of 33 CFR 208.11, and one copy of the signed tracings of diagrams are forwarded to HQDA (DAEN-CWE-HW) WASH DC 20314 for promulgation and filing. The office of the Chief of Engineers will forward the pertinent project data to the Liaison Officer with the Federal Register, requesting publication in the FEDERAL REGISTER.

g. Regulations developed in accordance with 33 CFR 208.11 and applicable to projects that are not under supervision of the Bureau of Reclamation are processed in substantially the manner described above. All coordination required between the Corps of Engineers and the operating agency will be accomplished at field level.

h. Upon completion of actions listed above, Division Commanders are responsible for informing the operating agencies at field level that regulations have been promulgated.

2. *Signature blocks:* Some 33 CFR 208.11 regulations contain diagrams of parameter curves that cannot be published in the FEDERAL REGISTER, but are made a part thereof by appropriate reference. Each diagram bears a title block with spaces for the signature of authenticating officials of the Corps of Engineers and the owner/operating agency of the project involved.

3. *Designation of Corps of Engineers Representatives.* Division Commanders are designated representatives of the Chief of Engineers in matters relating to development and processing of 33 CFR 208.11 regulations for eventual promulgation through publication of selected data specified in paragraph (d)(11) § 208.11. Division Commanders are designated as the Corps of Engineers signee on all letters of understanding, water control agreements and other documents which may become part of prescribed regulations for projects located in their respective geographic areas, and which are subject to the provisions of 33 CFR 208.11.

APPENDIX D TO § 222.5—SAMPLE TABULATION
BARDWELL LAKE, MONTHLY LAKE REPORT, MAY 1975

Day	Elevations 0800: 2,400 feet-MSL	Storage 2400 A-F	Evap DSF	Pump DSF	Release DSF	Inflow adj. DSF	Rain, inch
1	421.30 421.31	55979	28	2.0	0	84	0.00
2	421.32 421.37	56196	5	2.0	0	117	.00
3	421.43 421.44	56449	23	1.9	0	152	.14
4	421.45 421.47	56558	1	1.8	0	58	.00
5	421.49 421.34	56088	1	2.0	324	50	.00
6	421.20 421.01	54902	14	1.9	632	50	.00
7	420.88 420.89	54473	4	2.0	269	59	.09
8	420.89 420.91	54544	5	2.3	0	44	.00
9	420.90 420.89	54473	11	1.5	0	38	.00
10	420.90 420.90	54509	28	3.0	0	27	.00
11	420.91 421.35	56124	26	1.8	0	824	.00
12	421.54 421.65	57213	31	2.1	0	582	1.61
13	421.70 421.75	57578	29	2.2	0	216	.00
14	421.78 421.76	57614	34	1.9	249	303	.03
15	421.69 421.52	56739	22	1.9	643	225	.57
16	421.39 421.28	55871	39	2.1	535	138	.00
17	421.19 421.09	55188	10	2.2	393	119	.00
18	421.03 421.05	55045	46	2.0	143	60	.00
19	421.04 421.07	55116	17	2.3	0	55	.00
20	421.06 421.30	55943	21	2.1	0	440	.21
21	421.39 421.47	56558	20	2.1	0	332	.97
22	421.50 421.39	56268	42	2.1	247	145	.00
23	421.37 424.91	69726	31	2.0	328	7146	.22
24	425.61 426.15	74825	22	2.0	0	2595	2.38
25	426.15 426.55	76523	18	2.3	0	876	.11
26	426.72 426.80	77598	42	2.1	0	586	.00
27	426.95 427.00	78465	23	2.0	0	462	.00
28	427.14 427.15	79116	31	2.1	0	361	.19
29	427.31 427.70	81528	61	1.9	0	1279	.20
30	427.94 428.05	83082	11	2.0	0	796	1.02
31	428.20 428.22	83837	7	2.1	0	389	.00
Monthly total:							

Day	Elevations 0800: 2,400 feet-MSL	Storage 2400 A-F	Evap DSF	Pump DSF	Release DSF	Inflow adj. DSF	Rain, inch
(DSF)			700	64	3763	18626	7.74
(A-F)		27966	1389	126	7464	36945	

APPENDIX E TO § 222.5—LIST OF PROJECTS

Project name ¹	State/county	Stream ¹	Project purpose ²	Storage 1,000 AF	Elev limits feet M.S.L.		Area in acres		Auth legis ³
					Upper	Lower	Upper	Lower	
Lower Mississippi Valley Division									
Alligator—Catfish FG	MS Issaquena	Little Sunflower	F	0.0	0.0	0.0	0	0	FCA Jun 36.
Arkabutla Lk	MS Desoto	Coldwater	F	525.0	238.3	209.3	33,400	5,100	FCA Jun 36.
Ascalmore—Tippo FG & CS	MS Tallahatchie	Ascalmore	F	0.0	136.0	118.0	0	0	FCA Jun 36.
Bienvenue FG	LA St Bernard	Bayou Bienvenue	F	0.0	2.0	2.0	0	0	PL 298-89
Big Lk Ditch #81 CS	AR Mississippi	Ditch 81 Extension.	C	0.0	0.0	230.0	0	0	FCA Oct 65.
Big Lk Div CS	AR Mississippi	Little R	C	0.0	0.0	230.0	0	0	FCA Oct 65.
Big Lk North End CS	AR Mississippi	Little R	C	0.0	0.0	230.0	0	0	FCA Oct 65.
Big Lk South end CS	AR Mississippi	Ditch 28	C	0.0	0.0	230.0	0	0	FCA Oct 65.
Birds Point—New Madrid Div Floodway	MO New Madrid	Mississippi	F	0.0	330.5	328.5	131,000	71,000	FCA May 28.
Bodcau Lk	LA Bossier	Bayou Bodcau	F	35.3	199.5	157.0	21,000	110	PL 74-839.
Bonnet Carre Div Spillway	LA St Charles	Mississippi R	F	0.0	24.0	20.0	0	0	FCA May 28.
Bowman Lock	LA Vermilion	GIWW	I	0.0	1.2	1.2	0	0	PL 79-14.
Caddo Lk	LA Caddo	Cypress Bayou	N	128.6	182.7	168.5	59,000	26,800	FCA Oct 65.
Cairo 10th & 20th St PS	IL Pulaski	Ohio	F	0.0	310.5	299.0	0	0	PL 90-483.
Calcasieu SW Barrier & Lock	LA Calcasieu	Calcasieu R	I	0.0	1.2	1.2	0	0	RHA Oct 62. PL 79-525.
Calion L&D	AR Union	Ouachita	N	0.0	77.0	77.0	12,200	12,200	RHA 1950.
Calumet FG East & West	LA St Mary	Wax Lake Outlet Bayou Teche	FN	0.0	3.0	3.0	0	0	FCA Jun 36.
Cannon Re-reg	MO Ralls	Salt R	PCA	5.8	528.0	521.0	1,020	460	HD 507.
Carlyle Lk	IL Clinton	Kaskaskia R	F	699.0	462.5	445.0	50,440	24,580	SD 44.
			NMCAR	233.0	445.0	429.5	0	7,100	
Catahoula Lk CS	LA LaSalle	Catahoula Div	CR	118.0	34.0	27.0	25,000	94	RHA 1960.
Catfish Point CS	LA Cameron	Mermentau R	FN	0.0	1.2	1.2	0	0	FCA Aug 41, RHA Jul 64.

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					Upper	Lower	Upper	Lower	
Charenton FG	LA St Mary	Grand Lk	FN	0.0	0.0	0.0	0	0	RHA Jul 46, FCA May 28.
Cocodrie FG FG	LA Concorida	Bayou Cocodrie	F	0.0	46.0	13.0	0	0	FCA Aug 41.
Collins Cr	MS Warren	Collins Cr	F	0.0	84.0	67.0	0	0	FCA 1941.
Columbia L&D	LA Caldwell	Ouachita	N	0.0	52.0	52.0	7,070	7,070	RHA 1950.
Connerly CS	AR Chicot	Connerly Bayou	FCR	0.0	116.0	106.0	0	0	FCA Aug 68.
Courtableau Drainage CS	LA St Landry	Bayou Courtableau	F	0.0	18.0	16.0	0	0	FCA May 28, PL 391-70.
Darbonne CS	LA St. Landry	Bayou Darbonne	FI	0.0	18.0	16.0	0	0	FCA May 28, PL 391-70.
DeGray LK	AR Desoto	Caddo	FNPMRA	881.9	423.0	345.0	23,800	6,400	RHA 1950, WSA 1958.
DeGray Rereg. St	AR Clark	Caddo	NMRA	3.6	221.0	209.0	430	90	RHA 1950. WSA 1958.
Ditch Bayou Dam	AR Chicot	Ditch Bayou	FCR	0.0	106.0	93.0	0	0	FCA Aug 68.
Drainage Dist #17 PS	AR Mississippi	Ditch 71	F	3.0	236.0	228.0	4,100	0	FCA Aug 68, PL 90-483.
Drinkwater PS	MO Mississippi	Drinkwater Sewer	F	20.6	315.0	307.0	4,000	700	FCA May 50, PL 516.
Dupre FG	LA St Bernard	Bayou Dupre	F	0.0	2.0	2.0	0	0	PL 298-89.
East St Louis PS	IL St. Clair	IDD	F	0.0	0.0	0.0	0	0	FC Act 36.
Empire FG Hurr Prot & Lock	LA Plaque mines	Mississippi R	F	0.0	5.0	5.0	0	0	PL 874-87.
Enid Lk	MS Yalobusha	Yacona	F	660.0	268.0	230.0	28,000	6,100	FCA Jun 36.
Felsenthal L&D	AR Union	Ouachita	N	32.5	70.0	65.0	46,500	17,500	RHA 1950.
Finley Street PS	TN Dyer	Forked Deer	F	0.5	269.0	257.0	94	22	FCA 1948, PL 85-500.
Freshwater Lock	LA Vermilion	Freshwater Bayou	I NI	0.0	0.0	0.0	0	0	PL 86-645.
Graham Burke PS	AR Phillips	White	F	2,805.0	174.8	140.0	149,000	2,500	FCA May 28, PL 85-500.
Grenada Lk	MS Grenada	Yalobusha Skuna	F	1,357.4	231.0	193.0	64,600	9,800	FCA Jun 36.
Huxtable PS	AR Lee	St Francis	F	2,863.0	207.2	165.0	18,500	1,400	FCA May 50.
Jonesville L&D	LA Catahoula	Black	N	0.0	34.0	34.0	7,120	7,120	RHA 1950.
Kaskaskia L&D	IL Randolph	Kaskaskia R	N	1.1	368.0	363.0	1,300	1,200	SD 44.
L&D 1	LA Catahula	Red R	N	0.0	40.0	40.0	0	0	PL 90-483.
L&D 2	LA Rapides	Red R	N	0.0	71.2	64.0	0	0	PL 90-483.
L&D 3	LA Rapides	Red R	N	0.0	95.0	91.5	0	0	PL 90-483.

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					Upper	Lower	Upper	Lower	
L&D 4	LA Natchitoches	Red R	N	0.0	120.0	119.6	0	0	PL 90-483.
L&D 5	LA Red R	Red R	N	0.0	145.0	140.2	0	0	PL 90-483.
L&D 24	MO Pike	Mississippi R	N	29.7	449.0	445.0	13,000	12,000	R&H Act, Jul 3/30. R&H Act, Aug 30/35.
L&D 25	MO Lincoln	Mississippi R	N	49.7	434.0	429.7	18,000	16,600	R&H Act, Jul 3/30. R&H Act, 8/30/35.
L&D 26	IL Madison	Mississippi R	N	107.1	419.0	414.0	30,000	27,700	R&H Act, Jul 3/30. R&H Act, 8/30/1935.
Larose to Golden Meadow Hurr Prot FG	LA LaFourche	Bayou LaFourche	F	0.0	3.0	3.0	0	0	FCA Oct 65, PL 89-298.
Little Sun flower CS	MS Issaquena	Lit. Sunflower	F	0.0	85.0	60.0	0	0	FCA 1941.
Lk #9 Culvert & PS	KY Fulton	Mississippi	F	6.5	286.0	282.0	0	0	FCA Oct 65.
Lk Chicot PS	AR Chicot	Macon Lk	FCR	0.0	118.2	90.0	0	0	FCA Aug 68.
Lk Greeson	AR Pike	Little Missouri	P	0.0	563.0	436.9	0	0	FCA 1941.
			FP	407.9	563.0	504.0	9,800	2,500	
Lk Ouachita	AR Garland	Ouachita	P	0.0	592.0	480.0	0	0	FCA Dec 44.
Long Branch DS	LA Catahoula	Catahoula Div	F	0.0	32.5	32.5	0	0	FCA May 50.
Mark Twain Lk	MO Ralls	Salt R	F	894.0	638.0	606.0	38,400	18,600	HD 507.
			PMCAR	457.0	606.0	567.2	18,600	5,900	
Marked Tree Siphon	AR Poinsett	St. Francis	F	0.0	229.0	198.3	0	0	FCA Jun 30.
Morganza Div CS	LA Point Coupee	Morganza Floodway	F	0.0	59.5	49.0	0	0	FCA May 28.
Muddy Bayou CS	MS Warren	Muddy Bayou	FC	30.0	76.9	70.0	4,350	2,860	FCA Oct 65.
Old River Div CS Low Sill Overbank & Aux	LA W. Feliciana	Old R	F	0.0	70.0	5.0	0	0	PL 83-780.
Old River Lock	LA W Feliciana	Old R	N	0.0	65.4	10.0	0	0	FCA Sep 54, PL 780-83.
Port Allen Lock	LA Port Allen	GIWW	N	0.0	46.1	2.6	0	0	RHA Jul 46.
Prairie Dupont East & West PS	IL St Clair	IDD	F	0.0	0.0	0.0	0	0	FC Act 62.
Rapides-Boeuf Div Canal	LA Rapides	Bayou Rapides	F	0.0	66.0	62.2	0	0	FCA Aug 41, GD 359-77.

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					Upper	Lower	Upper	Lower	
CS									
Rend Lk	IL Franklin	Big Muddy R	F	109.0	405.0	410.0	24,800	18,900	HD 541.
			MA	160.0	405.0	391.3	18,900	5,400	
Sardis Lk	MS Panola	Little Sunflower	F	1,569.9	281.4	236.0	58,500	10,700	FCA Jun 36.
Schooner Bayou CS & Lock	LA Vermilion	Schooner Bayou	I	0.0	1.2	1.2	0	0	FCA Aug 41.
Shelbyville Lk	IL Shelby	Kaskaskia R	F	474.0	626.5	599.7	25,300	11,100	HD 232.
			NMCAR	180.0	599.7	573.0	11,100	3,000	
Sorrell Lock	LA Iberville	GIWW	N	0.0	29.7	3.5	0	0	FCA May 28.
St Francis Lk CS	AR Poinsett	Oak Donnick Floodway	C	0.0	0.0	210.0	0	2,240	FCA Oct 65.
Steele Bayou CS	MS Issaquena	Steele Bayou	F	0.0	68.5	60.0	0	0	FCA 1941.
Tchula Lk Lower FG	MS Humphreys	Tchula Lk	F	0.0	110.0	84.0	0	0	FCA Jun 36.
Tchula Lk Upper FG	MS Humphreys	Tchula Lk	F	0.0	108.0	92.0	0	0	FCA Jun 36.
Teche-Vermilion PS & CS	LA St Mary	Atchafalaya R	MI	0.1	18.0	16.0	0	0	PL 89-789, FCA May 28.
Tensas-Cocodrie PS	LA Cocordia	Bayou Corcodrie	F	0.0	37.0	23.0	0	0	FCA Oct 65.
Treasure Island PS	MO Dunklin	Little R	F	23.4	252.0	235.0	7,800	180	FCA Jul 46.
Wallace Lk	LA Caddo	Cypress Bayou	F	96.1	158.0	142.0	9,300	2,300	RHA Mar 45, PL 75-761.
Wappapello Lk	MO Wayne	St Francis R	F	613.2	394.7	354.7	23,200	5,200	HD 159.
Wasp Lk	MS Humphreys	Wasp Lk-Bear Cr	F	0.0	111.6	88.5	0	0	FCA Jun 36.
West Hickman PS	KY Fulton	Mississippi	F	0.0	302.0	296.0	9	4	FCA 1948.
Wood R PS	IL Madison	IDD	F	0.0	0.0	0.0	0	0	FC Act 38.
Yazoo City PS	MS Yazoo	Yazoo	F	0.0	96.0	69.0	0	0	FCA Jun 36.
Missouri River Division									
Bear Creek Dam & Res	CO Jefferson	Bear Cr	F	28.8	5,635.5	5,558.0	718	109	PL 90-483.
			FCR	1.9	5,558.0	5,528.0	109	17	SD 87-90.
Big Bend Dam & Lk Sharpe	SD Lyman Buffalo Hughes	Missouri R	F	61.0	1,423.0	1,422.0	61,000	60,000	PL 78-534.
			FNPIPCAR	117.0	1,422.0	1,420.0	60,000	57,000	SD 247-78.
Blue Springs Dam & Lk	MO Jackson	Little Blue R	F	15.8	820.0	802.0	982	722	PL 90-483.
			FRC	10.8	802.0	760.0	722	0	HD 169-90.

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					Upper	Lower	Upper	Lower	
Blue Stem Lake & Dam 4	NE Lancaster	Olive Br. Salt Creek	F	7.2	1,322.5	1,307.4	660	315	PL 85-500.
Bowman-Haley Dam & Res	ND Bowman	No Fk Grand River	FCR	3.0	1,307.4	1,277.0	315	1	HD 396-84.
			F	72.7	2,777.0	2,754.8	5,131	1,732	PL 87-874.
			FMCR	15.5	2,754.8	2,740.0	1,732	565	HD 574-87.
Branched Oak Lk & Dam 18	NE Lancaster	Oak Creek trib. Salt Creek	F	71.6	1,311.0	1,284.0	3,640	1,780	PL 85-500.
			FCR	26.0	1,284.0	1,250.0	1,780	0	HD 396-84.
Bull Hook Dam	MT Hill	Bull Hook Cr Scott Coulee	F	6.5	2,593.0	2,540.0	283	0	PL 78-534.
Cedar Canyon Dam	SD Pennington	Deadman's Gulch	F	0.1	3,545.0	3,526.0	11	2	PL 80-858.
Chatfield Dam & Res	CO Douglas	S Platte	F	204.7	5,500.0	5,432.0	4,742	1,412	PL 81-516.
			FQ	26.7	5,432.0	5,385.0	1,412	12	HD 669-80.
Cherry Cr Dam & Res	CO Araphahoe	Cherry Cr	F	80.0	5,598.0	5,550.0	2,637	852	PL 77-228.
			FR	14.0	5,550.0	5,504.0	852	0	HD 426-76, PL 78-534.
Clinton Dam & Lk	KS Douglas	Wakarusa R	F	267.8	903.4	875.5	12,891	7,006	PL 87-874.
			FMCAR	129.2	875.5	820.0	7,006	0	SD 122-87.
Cold Brook Dam & Res	SD Fall River	Cold Brook	F	6.7	3,651.4	3,585.0	198	36	PL 77-228.
			FR	0.5	3,585.0	3,548.0	36	0	HD 655-76.
Conestoga Lake & Dam 12	NE Lancaster	Holmes Cr Trib to Salt Cr	F	8.0	1,252.0	1,232.9	620	230	PL 85-500.
			FCR	2.6	1,232.9	1,197.0	230	1	HD 396-84.
Cottonwood Springs Dam & Res	SD Fall River	Cottonwood Springs Cr	F	7.7	3,936.0	3,875.0	214	44	PL 77-228.
			FR	0.2	3,875.0	3,868.0	44	30	HD 655-76.
Fort Peck Dam & Res	MT Valley, Mc Cone Garfield	Missouri R	F	977.0	2,250.0	2,246.0	249,000	240,000	PL 73-409.
			FNPMCAR	13,649.0	2,246.0	2,160.0	240,000	92,000	PL 75-529, HD 238-73. PL 78-534, SD 247-78.
Fort Randall Dam, Lk Francis Case	SD Gregory Charles	Missouri R	F	985.0	1,375.0	1,365.0	102,000	95,000	PL 78-534.

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					Upper	Lower	Upper	Lower	
Garrison Dam, Lk Sakakawea	ND Mercer McLean	Missouri R	FNPIMCAR	3,021.0	1,365.0	1,320.0	95,000	41,000	SD 247-78.
			F	1,494.0	1,854.0	1,850.0	382,000	365,000	PL 78-534.
Gavins Point Dam, Lewis & Clark Lk	SD Yankton	Missouri R	FNPIMCAR	17,440.0	1,850.0	1,775.0	365,000	129,000	SD 247-78.
			F	61.0	1,210.0	1,208.0	32,000	29,000	PL 78-534.
Glenn Cunningham Lk, Dam 11	NE Knox	Little Papillion Cr	FNPIMCAR	95.0	1,208.0	1,204.5	29,000	25,000	SD 247-78.
	NE Douglas		F	14.0	1,142.0	1,121.0	922	392	PL 90-483.
Harlan County Lk	NE Harlan	Republican R	FRCA	3.9	1,121.0	1,085.0	392	0	HD 349-90.
			F	498.0	1,973.5	1,946.0	23,064	13,249	PL 77-228.
Harry S Truman Dam & Res	MO Benton	Osage R	FI	342.6	1,946.0	1,875.0	13,249	0	HD 892-76, PL-78-534.
			F	4,005.9	739.6	706.0	209,300	55,600	PL 83-780.
Hillsdale Lk	KS Miami	Big Bull Cr	FPCR	1,203.4	706.0	635.0	55,600	0	HD 549-81, PL 87-874.
			F	83.6	931.0	917.0	7,410	4,580	HD 578-87.
Holmes Park Lk & Dam 17	NE Lancaster	Antelope Cr Trib to Salt Cr	F	76.3	917.0	852.4	4,580	0	PL 83-780.
			F	5.7	1,266.0	1,242.4	410	100	HD 642-81.
Kanopolis Lk	KS Ellsworth	Smoky Hill R	FCR	0.8	1,242.4	1,218.0	100	3	PL 85-500.
			F	370.0	1,508.0	1,463.0	13,999	3,560	HD 396-84.
Kelly Road Dam	CO Araphoe	Westerly Cr	FI	55.8	1,463.0	1,425.0	3,560	0	PL 75-761.
	MO Randolph	Little East Fk Chariton R	F	0.3	5,362.0	5,342.0	38	0	PL 78-534, HD 842-76.
Long Branch Lk	MO Jackson	Little Blue R	F	30.4	801.0	791.1	3,670	2,429	PL 80-858, PL 84-99.
			F	34.6	791.0	751.1	2,429	0	PL 89-298.
Longview Lk	MO Jackson	Little Blue R	FCAR	24.8	909.0	891.0	1,960	930	HD 238-89.
			F	22.1	891.0	810.0	930	0	PL 90-483.
Melvern Lk	KS Osage	Marais des Cygnes R	FCAR	208.4	1,057.0	1,036.0	13,948	6,928	HD 169-90.
			F	208.4	1,057.0	1,036.0	13,948	6,928	PL 83-780.

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					Upper	Lower	Upper	Lower	
Milford Lk	KS Geary	Republican R	FNMCAR	154.4	1,036.0	960.0	6,928	0	PL 75-761, HD 549-81.
			F	756.7	1,176.2	1,144.4	27,255	17,270	PL 83-780.
			FCA	388.8	1,144.4	1,080.0	15,709	0	HD 642-81, PL 75-761.
Oahe Dam & Lk	ND 4 Counties	Missouri R	F	1,097.0	1,620.0	1,617.0	373,000	359,000	PL 78-534.
			FNPMCAR	16,789.0	1,617.0	1,540.0	359,000	117,000	SD 247-78.
Olive Cr Lk & Dam 2	NE Lancaster	Olive Br of Salt Cr	F	4.0	1,350.0	1,335.0	355	174	HD 396-84.
			FCR	1.5	1,335.0	1,314.0	174	4	PL 85-500.
Papio Dam Site #18 & Lk	NE Douglas	Boxelder Cr Papio Cr	F	7.1	1,128.2	1,110.0	595	255	PL 90-483.
			FCAR	3.4	1,110.0	1,060.5	255	0	HD 349-90.
Papio Dam Site #20 & Lk	NE Sarpy	Trib South Branch Papio	F	6.1	1,113.1	1,096.0	493	246	PL 90-483.
			FCAR	2.7	1,096.0	1,069.0	246	10	HD 349-90.
Pawnee Lk & Dam 14	NE Lancaster	No. Middle Cr of Salt Cr	F	21.0	1,263.5	1,244.3	1,470	728	PL 85-500.
			FCR	8.5	1,244.3	1,206.0	728	1	HD 396-84.
Perry Lk	KS Jefferson	Delaware R	F	521.9	920.6	891.5	25,342	12,202	PL 83-780.
			FN	243.2	891.5	825.0	122	0	HD 642-81.
Pipestem Dam & Res	ND Stutsman	Pipestem Cr	F	137.0	1,496.3	1,442.4	4,754	885	PL 89-298.
			FRC	9.6	1,442.4	1,415.0	885	62	HD 266-89.
Pomme De Terre Lk	MO Polk	Pomme De Terre R	F	407.2	874.0	839.0	15,980	7,890	PL 75-761.
			FNPCAR	241.6	839.0	750.0	7,890	0	HD 549-81, PL 83-780.
Pomona Lk	KS Osage	110 Mile Cr	F	176.8	1,003.0	974.0	8,520	400	PL 83-780.
			FNMAR	70.6	974.0	912.0	4,000	0	HD 549-81.
Rathbun Lk	IA Appanoose	Chariton R	F	346.3	926.0	904.0	20,948	11,013	PL 83-780.
			FNM	205.4	904.0	844.0	11,013	0	HD 561-81.
Smithville Lk	MO Clay	Little Platte R	F	101.8	876.2	864.2	9,995	7,192	PL 89-298.
			FMCAR	144.6	864.2	799.0	7,192	0	HD 262-89.
Spring Gulch Imbankment	CO Douglas	Spring Gulch	F	1.8	5,600.00	5,535.0	88	0	PL 81-516, HD 669-80.

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					Upper	Lower	Upper	Lower	
Stagecoach Lk & Dam 9	NE Lancaster	Hickman Br of Salt Cr	F	4.7	1,285.0	1,271.1	490	196	PL 85-500.
>			FRC	1.9	1,271.1	1,246.0	196	0	HD 396-84.
Standing Bear Lk & Dam 16	NE Douglas	Trib Big Papillion Cr	F	3.7	1,121.0	1,104.0	302	137	PL 90-483.
			FRC	1.5	1,104.0	1,060.0	137	0	HD 349-90.
Stockton Lk	MO Cedar	Sac R	F	779.6	892.0	867.0	38,288	24,777	PL 83-780.
			FARPN	887.1	867.0	760.0	24,777	0	HD 549-89.
Tuttle Creek Lk	KS Riley	Big Blue R	F	1,937.4	1,136.0	1,075.0	54,179	14,875	PL 75-761.
			FN	177.1	1,075.0	1,061.0	14,875	0	HD 842-76.
Twin Lakes & Dam 13	NE Seward	Middle Cr Salt Cr	F	5.3	1,355.0	1,341.0	505	255	PL 85-500.
			CFR	2.8	1,341.0	1,306.0	255	1	HD 396-84.
Wagon Train Lk & Dam 8	NE Lancaster	Hickman Br of Salt Cr	F	6.8	1,302.0	1,287.8	660	303	PL 85-500.
			FCR	2.5	1,287.8	1,260.0	303	4	HD 396-84.
Wehrspann Lk & Dam 20	NE Sarpy	Trib South Branch Papio	F	6.1	1,113.1	1,096.0	493	246	PL 90-483.
			FCAR	2.7	1,096.0	1,069.0	246	10	HD 349-90.
Wilson Lk	KS Russell	Saline R	F	530.7	1,554.0	1,516.0	19,980	9,040	PL 78-534.
			FRC	247.8	1,516.0	1,440.0	9,040	0	SD 191-78, SD 247-78.
Yankee Hill Lk & Dam 10	NE Lancaster	Cardwell Br of Salt Cr	F	5.6	1,262.0	1,244.9	475	208	PL 85-500.
			FCR	2.0	1,244.9	1,218.0	208	0	HD 396-84.
North Atlantic Division									
Almond Lake	NY Steuben	Canacadea Cr	F	14.6	1,300.0	1,255.0	489	124	PL 74-738.
Alvin R. Bush Dam	PA Clinton	Kettle Cr	F	73.4	937.0	840.0	1,430	160	FCA Sep 54.
Arkport Dam	NY Steuben	Canisteo R	F	8.0	1,304.0	1,218.0	192	0	PL 74-738.
Aylesworth Cr Lk	PA Lackawanna	Aylesworth Cr	F	1.7	1,150.0	1,108.0	87	7	PL 87-874.
Beltzville Dam & Lk	PA Carbon, Monroe	Pohopoco Cr	F	27.0	651.0	628.0	1,411	947	PL 87-874.
			FMA	39.8	628.0	537.0	947	113	

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					Upper	Lower	Upper	Lower	
Bloomington Lk	MD Garret	North Branch Potomac R	F	36.2	1,500.0	1,466.0	1,184	952	PL 87-874.
Blue Marsh Dam & Lk	PA Lebanon Berks	Tulpehocken CR	FMA	92.0	1,466.0	1,255.0	952	42	PL 87-874.
			F	27.1	307.0	290.0	2,159	1,147	
Cowanessque Lk	PA Tioga	Cowanessque R	FMA	19.9	290.0	261.0	1,147	323	PL 85-500.
			F	82.0	1,117.0	1,045.0	2,060	410	
Curwensville Lk	PA Clearfield	West Branch Susquehanna R	F	114.7	1,228.0	1,162.0	3,020	790	FCA Sep 54.
East Sidney Lk	NY Delaware	Ouleout Cr	F	30.2	1,203.0	1,150.0	1,100	210	PL 74-738.
Foster Joseph Sayers Dam	PA Centre	Bald Eagle Cr	F	70.2	657.0	630.0	3,450	1,730	FCA Sept 54.
Francis E. Walter Dam & Res	PA Carbon, Luzerne, Monroe	Lehigh R	F	107.8	1,450.0	1,300.0	1,830	80	PL 79-526.
Gathright Dam & Lk Moomaw	VA Alleghany, Bath	Jackson R	F	79.9	1,610.0	1,582.0	3,160	2,530	PL 79-526.
General Edgar Jadwin Dam Prompton Dam & Res	PA Wayne	Dyberry Cr W Br Lackawaxen R	AR	60.7	1,582.0	1,554.0	2,530	1,780	PL 80-858.
			F	24.5	1,053.0	973.0	659	0	
Raystown Lk	PA Huntingdon	Raystown Br	F	48.5	1,205.0	1,125.0	910	290	PL 80-858.
			FR	248.0	812.0	786.0	10,800	8,300	
Stillwater Lk	PA Susquehanna	Lackawanna R	F	514.0	786.0	622.8	8,300	150	PL 87-874.
Tioga-Hammond Lakes	PA Tioga	Crooked Cr	F	11.6	1,621.0	1,572.0	422	83	PL 77-228.
Hammond	PA Tioga		F	54.2	1,131.0	1,086.0	1,770	680	PL 85-500.
Tioga-Hammond Lakes	PA Tioga	Tioga R	F	52.5	1,131.0	1,081.0	1,630	470	PL 85-500.
Tioga									
Whitney Piont Lk	NY Broome	Otselic R	F	66.5	1,010.0	973.0	3,340	1,200	PL 74-738.
York Indian Rock Dam	PA York	Codorus Cr	F	28.0	435.0	370.0	1,430	0	PL 74-738.
North Central Division									
Badhill Dam & Res	ND Barnes	Sheyenne R	FM	68.6	1,266.0	1,257.2	5,430	4,430	FCA Dec 44.
Brandon Road L&D	IL Will	Illinois R	N	8.0	539.0	538.0	300	250	PL 71-126.

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					Upper	Lower	Upper	Lower	
Cedars L&D	WI Outagamie	Fox R	N	1.8	703.6	698.7	255	140	RHA of 1882, 1885.
Coralville Dam & Res	IA Johnson	Iowa R	F	439.0	712.0	680.0	24,800	3,580	PL 75-761.
			C	40.3	680.0	652.0	3,580	0	PL 75-761.
Depree L&D	WI Brown	Fox R	N	9.4	591.0	586.7	926	0	PL 71-126.
Dresden Island L&D	IL Grundy	Illinois R	N	1.0	505.0	504.0	1,690	1,550	FCA 1958.
Eau Galle Dam & Res	WI Pierce	Eau Galle R	FCR	1.6	940.0	938.5	1,500	1,350	PL 78-534.
Farmdale Dam	IL Tazwell	Farm Cr	F	11.3	616.0	551.0	385	0	PL 78-534.
Fondulac Dam	IL Tazwell	Fondulac Cr	F	2.3	579.0	530.0	97	0	PL 78-534.
Gull Lk Dam & Res	MN Cass	Gull R	N	70.4	1,194.0	1,192.7	13,100	12,700	RHA 1899.
Highway 75 Dam & Res	MN Bigstone, Lacqui, Parle	Minnesota R	FC	11.1	952.3	947.3	2,790	910	FCA Oct 65.
Homme Dam & Res	ND Walsh	Park R	FM	3.7	1,080.0	1,074.0	190	176	FCA of 22 Dec 44.
L&D 1	MN Hennepin, Ramsey	Mississippi R	N	13.0	725.1	722.8	5,800	5,500	RHA 1910.
L&D 2	MN Dakota, Wash	Mississippi R	N	8.0	687.2	686.5	11,810	11,000	RHA 1927.
L&D 3	MN Goodhue, Pierce	Mississippi R	N	17.8	675.0	674.0	17,950	17,650	RHA 1930.
L&D 4	WI Wabasha, Buffalo	Mississippi R	N	18.0	667.0	666.5	38,820	36,600	RHA 1930.
L&D 5	MN Winona, Buffalo	Mississippi R	N	6.2	660.0	659.5	12,680	12,000	RHA 1930.
L&D 5A	MN Winona, Buffalo	Mississippi R	N	7.2	651.0	650.0	7,500	7,000	RHA 1930.
L&D 6	MN Winona	Mississippi R	N	8.4	645.5	644.5	8,870	8,000	RHA 1930.
L&D 7	MN Winona	Mississippi R	N	2.6	639.0	639.0	13,440	13,400	RHA 1930.
L&D 8	WI LaCrosse	Mississippi R	N	20.4	631.0	630.0	20,800	20,000	RHA 1930.
	MN Houston								
L&D 9	WI Vernon	Mississippi R	N	28.7	620.0	619.0	29,125	28,300	RHA 1930.
	WI Crawford								
L&D 10	IA Allamakee	Mississippi R	N	16.8	611.0	610.0	17,070	16,500	RHA 1930.
	IA Clayton								
L&D 11	WI Grant	Mississippi R	N	19.1	603.1	602.0	21,100	20,000	PL 71-520.
	IA Dubuque								
L&D 12	IA Jackson	Mississippi R	N	12.2	592.1	591.0	13,000	12,400	PL 71-520.
L&D 13	IL Whiteside	Mississippi R	N	24.2	583.1	582.0	30,000	28,500	PL 71-520.

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					Upper	Lower	Upper	Lower	
L&D 14	IA Scott	Mississippi R	N	9.0	572.1	571.0	10,500	9,980	PL 71-520.
L&D 15	IL Rock Island	Mississippi R	N	5.5	561.1	559.0	3,725	3,540	PL 71-520.
L&D 16	IL Rock Island	Mississippi R	N	12.1	545.1	544.0	13,000	12,400	PL 71-520.
L&D 17	IL Mercer	Mississippi R	N	7.5	537.1	536.0	7,580	7,200	PL 71-520.
L&D 18	IL Henderson	Mississippi R	N	11.0	529.1	528.0	13,300	12,600	PL 71-520.
L&D 19	IA Lake	Mississippi R	N	55.0	518.2	517.2	33,500	31,800	PL 71-520.
L&D 20	MO Lewis	Mississippi R	N	5.8	481.5	476.5	7,960	7,550	PL 71-520.
L&D 21	IL Adams	Mississippi R	N	8.6	470.1	469.6	9,390	8,910	PL 71-520.
L&D 22	MO Polke	Mississippi R	N	8.4	459.6	459.1	8,660	8,230	PL 71-520.
Lac qui Parle Dam & Res	MN Chippewa Swift	Minnesota R	FC	119.3	941.1	931.2	13,500	6,400	FCA of 22 Jun 36.
Lagrange L&D	IL Brown	Illinois R	N	0.0	429.0	429.0	10,500	10,500	PL 73-184.
Leech Lake Dam & Res	MN Cass	Leech R	N	300.2	1,295.7	1,293.2	139,000	107,200	RHA of 1882 1895.
Little Kaukauna L&D	WI Brown	Fox R	N	3.6	601.0	592.8	447	42.0	RHA of 1882 1885.
Little Chute L&D	WI Outagamie	Fox R	N	0.4	694.2	688.9	74	67	RHA of 1882 1885.
Lockport Lock	IL Will	Chicago San Ship Canal	FNP	2.7	579.0	577.5	1,850	1,800	RHA 1930.
Lower Appleton L&D	WI Outagamie	Fox R	N	0.2	710.9	706.3	43	40	RHA of 1882 1895.
Marseilles Lk & Dam	IL LaSalle	Illinois R	N	0.7	483.0	482.8	1,400	1,320	PL 71-126.
Marsh Lake Dam & Res	MN Swift, Lacqui, Parle	Minnesota R	FC	23.9	941.1	937.6	8,650	5,150	FCA Jun 36.
Menasha Dam Lk Winnebago	WI Winnebago	Fox R	FN	452.0	746.8	743.5	181,120	168,500	
Mount Morris Dam	NY Livingston	Genesee R	F	337.4	760.0	585.0	3,300	0	PL 74-738.
O'Brien L&D	IL Cook	Calumet	N	0.3	581.9	578.2	50	50	RHA of 1946.
Peoria L&D	IL Peoria	Illinois R	N	0.0	440.0	440.0	27,800	27,800	PL 73-184.
Pine Dam & Res	MN Crow Wing	Pine R	N	40.4	1,230.3	1,227.3	13,900	13,000	RHA of 1899.
Pokegama Dam & Res	MN Itasca	Mississippi R	N	52.4	1,274.4	1,270.3	13,700	12,000	RHA of 1899.
Rapid Croche L&D	WI Outagamie	Fox R	N	3.4	608.5	602.1	568	0	RHA 1885.
Red Lake Dam & Res	MN Clearwater	Red Lake R	FA	1,810.0	1,174.0	1,173.5	288,800	287,300	FCA Dec 44.
Red Rock Dam & Res	IA Marion	Des Monies R	F	1,670.0	780.0	728.0	65,400	8,000	PL 75-761.

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Reservation Control Res	MN Traverse		R	72.0	728.0	690.0	8,000	0	PL 75-761.
	SD Roberts		FC	58.8	981.0	976.0	12,400	10,950	FCA 1936.
Sandy Lake Dam & Res	MN Aitkin	Sandy R	N	37.5	1,218.3	1,214.3	10,600	8,200	RHA of 1899.
Saylorville Dam & Res	IA Polk	Des Moines R	F	586.0	890.0	836.0	16,700	5,950	FCA 1936.
			P	90.0	836.0	810.0	5,950	0	FCA.
St Anthony Falls Lwr L&D	MN Hennepin	Mississippi R	N	0.0	750.0	750.0	50	50	RHA of 1937 1945.
St Anthony Falls Upr L&D	MN Hennepin	Mississippi R	N	17.4	801.0	799.0	8,800	8,600	RHA of 1937 1945.
Starved Rock L&D	IL LaSalle	Illinois R	N	1.0	459.0	458.0	1,155	1,020	PL 69-100.
Upper Appleton L&D	WI Outagamie	Fox R	N	7.4	738.7	735.4	1,171	1,040	RHA of 1882 1885.
Upper Kaukauna L&D	WI Outagamie	Fox R	N	1.1	656.8	652.8	134	115	RHA of 1882 1885.
White Rock Dam & Res	MN Traverse	Bois De Souix	FC	78.6	981.0	972.0	10,500	4,000	FCA 1936.
	SD Roberts								
Winnibigoshish Dam & Res	MN Cass Itasca	Mississippi R	N	98.7	1,300.9	1,296.9	98,700	62,000	RHA of 1899.
New England Division									
Ball Mountain Lk	VT Windham	West R	F	52.4	1,017.0	830.5	810	20	PL 78-534, 83-780.
Barre Falls Dam	MA Worcester	Ware R	F	24.0	807.0	761.0	1,400	0	PL 78-228.
Birch Hill Dam	MA Worcester	Millers R	F	49.9	852.0	815.0	3,200	0	PL 75-761.
Black Rock Lk	CT Litchfield	Branch Brook	F	8.5	520.0	437.0	190	21	PL 86-45.
Blackwater Dam	NH Merrimack	Blackwater R	F	46.0	566.0	515.0	3,280	0	PL 75-111.
Buffumville Lk	MA Worcester	Little R	F	11.3	524.0	492.5	530	200	PL 77-228.
Colebrook River Lk	CT Litchfield	West Branch	F	50.2	761.0	708.0	1,185	750	PL 86-645.
	MA Bekshire	Farmington R							
Conant Brook Dam	MA Hampden	Conant Brook	F	3.7	757.0	694.0	158	0	PL 86-645.
East Brimfield Lk	MA Hampden, Worcester	Quinebaug R	F	29.9	653.0	632.0	2,300	360	PL 77-228.
Edward MacDowell Lk	NH Hillsboro	Nubanusit Brook	F	12.8	946.0	911.0	840	165	PL 75-111.
Everett Lk	NH Hillsboro, Merrimack	Piscataquog R	F	91.5	418.0	340.0	2,900	130	PL 75-761.
Franklin Falls Dam	NH Belknap, Merrimack	Pemigewasset R	F	150.6	389.0	307.0	2,800	440	PL 75-111.
Hancock Brook Lk	CT Litchfield	Hancock Brook	F	3.9	484.0	460.0	266	40	PL 86-645.

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Hodges Village Dam	MA Worcester	French R	F	13.3	501.0	465.5	740	0	PL 77-228.
Hop Brook Lk	CT New Haven	Hop Brook	F	6.9	364.0	310.0	270	21	PL 86-645.
Hopkinton Lk	NH Merrimack	Contoocook R	F	70.1	416.0	380.0	3,700	220	PL 75-761.
Knightville Dam	MA Hampshire	Westfield R	F	49.0	610.0	480.0	960	0	PL 75-761.
Littleville Lk	MA Hampden, Hampshire	Middle Br, Westfield R	F	23.0	576.0	518.0	510	275	PL 85-500.
Mansfield Hollow Lk	CT Tolland	Natchaug R	F	49.2	257.0	205.5	1,880	200	PL 77-228.
New Bedford-Fairhaven Hurr Barrier	MA Bristol		F	0.0	0.0	0.0	0	0	PL 85-500.
North Hartland Lk	VT Windsor	Ottauquechee R	F	68.8	546.5	425.0	1,100	215	PL 75-761.
North Springfield Lk	VT Windsor	Black R	F	50.0	545.5	467.0	1,200	100	PL 75-761.
Northfield Br Lk	CT Litchfield	Northfield Br	F	2.4	576.0	500.0	67	7	PL 86-645.
Otter Br Lk	NH Cheshire	Otter Brook	F	17.6	781.0	701.0	374	70	PL 83-780.
Stamford Hurr Barrier	CT Fairfield		F	0.0	0.0	0.0	0	0	PL 86-645.
Surry Mountain Lk	NH Cheshire	Ashuelot R	F	31.7	550.0	500.0	970	260	PL 75-761.
Thomaston Dam	CT Litchfield	Naugatuck R	F	42.0	494.0	380.0	960	0	PL 78-534.
Townshend Lk	VT Windham	West R	F	32.9	553.0	478.0	735	95	PL 78-534, PL 83-780.
Tully Lk	MA Worcester	East Br Tully R	F	20.5	668.0	636.0	1,130	78	PL 75-761.
Union Village Dam	VT Orange	Ompompanoosuc R	F	38.0	564.0	420.0	740	0	PL 74-738.
West Hill Dam	MA Worcester	West R	F	12.4	264.0	234.0	1,025	0	PL 78-534.
West Thomspson	CT Windham	Quinebaug R	F	25.6	342.5	305.0	1,250	200	PL 86-645.
Westville Lake	MA Worcester	Quinebaug R	F	11.0	572.0	525.0	913	23	PL 77-228.

North Pacific Division

Albeni Falls Dam, Lk Pend, Oreille	ID Bonner	Pend Oreille R	FNP	1,155.0	2,062.5	2,049.7	95,000	86,000	PL 81-516.
Applegate Lk	OR Jackson	Applegate R	FIR	75.2	1,987.0	1,854.0	988	221	FCA 1962, PL 87-874, PL 87-874.
Big Cliff Dam	OR Marion, Linn	N Santiam R	P	3.5	1,206.0	1,182.0	130	98	HD 544, PL 75-761, PL 87-874.
Blue River Lk	OR Lane	Blue R	F	6.5	1,357.0	1,350.0	975	940	HD 531.

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Project name ¹	State/county	Stream ¹	Project purpose ²	Storage 1,000 AF	Elev limits feet M.S.L.		Area in acres		Auth legis ³
					Upper	Lower	Upper	Lower	
Bonneville L&D Lk	WA Skamania	Columbia R	FNI	78.8	1,350.0	1,180.0	940	133	PL 81-516.
Chena River Lakes	AK North Star Borough	Chena R	NP	138.0	77.0	70.0	20,800	19,850	RHA 1935.
Chief Joseph Dam Rufus Woods Lk	WA Douglas, Okanogan	Columbia R	F	34.0	506.7	490.0	5,400	400	PL 90-483.
Cottage Grove Lk	OR Lane	Coast Fk, Willamette R	P	192.3	956.0	930.0	8,400	6,800	HD 693, PL 79-525.
Cougar Lk	OR Lane	South Fk	F	29.8	791.0	750.0	1,155	295	HD 544, PL 75-761.
Detroit Lk	OR Marion	North Santiam	FNPI	11.3	1,699.0	1,690.0	1,280	1,235	HD 531.
			P	143.9	1,690.0	1,532.0	1,235	635	PL 81-516.
			F	9.9	1,532.0	1,516.0	635	602	PL 83-870.
			FNPI	19.1	1,569.0	1,563.0	3,490	3,455	HD 544, PL 75-761.
Dexter Dam	OR Lane	Middle Fk, Willamette R	P	281.6	1,563.5	1,450.0	3,455	1,725	
Dorena Lk	OR Lane	Cow R	FNPI	40.3	1,450.0	1,425.0	1,725	1,415	HD 544, PL 75-761.
Dworshak Dam and Res	ID Clearwater	North Fk, Clearwater R	F	4.8	695.0	690.0	990	940	HD 544.
Fall Cr Dam and Lk	OR Lane	Fall Cr	FNPI	5.5	835.0	832.0	1,885	1,815	PL 75-761.
Fern Ridge Lk	OR Lane	Long Tom R	FNI	65.0	832.0	770.5	1,815	520	HD 403, PL 87-874.
Foster Lake	OR Linn	South Santiam R	F	7.5	834.0	830.0	1,865	1,760	HD 531.
Green Peter Lk	OR Linn	Middle Fk, Santiam R	FNPI	107.5	830.0	728.0	1,760	460	PL 81-516
Hills Creek Lk	OR Lane	Middle Fk, Willamette R	F	15.7	375.1	373.5	10,305	9,340	HD 544.
			FNPI	93.9	373.5	353.0	9,340	1,515	PL 75-761
			F	4.9	641.0	637.0	1,260	1,195	HD 544
			FNPI	24.9	637.0	613.0	1,195	895	PL 86-645
			F	18.3	1,015.0	1,010.0	3,705	3,605	HD 531.
			FNPI	249.9	1,010.0	992.0	3,605	2,072	PL 81-516, PL 83-780.
			F	5.6	1,543.0	1,541.0	2,850	2,710	HD 531.

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					Upper	Lower	Upper	Lower	
Howard Hanson Dam	WA King	Green R	FNPI	194.6	1,541.0	1,448.0	2,710	1,575	PL 81-516.
			F	80.0	1,206.0	1,141.0	1,750	763	HD 531.
			FA	25.6	1,141.0	1,040.0	763	13	PL 81-516.
Ice Harbor Dam Lk	WA Walla, Walla, Franklin	Snake R	NP	24.9	440.0	437.0	8,370	8,210	HD 704, PL 79-14.
Sacajawea									
John Day Dam Lk Umatilla	OR Sherman	Columbia R	F	158.0	268.0	265.0	55,000	52,000	HD 531.
			FNP	150.0	265.0	262.0	52,000	49,000	PL 81-516.
			F	192.0	262.0	257.0	49,000	42,000	
Libby Dam Lk Koocanusa	MT Lincoln	Kootenai R	FP	4,979.5	2,459.0	2,287.0	46,365	14,391	HD 531, PL 81-516.
Little Goose L&D Lk Bryan	WA Columbia, Whitman	Snake R	PN	49.0	638.0	633.0	10,030	9,620	HD 704, PL 79-14.
Lookout Point Lk	OR Lane	Middle Fk, Willamette R	P	12.2	825.0	819.0	2,090	1,860	HD 544.
			FNPI	324.2	926.0	825.0	4,255	2,090	PL 75-761.
Lost Creek Lk	OR Jackson	Rogue R	FPIR	315.0	1,872.0	1,751.0	3,430	1,800	HD 566, PL 87-874.
Lower Granite L&D	WA Garfield, Whitman	Snake R	NPI	43.6	738.0	733.0	8,900	8,540	HD 704, PL 79-14.
Lucky Peak Dam and Lk	ID Ada	Boise R	F	13.9	3,060.0	3,055.0	2,817	2,745	PL 79-526.
			FI	264.4	3,055.0	2,905.0	2,817	802	
Lwr Monumental L&D Lk	WA Walla, Walla, Franklin	Snake R	NP	20.0	540.0	537.0	6,700	6,550	HD 704, PL 79-14.
HG West									
McNary L&D, Dam Lk	WA Benton	Columbia R	NP	185.0	340.0	335.0	38,800	36,000	HD 704, PL 79-14.
Wallula									
	OR Umatilla								
Mill Creek Dam Lk	WA Walla, Walla	Mill Cr	F	7.5	1,265.0	1,205.0	225	53	HD 578, PL 75-761.
Mud Mountain Dam	WA King, Pierce	White R	F	106.3	1,215.0	895.0	963	0	PL 74-738.
The Dalles L&D Lk Celilo	WA Klickitat	Columbia R	NP	52.5	160.0	155.0	11,200	10,350	HD 531, PL 81-516.
	OR Wasco								
Willow Creek Lk	OR Morrow	Willow Cr	F	11.6	2,113.5	2,047.0	269	96	PL 89-298.
Wynoochee Dam and Lk	WA Grays, Harbor	Wynoochee R	FMCA	65.4	800.0	700.0	1,170	193	HD 601, PL 93-251.

Ohio River Division

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Project name ¹	State/county	Stream ¹	Project purpose ²	Storage 1,000 AF	Elev limits feet M.S.L.		Area in acres		Auth legis ³
					Upper	Lower	Upper	Lower	
Allegheny L&D 2	PA Allegheny	Allegheny R	N	0.0	721.0	710.0	0	0	RHA 1935.
Allegheny L&D 3	PA Allegheny	Allegheny R	N	0.0	734.5	721.0	0	0	RHA 1935.
Allegheny L&D 4	PA Allegheny Westmoreland	Allegheny R	N	0.0	745.0	734.5	0	0	RHA 1912.
Allegheny L&D 5	PA Armstrong	Allegheny R	N	0.0	756.8	745.0	0	0	RHA 1912.
Allegheny L&D 6	PA Armstrong	Allegheny R	N	0.0	769.0	756.8	0	0	RHA 1912.
Allegheny L&D 7	PA Armstrong	Allegheny R	N	0.0	782.1	769.0	0	0	RHA 1912.
Allegheny L&D 8	PA Armstrong	Allegheny R	N	0.0	800.0	782.1	0	0	RHA 1912, 1935.
Allegheny L&D 9	PA Armstrong	Allegheny R	N	0.0	822.0	800.0	0	0	RHA 1935.
Allegheny Res Kinzua Dam	PA Warren	Allegheny R	F	607.0	1,365.0	1,328.0	21,180	12,080	PL 74-738.
			FPCAR	549.0	1,328.0	1,240.0	12,080	1,900	
Alum Cr Lk	OH Delaware	Alum Cr	F	53.1	901.0	888.0	4,852	3,387	PL 87-874.
			FMCR	79.2	888.0	885.0	3,387	3,105	
Atwood Lk	OH Tuscarawas	Indian Fk Cr	F	26.1	941.0	928.0	2,460	1,540	PW 1933.
			FCR	7.6	928.0	922.5	1,540	1,250	
Barkley Dam Lk Barkley	Ky Lyon, Livgst	Cumberland R	F	1,213.0	375.0	359.0	93,430	57,920	PL 79-525.
			FP	259.0	359.0	354.0	57,920	45,210	
			N	610.0	354.0	233.0	45,210	0	
Barren River Lk	KY Allen, Barren	Barren R	F	558.8	590.0	552.0	20,150	10,000	PL 75-261.
			FMR	190.3	552.0	525.0	10,000	4,340	
Beach City Lk	OH Tuscarawas	Sugar Cr	F	69.9	976.5	948.0	6,150	420	PW 1933.
			FCR	0.0	0.0	0.0	0	420	
Beech Fk Lk	WV Wayne	Beech Fk Cr	F	28.3	614.5	592.0	1,847	725	PL 87-874.
			FCR	5.0	592.0	583.5	725	460	
Belleville L&D	WV Wood	Ohio R	N	0.0	582.0	560.0	0	0	RHA 1909.
	OH Meigs								
Berlin Lk	OH Mahoning, Portage	Mahoning R	F	38.3	1,032.0	1,024.7	5,500	3,590	PL 75-761.
			FMCAR	56.6	1,024.7	1,016.5	3,590	2,200	
Bluestone Lk	WV Summers	New R	F	592.6	1,520.0	1,410.0	9,180	2,040	PL 74-738.
			FCR	7.5	1,410.0	1,406.0	2,040	1,800	PL 75-761.

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					Upper	Lower	Upper	Lower	
Bolivar Dam	OH Stark, Tuscarawas	Sandy Cr	F	149.6	962.0	895.0	6,500	0	PW 1933.
Brookville Lk	IN Franklin	E Fork of Whitewater R	FMR	128.4	748.0	713.0	5,260	2,430	PL 75-761.
Buckhorn Lk	KY Leslie	Middle Fk of Kentucky R	F	135.8	840.0	782.0	3,610	1,230	PL 75-761.
Burnsville Lk	WV Braxton	L Kanawha R	FR	21.8	782.0	757.0	1,230	550	PL 75-761.
			F	51.5	825.0	789.0	1,902	965	
			FCAR	10.2	789.0	776.0	965	553	
CJ Brown Dam & Res	OH Clark	Buck Cr	F	26.8	1,023.0	1,012.0	2,720	2,120	PL 87-874.
CM Harden Lk	IN Parke	Raccoon Cr	F	83.5	690.0	661.0	3,910	2,060	PL 75-761
			FAR	33.1	661.0	640.0	2,060	1,100	
Caesar Cr Lk	OH Warren	Caesar Cr	F	140.2	883.0	849.0	6,110	2,830	PL 75-761.
			FMAR	88.7	849.0	800.0	2,830	700	
Cagles Mill Lk	IN Putman	Mill Cr	F	201.0	704.0	636.0	4,840	1,400	PL 75-761.
Cannelton L&D	KY Hancock	Ohio R	N	0.0	383.0	358.0	0	0	RHA 1909
	IN Perry								
Carr Fk Lk	KY Knott	Carr Cr	F	25.1	1,055.0	1027.0	1,120	710	PL 87-874.
			FAR	10.8	1,027.0	1009.0	710	530	
Cave Run Lk	KY Rowan	Licking R	F	391.5	765.0	730.0	14,870	8,270	PL 74-738
			FAR	75.3	730.0	720.0	8,270	6,790	
Center Hill Lk	TN Dekalb	Caney FK	F	762.0	685.0	648.0	23,060	18,220	PL 75-761.
			P	492.0	648.0	618.0	18,220	14,590	
Charles Mill Lk	OH Ashland	Black Fk	F	80.6	1,020.0	997.0	6,050	1,350	PW 1933.
			FCR	4.5	997.0	993.0	1,350	827	
Cheatham L&D	TN Cheatham	Cumberland R	P	19.8	385.0	382.0	7,450	5,630	RHA 1946, PL 396.
			N	84.2	382.0	345.0	5,630	0	PL 396.
Clendening Lk	OH Harrison	Brush Fk	F	27.5	910.5	898.0	2,620	1,800	PW 1933.
			FCR	8.0	898.0	893.0	1,800	1,430	
Conemaugh River Lk	PA Indiana, Westmoreland	Conemaugh R	F	270.0	975.0	880.0	6,820	300	PL 74-738, PL 75-761.

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					Upper	Lower	Upper	Lower	
Cordell Hull Dam & Res	TN Smith	Cumberland R	PR	17.8	504.5	499.0	12,200	9,820	RHA 1946.
			NR	0.0	499.0	424.0	9,820	0	
Crooked Cr Lk	PA Armstrong	Crooked Cr	F	89.4	920.0	840.0	1,940	350	PL 74-738, PL 75-761.
Dale Hollow Lk	TN Clay	Obey R	F	353.0	663.0	651.0	30,990	27,700	PL 75-761.
			P	496.0	651.0	631.0	27,700	21,880	
Dashields L&D	PA Allegheny	Ohio R	N	0.0	692.0	682.0	0	0	RHA 1909.
Deer Cr Lk	OH Pickaway	Deer Cr	F	81.5	844.0	810.0	4,046	1,277	PL 75-761.
			FCR	14.6	810.0	796.0	1,277	727	
Delaware Lk	OH Delaware	Olentangy R	F	118.0	947.0	915.0	8,550	1,270	PL 75-761
			FCAR	5.6	915.0	910.0	1,270	950	
Dewey Lk	KY Floyd	Johns Cr	F	76.1	686.0	650.0	3,340	1,100	PL 75-761
			FCR	4.9	650.0	645.0	1,100	880	
Dillon Lk	OH Muskingum	Licking R	F	256.5	790.0	737.0	10,280	1,560	PL 75-761.
			FCR	4.4	737.0	734.0	1,560	1,330	
Dover Dam	OH Tuscarawas	Tuscarawas R	F	203.0	916.0	858.0	10,100	0	PW 1933.
E Br Clarion River Lake	PA Elk	E Br Clarion R	F	19.0	1,685.0	1,670.0	1,370	1,160	PL 78-526.
			FCAR	19.8	1,670.0	1,651.0	1,160	920	
E Fk Res Wm H Harsha Lk	OH Clermont	E Fk Little Miami R	F	202.2	795.0	733.0	4,600	2,160	PL 75-761.
			FMCAR	73.6	733.0	683.0	2,160	820	
East Lynn Lk	WV Wayne	E Fk Twelvepole	F	65.3	701.0	662.0	2,351	1,005	PL 75-761.
			FCR	5.5	662.0	656.0	1,005	823	
Emsworth L&D	PA Allegheny	Ohio R	N	0.0	710.0	692.0	0	0	RHA 1909.
Fishtrap Lk	KY Pike	Levisa Fk	F	126.7	825.0	757.0	2,681	1,131	PL 75-761.
			FCAR	27.2	757.0	725.0	1,131	569	
Gallipolis L&D	WV Mason	Ohio R	N	0.0	538.0	515.0	0	0	RHA 1935.
	OH Gallia								
Grayson Lk	KY Carter	L Sandy R	F	89.6	681.0	645.0	3,633	1,509	PL 86-645.
			FCAR	10.7	645.0	637.0	1,509	1,159	

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					Upper	Lower	Upper	Lower	
Green R L&D 1	KY Henderson	Green R	N	0.0	349.1	337.3	0	0	RHA 1888.
Green R L&D 2	KY McLean	Green R	N	0.0	363.4	349.1	0	0	RHA 1888.
Green River Lk	KY Taylor	Green R	F	479.1	713.0	675.0	19,100	8,210	PL 75-761.
			FAR	81.5	675.0	664.0	8,210	6,650	
Greenup L&D 3	KY Greenup	Ohio R	N	0.0	515.0	485.0	0	0	RHA 1909.
	OH Scioto								
Hannibal L&D	WV Wetzel	Ohio R	N	0.0	623.0	602.0	0	0	RHA 1909.
	OH Monroe								
Hildebrand L&D	WV Monongalia	Monongahela	N	0.0	835.0	814.0	0	0	RHA 1950.
Huntington Lk	IN Hunt	Wabash R	F	140.6	798.0	749.0	7,900	900	PL 85-500.
			FR	8.4	749.0	737.0	900	500	
J Percy Priest Dam & Res	TN Davidson	Stones R	F	252.0	504.5	490.5	22,720	14,400	PL 75-761.
			FP	15.0	490.5	489.5	14,400	14,000	
			FPR	0.0	489.5	483.0	14,000	11,630	
			PR	0.0	483.0	480.0	11,630	10,570	
JW Flannagan Dam & Res	VA Dickenson	Pound R	F	78.6	1,446.0	1,396.0	2,098	1,143	PL 75-761.
			FMCR	16.5	1,396.0	1,380.0	1,143	310	
Kentucky R L&D 1	KY Carroll	Kentucky R	N	0.0	430.0	421.8	0	0	RHA 1879.
Kentucky R L&D 2	KY Henry Owen	Kentucky R	N	0.0	444.0	430.0	0	0	RHA 1879.
Kentucky R L&D 3	KY Henry Owen	Kentucky R	N	0.0	457.1	444.0	0	0	RHA 1879.
Kentucky R L&D 4	KY Franklin	Kentucky R	N	0.0	470.4	457.1	0	0	RHA 1879.
Laurel River Lk	KY Laurel, Whitley	Laurel R	P	185.0	1,018.5	982.0	6,060	4,200	PL 86-645.
			R	250.6	982.0	760.0	4,200	0	
Leesvillie Lake	OH Carroll	McGuire Cr.	F	17.9	977.5	963.0	1,470	1,000	PW 1933.
			FCR	5.5	963.0	957.0	1,000	829	
London L&D	WV Kanawha	Kanawha R	N	0.0	614.0	590.0	0	0	RHA 1930.
Loyalhanna Lk	PA Westmoreland	Loyalhanna Cr	F	93.3	975.0	910.0	3,280	210	PL 74-738.
			FC	0.0	0.0	0.0	0	0	PL 75-761.
M J Kirwan Dam & Res	OH Portage	W. Br Mahoning R	F	22.0	993.0	985.5	3,240	2,650	PL-74-738

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Project name ¹	State/county	Stream ¹	Project purpose ²	Storage 1,000 AF	Elev limits feet M.S.L.		Area in acres		Auth legis ³
					Upper	Lower	Upper	Lower	
Mahoning Cr Lk	PA Armstrong	Mahoning Cr	FCAR	52.9	985.5	951.0	2,650	570	PL 75-761.
			F	64.7	1,162.0	1,098.0	2,370	280	PL 74-738.
			FRC	5.1	1,098.0	1,075.0	280	170	PL 75-761.
Markland L&D	IN Switzerland	Ohio R	N	0.0	455.0	420.0	0	0	RHA 1909
	KY Gallatin								
Marmet L&D	WV Kanawha R	Kanawha	N	0.0	590.0	566.0	0	0	RHA 1930.
Martins Fk Lk	KY Harlan	Martins Fk of Clover R	F	14.3	1,341.0	1,310.0	578	340	PL 89-298.
			FAR	3.1	1,310.0	1,300.0	340	274	
			R	3.7	1,300.0	1,265.0	274	0	
Maxwell L&D	PA Fayette Washington	Monongahela R	N	0.0	763.0	743.5	0	0	RHA 1909.
McAlpine L&D	KY Jefferson	Ohio R	N	0.0	420.0	383.0	0	0	RHA 1909.
	IN Clark								
Meldahl L&D	KY Bracken	Ohio R	N	0.0	485.0	455.0	0	0	RHA 1909.
	OH Clermont								
Mississinewa Lk	IN Miami	Mississinewa R	F	293.2	779.0	737.0	12,830	3,180	PL 85-500.
			FR	51.9	737.0	712.0	3,180	1,280	.
Mohawk Dam	OH Coshocton	Walhonding R	F	285.0	890.0	799.2	7,950	0	PW 1933.
Mohicanville Dam	OH Ashland	Lk Fork	F	102.0	963.0	932.0	8,800	0	PW 1933.
Monongahela R L&D 2	PA Allegheny	Monongahela R	N	0.0	718.7	710.0	0	0	RHA 1902.
Monongahela R L&D 3	PA Allegheny	Monongahela R	N	0.0	726.9	718.7	0	0	RHA 1905.
Monongahela R L&R 4	PA Washington	Monongahela R	N	0.0	743.5	726.9	0	0	RHA 1909.
	Westmoreland								
Monongahela R L&D 7	PA Greene, Fayette	Monongahela R	N	0.0	778.0	763.0	0	0	RHA 1922.
Monongahela R L&D 8	PA Greene, Fayette	Monongahela R	N	0.0	797.0	778.0	0	0	RHA 1922, 1950, 1973.
Monroe Lk	IN Monroe	Salt Cr	F	258.8	556.0	538.0	18,450	10,750	FCA 1958.
			FMA	159.9	538.0	515.0	10,750	3,280	.
Montgomery Island L&D	PA Beaver	Ohio R	N	0.0	682.0	664.5	0	0	RHA 1909.
Morgantown L&D	WV Monongalia	N	0.0	814.0	797.0	0	0	RHA	

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Project name ¹	State/county	Stream ¹	Project purpose ²	Storage 1,000 AF	Elev limits feet M.S.L.		Area in acres		Auth legis ³
					Upper	Lower	Upper	Lower	
Mosquito Cr Lk	Monongahela R OH Trumbull	Mosquito Cr	F FMCAR	21.7 80.4	904.0 901.4	901.4 899.9	8,900 7,850	1909. 7,220	PL 75-761.
N Br Kokosing River Lk	OH Knox	North Br of Kokosing R	F	13.9	1,146.0	1,121.0	1,140	154	PL 87-874.
N Fk Pound Lk	VA Wise	N Fk Pound R	F FMCR	8.0 1.3	1,644.0 1,611.0	1,611.0 1,601.0	349 154	154 106	PL 86-645. .
New Cumberland L&D	WV Hancock	Ohio R	N	0.0	664.5	644.0	0	0	RHA 1909.
Newburgh L&D	OH Jefferson	Ohio R	N	0.0	358.0	342.0	0	0	RHA 1909.
Nolin Lk	KY Henderson IN Warrick	Ohio R	N	0.0	358.0	342.0	0	0	RHA 1909.
Ohio R L&D 52	KY Edmonson	Nolin R	F FR	439.2 106.4	560.0 515.0	515.0 490.0	14,530 5,790	5,790 2,890	PL 75-761.
Ohio R L&D 53	KY McCracken IL Massac	Ohio R	N	0.0	302.0	290.0	0	0	RHA 1909, 1910, 1918.
Old Hickory L&D	KY Ballard IL Pulaski	Ohio R	N	0.0	290.0	276.6	0	0	RHA 1909, 1910, 1918.
Opekiska L&D	TN Davidson Sumner	Cumberland R	P N	63.0 357.0	445.0 442.0	442.0 375.0	22,500 19,550	19,550 0	RHA 1946.
Paint Cr Lk	WV Monongahela	Monongahela R	N	0.0	857.0	835.0	0	0	RHA 1950.
Paintsville Lk	OH Ross, Highland	Paint Cr	F FMCAR	124.7 11.4	845.0 798.0	798.0 787.5	4,761 1,190	1,190 770	PL 75-761.
Patoka Lk	KY Johnson	Paint Cr	F FCAR	32.8 36.3	731.0 709.0	709.0 650.0	1,867 1,139	1,139 261	PL 89-298.
Piedmont Lk	IN DuBois	Patoka R	F FMCAR	121.1 167.3	548.0 536.0	536.0 506.0	11,300 8,880	8,880 2,010	PL 89-298.
Pike Island L&D	OH Harrison	Stillwater Cr	F FCR	32.2 8.6	924.6 913.0	913.0 909.0	3,170 2,310	2,310 1,987	PW 1933.
	WV Ohio	Ohio R	N	0.0	644.0	623.0	0	0	RHA 1909.

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Project name ¹	State/county	Stream ¹	Project purpose ²	Storage 1,000 AF	Elev limits feet M.S.L.		Area in acres		Auth legis ³
					Upper	Lower	Upper	Lower	
Pleasant Hill Lk	OH Belmont	Clear Fk	F	74.2	1,065.0	1,020.0	2,600	850	PW 1933.
	OH Ashland		FCR	5.5	1,020.0	1,012.5	850	627	
R D Bailey Lk	WV Mingo, Wyoming	Guyandot R	F	169.5	1,155.0	1,035.0	2,850	630	PL 87-874.
			FCAR	12.2	1,035.0	1,012.0	630	440	
Racine L&D	WV Mason	Ohio R	N	0.0	560.0	538.0	0	0	RHA 1909.
	OH Meigs								
Rough River Lk	Grayson, Breckinridge	Rough R	F	214.4	524.0	495.0	10,260	5,100	PL 75-761.
	Ridge		FMR	90.2	495.0	470.0	5,100	2,180	
Salamonie Lk	IN Wabash	Salamonie R	F	202.9	793.0	755.0	9,340	2,860	PL 85-500.
			FR	47.6	755.0	730.0	2,860	976	
Senecaville Lk	OH Guernsey	Seneca Fk	F	45.1	842.5	832.2	5,170	3,550	PW 1933.
			FCR	12.8	832.2	828.2	3,550	2,912	
Shenango River Lk	PA Mercer	Shenango R	F	151.0	919.0	896.0	11,090	3,560	PL 75-761.
			FCAR	29.9	896.0	885.0	3,560	1,910	
Smithland L&D	KY Livingston	Ohio R	N	0.0	324.0	302.0	0	0	RHA 1909.
	IL Pope								
Summersville Lk	WV Nicholas	Gauley R	F	221.9	1,710.0	1,1652.0	4,913	2,790	PL 75-761.
			FRCA	161.8	1,652.0	1,535.0	2,790	514	
Sutton Lk	WV Braxton	Elk R	FCAR	60.0	925.0	850.0	1,520	270	PL 75-761.
Tappan Lk	OH Harrison	L Stillwater Cr	F	26.5	909.0	899.3	3,100	2,350	PW 1933.
			FCR	11.4	899.3	894.0	2,350	1,960	
Tionesta Lk	PA Forest	Tionesta Cr	F	125.6	1,170.0	1,085.0	2,770	480	PL 74-738. PL 75-761.
Tom Jenkins Dam, Burr Oak, Lk	OH Athens	E Br Sandy Cr	F	17.6	740.0	721.0	1,192	664	FCA 1944.
			FRM	5.8	721.0	710.0	664	394	PL 78-534.
Tygart Lake	WV Taylor	Tygart R	F	178.1	1,167.0	1,094.0	3,430	1,740	PWA 1934.
			FMACR	99.9	1,094.0	1,010.0	1,740	620	
Union City Res	PA Erie	French Cr	F	47.6	1,278.0	1,210.0	2,290	0	PL 87-874.

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					Upper	Lower	Upper	Lower	
Uniontown L&D	KY Union	Ohio R	N	0.0	342.0	324.0	0	0	RHA 1909.
W FK of Mill Cr Winton Woods Lk	IN Posey								
Willow Island L&D	OH Hamilton	W Fk Mill Cr	F	9.8	702.0	675.0	557	183	PL 79-526.
	WV Pleasants	Ohio R	N	0.0	602.0	582.0	0	0	RHA 1909.
	OH Washington								
Wills Cr Lk	OH Coshockton Wills Cr, Muskingum		F	190.0	779.0	742.0	11,450	900	PW 1933.
			CR	0.0	0.0	0.0	0	0	
Winfield L&D	WV Putnam	Kanawha R	N	0.0	566.0	538.0	0	0	RHA 1935.
Wolf Cr Dam, Lk Cumberland	KY Russell	Cumberland R	P	2,142.0	723.0	673.0	50,250	35,820	
			F	2,094.0	760.0	723.0	63,530	50,250	PL 75-761.
Woodcock Cr Lk	PA Crawford	Woodcock Cr	F	15.0	1,209.0	1,181.0	775	325	FCA 1962.
			FCAR	5.0	1,181.0	1,162.5	325	100	
Youghiogheny R Lk	PA Fayette	Youghiogheny R	F	99.5	1,470.0	1,439.0	3,570	2,840	FCA 1938.
			FCAR	149.3	1,439.0	1,419.0	2,840	2,300	
South Atlantic Division									
Aberdeen L&D and Res	MS Monroe	Tombigbee R	N	3.9	190.5	189.5	4,359	3,883	PL 79-525.
Aliceville Lock Dam & Res	AL Pickens	Tombigbee R	N	7.6	136.5	135.5	8,655	7,945	PL 79-525.
Allatoona Dam & Res	GA Bartow	Etowah R	F	302.6	860.0	840.0	19,201	11,862	PL 77-228.
			PMAR	284.6	840.0	800.0	11,862	3,251	
B Everett Jordan Dam & Lk	NC Chatham	Haw R	F	538.4	240.0	216.0	31,811	13,942	PL 88-253.
			FMCAR	140.4	216.0	202.0	13,942	6,658	
Bay Springs Lock Dam & Res	MS Tishomingo	Tombigbee R	N	37.0	414.0	408.0	6,700	5,740	PL 79-525.
Buford Dam Lk, Sidney Lanier	GA Forsyth, Gwinnett	Chattahoochee R	F	598.8	1,085.0	1,071.0	47,182	38,542	PL 79-14.
			PNMR	1,087.6	1,071.0	1,035.0	38,542	22,442	
Carters Dam & Res	GA Murray	Coosawattee R	F	89.2	1,099.0	1,074.0	3,880	3,275	PL 79-14.
			PRA	41.4	1,074.0	1,022.0	3,275	2,196	

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					Upper	Lower	Upper	Lower	
Claiborne Lock Dam & Res	AL Monroe	Alabama R	N	16.6	35.0	32.0	5,930	5,210	PL 79-14.
Clarks Hill Dam & Lk	GA Columbia	Savannah R	F	390.0	335.0	330.0	78,500	71,100	PL 78-534.
	SC McCormick		FP	1,045.0	330.0	312.0	71,100	45,000	
Coffeeville Lock Dam & Res	AL Clark, Choctaw	Tombigbee R	N	19.9	32.5	30.0	8,500	7,500	PL 60-317.
Columbus Lock Dam & Res	MS Lowndes	Tombigbee R	N	8.5	163.5	162.5	9,400	8,500	PL 79-525.
Demopolis Lock Dam & Res	AL Sumter, Marengo	Tombigbee R	N	0.0	73.0	73.0	10,000	10,000	PL 60-317.
Falls Dam & Lk	NC Wake	Neuse R	F	220.9	264.0	250.1	20,810	11,310	PL 89-298.
			FMCAR	89.7	250.1	236.5	11,310	2,600	
G W Andrews L&D and Res	AL Houston	Chattahoochee R	N	8.2	102.0	96.0	1,540	1,190	PL 79-14.
	GA Early								
Gainesville L&D and Res	AL Sumter, Greene	Tombigbee R	N	5.8	109.5	108.5	6,920	5,900	PL 79-525.
Hartwell Dam & Lk	GA Hart	Savannah R	F	293.0	665.0	660.0	61,400	55,950	PL 81-516.
	SC Anderson		FP	1,416.0	660.0	625.0	55,950	27,650	
Holt Lock Dam & Res	AL Tuscaloosa	Black-Warrior R	NP	3.3	187.0	186.0	3,296	3,252	PL 60-317.
Inglis Dam Lk Rousseau	FL Levy, Marion, Citrus	Cross FL Barge Canal	N	13.0	27.5	24.0	4,030	2,040	PL 77-675.
Jim Woodruff L&D	FL Gadsden, Jackson	Apalachicola R	NP	20.0	77.5	76.5	38,850	36,000	PL 79-14.
John H Kerr Dam & Res	VA Mecklenburg	Roanoke R	F	1,281.4	320.0	300.0	83,200	48,900	PL 78-534.
			FP	1,027.0	300.0	268.0	48,900	19,700	
John Hollis Bankhead L&D and Res	AL Tuscaloosa	Black-Warrior R	NP	27.1	255.0	252.0	9,245	8,730	PL 60-168.
Lk Okeechobee	FL Okeechobee, Glades, Hendry, Palm Beach, Martin	Central and Southern FL	FNIMC	2,859.0	17.5	10.5	454,900	326,000	PL 71-520, PL 75-392, PL 79-14, PL 80-858, PL 83-780, PL 90.
Lock A	MS Monroe	Tombigbee R	N	0.9	220.5	219.5	980	850	PL 79-525.
Lock B	MS Monroe	Tombigbee R	N	2.7	245.5	244.5	2,841	2,615	PL 79-525.
Lock C	MS Itawamba	Tombigbee R	N	1.6	270.5	269.5	1,699	1,586	PL 79-525.
Lock D	MS Itawamba	Tombigbee R	N	2.0	300.5	299.5	2,021	1,959	PL 79-525.
Lock E	MS Itawamba, Prentiss	Tombigbee R	N	0.9	330.5	329.5	889	821	PL 79-525.

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					Upper	Lower	Upper	Lower	
Millers Ferry L&D	AL Wilcox	Alabama R	NP	16.7	80.0	79.0	17,201	16,160	PL 79-14.
Okatibbee Dam & Res	MS Lauderdale	Okatibbee Cr	F	46.5	352.0	343.0	6,580	3,800	PL 87-874.
Philpott Dam & Lk	VA Henry	Chickasawbay R	RMA	34.3	343.0	328.0	3,800	1,275	PL 78-534.
		Smith R	F	34.2	985.0	974.0	3,370	2,880	
			FP	111.2	974.0	920.0	2,880	1,350	
R B Russell Dam and Lk	GA Elbert	Savannah R	F	140.0	480.0	475.0	29,340	26,653	PL 89-789.
	SC Abbeville		FP	126.8	475.0	470.0	26,653	24,117	
Robert F Henry Lock Dam & Res	AL Autauga, Lowndes	Alabama R	NP	44.6	125.0	124.0	13,300	10,470	PL 79-14.
Rodman Dam & Lk	FL Putman & Marion	Cross FL Barge Canal	N	48.0	23.2	20.0	17,350	12,950	PL 77-675.
S-10 & Water Cons Area 1	FL Palm Beach	Central and Southern FL	F	181.9	18.3	17.0	141,250	141,250	PL 80-858.
S-11 & Water Cons Area 2A	FL Palm Beach Broward	Central and Southern FL	FIMC	273.2	17.0	14.0	141,250	26,00	PL 80-858.
			F	236.3	16.6	14.5	110,500	110,500	
			FIMC	165.0	14.5	13.0	110,500	107,500	
S-12 & Water Cons Area 3A	FL Broward & Dade	Central and Southern FL	F	1,661.0	14.5	10.5	487,200	385,000	PL 80-858.
Selden Lock and Res	AL Hale, Greene	Black-Warrior R	FIMC	465.0	10.5	9.5	385,000	316,000	PL 83-780.
			N	9.1	95.5	94.0	8,200	6,900	
			F	112.0	1,075.0	1,030.0	4,000	1,475	
W Kerr Scott Dam & Res	NC Wilkes	Yadkin R	FM	33.0	1,030.0	1,000.0	1,475	675	PL 79-526.
Walter F George L&D	GA Clay	Chattahoochee R	NP	244.0	190.0	184.0	45,181	36,375	PL 81-516.
West Point Dam & Res	AL Henry								PL 87-874.
	GA Troup	Chattahoochee R	NPMAR	306.1	635.0	620.0	25,864	15,512	
William Bacon Oliver L&D and Res	AL Tuscaloosa	Black Warrior R	N	0	122.9	122.9	790	790	PL 60-317.

South Pacific Division

Alamo Dam & Lk	AZ Mohave, Yuma	Bill Williams R	F	1,046.2	1,235.0	1,174.0	13,307	7,045	PL 78-534.
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					Upper	Lower	Upper	Lower	
Bear Dam	CA Mariposa	Bear Cr	F	7.7	413.5	344.0	265	0	PL 78-534.
Black Butte Lk	CA Tehama	Stony Cr	FI	137.1	473.5	414.6	4,453	577	PL 78-534.
Brea Dam & Res	CA Orange	Brea Cr	F	4.0	279.0	208.0	163	0	FCA 1936.
Buchanan Dam H.V. Eastman Lk	CA Madera	Chowchilla R	F	45.0	587.0	559.0	1,785	1,482	PL 78-874.
			FI	140.0	587.0	466.0	1,785	484	
Burns Dam	CA Merced	Burns Cr	F	6.8	300.0	266.0	662	0	PL 78-534.
Carbon Canyon Dam & Res	CA Orange	Carbon Cr	F	6.6	475.0	403.0	225	0	PL 74-738.
Coyote Valley Dam Lk Mendocino	CA Mendocino	East Fork, Russian R	F	50.1	764.8	737.5	1,922	1,740	PL 75-761.
			IM	72.3	737.5	637.0	1,740	20	
Dry Cr (Warm Springs) Lk & Channel	CA Sonoma	Dry Cr	F	136.0	495.0	451.1	3,600	2,600	PL 87-874.
			MR	225.0	451.1	291.0	2,600	500	
Farmington Dam	CA San Joaquin, Stanislaus	Littlejohn Cr	F	52.0	156.5	120.0	4,107	0	PL 78-534.
Fullerton Dam & Res	CA Orange	Fullerton Cr	F	0.8	290.0	261.0	62	0	FCA 1936.
Hansen Dam Res	CA Los Angeles	Tujunga Wash	F	25.4	1,060.0	990.0	781	0	FCA 1936.
Hidden Dam Hensley Lk	CA Madera	Fresno R	F	65.0	540.0	485.8	1,567	811	PL 87-874.
			FI	85.0	540.0	448.0	1,567	280	
Isabella Lk	CA Kern	Kern R	FI	568.1	2,605.5	2,470.0	11,454	26	PL 78-534.
Lopez Dam Res	CA Los Angeles	Pocoima Wash	F	0.4	1,272.9	1,253.7	40	0	FCA 1936.
Mariposa Dam	CA Mariposa	Mariposa Cr	F	15.0	439.5	370.0	512	0	PL 78-534.
Martis Cr Lk	CA Nevada	Martis Cr	F	19.6	5,838.0	5,780.0	762	61	PL 87-874.
Mathews Canyon Dam & Res	NV Lincoln	Mathews Canyon	F	6.3	5,461.0	5,420.0	300	0	PL 81-516.
Mojave River Dam & Res	CA San Bernardino	Mojave R	F	89.7	3,134.0	2,988.0	1,978	0	PL 86-645.
New Hogan Lk	CA Calaveras	Calaveras R	F	165.0	713.0	666.2	4,333	2,818	PL 78-534.
			FI	302.2	713.0	586.0	4,333	702	
Owens Dam	CA Mariposa	Owens Cr	F	3.6	407.5	347.0	174	0	PL 78-534.
Painted Roc Dam & Res	AZ Maricopa	Gila R	F	2,491.5	661.0	524.0	53,200	0	PL 81-516.

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					Upper	Lower	Upper	Lower	
Pine Canyon Dam & Res	NV Lincoln	Pine Canyon	F	7.8	5,675.0	5,604.0	254	0	PL 81-516.
Pine Flat Lk Kings R	CA Fresno	Kings R	F	1,000.0	951.5	565.5	5,956	0	PL 78-534.
Prado Dam & Res	CA Riverside	Santa Ana R	F	196.2	543.0	460.0	6,630	0	FCA 1936.
San Antonio Dam & Res	CA Los Angeles	San Antonio Cr	F	7.7	2,238.0	2,125.0	145	0	FCA 1936.
Santa Fe Dam & Res	CA Los Angeles	San Gabriel R	F	32.1	496.0	421.0	1,084	0	FCA 1936, 1941.
Sepolveda Dam & Res	CA Los Angeles	Los Angeles R	F	17.4	710.0	668.0	1,335	0	FCA 1936.
Success Lk	CA Tulare	Tule R	FI	75.0	652.5	588.9	2,477	409	PL 78-534.
Terminus Dam Lk Kaweah	CA Tulare	Kaweah R	FI	136.1	694.0	570.0	1,913	276	PL 78-534.
Whitlow Ranch Dam & Res	AZ Pinal	Queen Cr	F	35.6	2,166.0	2,056.0	828	0	PL 79-526.
Whittler Marrows Dam & Res	CA Los Angeles	San Gabriel Rio Hondo R	F	34.9	228.5	184.0	2,411	0	FCA 1936.
Southwestern Division									
Abiquiu Dam	NM Rio Arriba	Rio Chama	F	572.2	6,283.5	6,220.0	7,469	4,120	PL 80-858.
			FM	191.3	6,220.0	6,060.0	4,120	0	
Addicks Res	TX Harris	Buffalo Bayou	F	200.8	112.0	71.1	16,423	0	HD250-83-2.
Aquilla Lk	TX Hill	Aquilla Cr	F	161.4	564.5	537.5	8,980	3,280	PL 90-483.
			MR	93.6	537.5	478.6	3,280	26	
Arcadia Lk	OK Oklahoma	Deep Fork R	F	64.4	1,029.5	1,006.0	3,820	1,820	PL 91-611.
			FMCR	27.4	1,006.0	970.0	1,820	20	
B A Steinhagen Lk	TX Taylor, Jasper	Neches R	F	24.5	83.0	81.0	13,700	10,950	SD98-76-1.
Bardwell Lk	TX Ellis	Waxahachie Cr	F	79.6	439.0	421.0	6,040	3,570	PL 86-399.
			M	42.8	421.0	372.6	3,570	0	
Barker Res	TX Harris Ft Bend	Buffalo Bayou	F	209.0	106.0	73.2	16,734	0	HD250-83-2, RHA 1938.
Beaver Lk	AR Carrol, Benton, Washington	White R	F	299.6	1,130.0	1,120.0	31,700	28,220	PL 83-780.
			FPM	925.1	1,120.0	1,077.0	28,220	15,540	PL 85-500.
Belton Lk	TX Bell	Leon R	F	640.0	631.0	594.0	23,600	12,400	PL 79-526.
			MI	372.7	594.0	470.0	12,400	42	HD88-81-1.
Benbrook Lk	TX Tarrant, Parker	Clear Fk Trinity R	F	170.4	724.0	694.0	7,630	3,770	HD103-771.

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					Upper	Lower	Upper	Lower	
Big Hill LK	KN Labette	Big Hill Cr	NM	72.5	694.0	656.0	3,770	730	
			F	13.1	867.5	858.0	1,520	1,240	PL 87-874.
			FMR	27.2	858.0	814.0	1,240	70	HD572-87-2.
Birch Lk	OK Osage	Birch Cr	F	39.0	774.0	750.5	2,340	1,140	PL 87-874.
			FMCAR	15.8	750.5	730.0	1,140	384	HD563-87-2.
Blue Mountain Lk	AR Yell, Logan	Petit Jean R	F	233.3	419.0	384.0	11,000	2,910	PA 75-761.
Broken Bow Lk	OK McCurtain	Mountain Fk R	F	450.2	627.5	599.5	18,000	14,200	PL 85-500.
			FRPMAC	469.8	599.5	559.5	14,200	9,200	
Bull Shoals Lk	AR Baxter, Marion, Boone	White R	F	2,360.0	695.0	654.0	71,240	45,440	PL 77-228.
			PF	1,003.0	654.0	628.5	45,440	33,800	
Canton Lk	OK Blain	N Canadian R	F	265.8	1,638.0	1,615.4	15,710	7,910	PL 75-761.
Canyon Lk	TX Comal	Guadalupe R	FMI	97.2	1,615.4	1,596.5	7,910	2,710	HD56-/75-3.
			F	346.4	934.0	909.0	12,890	8,240	PL 79-14.
			M	366.4	909.0	75.0	8,240	0	
Clearwater Lk	MO Reynolds, Wayne	Black R	F	391.8	567.0	494.0	10,400	1,630	PL 75-761.
Cochiti Lk	NM Sandoval, Sante Fe, Los Alamos	Rio Grande	F	545.0	5,460.5	5,356.6	9,361	1,200	PL 86-645.
			FRC	43.0	5,356.6	5,330.0	1,200	0	
			F	198.8	4,218.0	4,201.0	13,664	9,692	HD 308-74.
Conchas Lk	NM San Miguel	Candian R	FI	259.6	4,201.0	4,155.0	9,692	3,000	
			F	184.3	732.0	710.0	13,380	4,850	PL 87-874.
Copan Lk	OK Washington	L Caney R	FMCA	42.8	710.0	687.5	4,850	110	HD563-87-2.
			F	63.8	1,289.0	1,274.0	5,400	3,230	PL 81-516.
Council Grove Lk	KS Morris	Neosho R	FMAR	48.5	1,274.0	1,240.0	3,230	42	
			F	101.3	473.5	437.0	4,050	1,680	PL 85-500.
DeQueen Lk	AR Sevier	Rolling Fork R	FMCRQ	25.5	437.0	415.0	1,680	710	
			F	67.1	557.5	526.0	2,970	1,360	PL 85-500.
Dierks Lk	AR Sevier, Howard	Saline R	FMCR	15.1	526.0	512.0	1,360	810	
			F	79.2	1,347.5	1,339.0	10,740	8,000	PL 89-298.
Eldorado Lk	KS Butler	Walnut R	F	79.2	1,347.5	1,339.0	10,740	8,000	PL 89-298.

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					Upper	Lower	Upper	Lower	
Elk City Lk	KS Montgomery	Elk R	FMAR	154.0	1,339.0	1,296.0	8,000	420	HD232-89-1.
			F	239.5	825.0	796.0	13,150	4,450	HD440-76-1.
			FMA	44.8	796.0	764.0	4,450	64	
Eufaula Lk	OK McIntosh, Pittsburg, Haskell	Candian R	F	1,510.9	597.0	585.0	147,960	105,480	PL 79-525.
Fall River Lk	KS Greenwood	Fall R	FNPM	1,463.0	585.0	565.0	105,480	46,120	
			F	234.5	987.5	948.5	10,400	2,350	HD440-76-1.
			FA	15.0	948.5	940.0	2,350	1,170	
Fort Gibson Lk	OK Wagoner	Neosho (Grand) R	F	919.2	582.0	554.0	51,000	19,900	FEC 1941.
			FP	53.9	554.0	551.0	19,100	16,950	RHA 1946.
Fort Supply Lk	OK Woodward	Wolf Cr	F	86.8	2,028.0	2,004.0	5,690	1,820	PL 74-738.
			FM	13.9	2,004.0	1,988.0	1,820	0	
Galisteo Dam	NM Santa Fe	Galisteo Cr	F	79.4	5,608.0	5,496.0	2,060	0	PL 86-645.
Georgetown Lk	TX Williamson	N.F. San Gabriel R	F	87.6	834.0	791.0	3,220	1,310	PL 87-874.
			MC	29.2	791.0	699.0	1,310	0	HD 591-82-2.
Gillham Lk	AR Howard, Polk	Cossatot R	F	188.7	569.0	502.0	4,680	1,370	PL 85-500.
			FMCQ	29.3	502.0	464.5	1,370	310	
Granger Lk	TX Williamson	San Gabriel R	F	162.2	528.0	504.0	11,040	4,400	PL 87-874.
			M	37.9	504.0	440.0	4,400	0	
Grapevine Lk	TX Denton, Tarrant	Denton Cr	F	243.1	560.0	535.0	12,710	7,280	HD103-77-1.
			M	154.3	535.0	451.0	7,380	41	
Great Salt Plains Lk	OK Alfalfa	Salt Fk	F	240.0	1,138.5	1,125.0	27,730	8,693	PL 74-738.
		Arkansas R	FC	31.4	1,125.0	1,115.0	8,690	0	
Greers Ferry Lk	AR Cleburne, Van Buren	Little Red R	F	934.0	487.0	461.0	40,480	31,460	PL 75-761.
			FP	716.5	461.0	435.0	31,460	23,740	PL 83-780.
Heyburn Lk	OK Creek	Polecat Cr	F	48.4	784.0	761.5	3,700	917	PL 79-526.
			FM	3.8	761.5	55.5	917	394	
Hords Cr Lk	TX Coleman	Hords Cr	F	16.7	1,920.0	1,900.0	1,260	510	PL 77-228.
			M	5.8	1,900.0	1,848.0	510	0	

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Hugo Lk	OK Choctaw	Kiamichi R	F	809.1	437.5	404.5	34,490	13,250	PL 79-526.
			FMCAR	127.2	404.5	390.0	13,250	4,500	
Hulah Lk	OK Osage	Caney R	F	257.9	765.0	733.0	13,000	3,570	PL 74-738.
	KS Chautaugua		FMA	31.1	733.0	710.0	3,570	0	PL 84-843.
Jemez Canyon Dam	NM Sandoval	Jemez R	F	73.0	5,232.0	5,196.1	2,877	1,370	PL 80-858 PL 81-516.
Joe Pool Lk	TX Dalla, Ellis, Tarrant	Mountain Cr	F	1,238.0	536.0	522.0	10,940	7,470	PL 89-298.
			M	176.9	522.0	456.0	7,470	10	
John Martin Res	CO Bent	Arkansas R	F	270.3	3,870.0	3,851.0	17,630	11,655	PL 74-738.
			FRC	350.9	3,851.0	0.0	11,655	0	
John Redmond Dam & Res	KS Coffee	Neosho R	F	559.0	1,068.0	1,039.0	31,700	9,300	PL 81-516.
			FMAR	70.8	1,039.0	1,020.0	9,300	108	
Kaw Lk	OK Kay, Osage	Arkansas R	F	919.4	1,044.5	1,010.0	38,020	17,040	PL 87-874.
	KS Cowley		FMARC	343.5	1,010.0	978.0	17,040	5,590	
Keystone Lk	OK Tulsa	Arkansas R	F	1,180.0	754.0	723.0	54,300	23,600	PL 81-516.
			FNPMC	296.7	723.0	706.0	23,600	13,300	
L&D 01, Norrell	AR Arkansas	Arkansas Post Canal	N	0.0	142.0	142.0	140	140	HD 758-79, RHA 1946.
L&D 02, Wilbur D. Mills Dam	AR Desha, Arkansas	Arkansas R	N	18.7	162.3	160.5	10,700	9,400	HD 758-79, RHA 1946.
L&D 03	AR Jefferson, Lincoln	Arkansas R	N	8.3	182.3	180.0	3,750	3,180	HD 758-79, RHA 1946.
L&D 04	AR Jefferson	Arkansas R	N	12.9	196.3	194.0	5,820	5,200	HD 758-79, RHA 1946.
L&D 05	AR Jefferson	Arkansas R	N	14.4	213.3	211.0	6,900	5,550	HD 758-79, RHA 1946.
L&D 06, David D. Terry	AR Pulaski	Arkansas R	N	9.6	231.3	229.0	4,830	4,130	HD 758-79.
L&D 07, Murray	AR Pulaski	Arkansas R	N	24.7	249.7	247.0	10,350	8,100	RHA 1946.
L&D 08, Toad Suck Ferry	AR Faulkner, Perry	Arkansas R	N	8.7	265.3	263.0	4,130	3,600	RHA 1946.
L&D 09, Arthur V. Ormond	AR Conway	Arkansas R	N	15.8	287.0	284.0	5,660	4,910	HD 758-79.
L&D, W. Rockefeller Lk									
L&D 10, Lk Dardanelle	AR Pope Yell	Arkansas R	NP	72.3	338.2	336.0	34,700	31,140	HD 758-79, RHA 1946.
L&D 11, Ozark-Jetta Taylor	AR Franklin	Arkansas	NPR	25.3	372.5	370.0	11,100	8,800	RHA 1946, HD 758-79.

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L&D 13, James W. Trimble	AR Sebastian, Crawford	Arkansas R	N	18.1	392.0	389.0	6,820	5,200	RHA 1946.
L&D 14, W. D. Mayo	OK Sequoyah, Leflore	Arkansas R	N	0.0	413.0	0.0	1,600	0	PL 79-525.
L&D 15, Robert S. Kerr Res	OK Leflore, Sequoyah	Arkansas R	NP	84.7	460.0	458.0	43,800	40,760	PL 79-525.
L&D 16, Webbers Falls Res	OK Muskogee	Arkansas R	NP	32.4	490.0	487.0	10,900	9,300	PL 79-525.
L&D 17, Chouteau	OK Wagoner	Verdigris R	N	0.0	511.0	511.0	2,270	2,270	PL 79-525, HD 758-79-2.
L&D 18, Newt Graham	OK Wagoner	Verdigris R	N	0.0	532.0	532.0	1,490	1,490	PL 97-525.
Lake O' The Pines	TX Marion	Cypress Cr	F	579.5	249.5	228.5	38,200	18,700	PL 79-526.
			M	250.0	228.5	201.0	18,700	1,100	
Lavon Lk	TX Collin	East Fork, Trinity R	F	275.6	503.5	492.0	29,450	21,400	HD 533-78-2.
			M	380.0	492.0	433.0	21,400	2,87	
Lewisville Lk Garza-Little Elm Dam	TX Denton	Elm Fork Trinity R	F	525.2	532.0	515.0	39,080	23,280	HD 403-77-1.
			M	436.0	515.0	433.0	23,280	12	
Marion Lk	KS Marion	Cottonwood R	F	60.2	1,358.5	1,350.5	9,050	6,200	PL 81-516.
			FMAR	83.3	1,350.5	1,320.0	6,200	170	
Millwood Lk	AR Little R Hempstead	Little R	F	1,650.0	287.0	259.2	95,200	29,200	PL 79-526.
			FMC	153.3	259.2	252.0	29,200	13,100	HD 785-79.
Navarro Mills Lk	TX Navarro Hill	Richland Cr	F	143.2	443.0	424.5	11,700	5,070	HD 498-83-2.
			M	53.2	424.5	375.3	5,070	0	
Nimrod Lk	AR Perry, Yell	Fourche La Fave R	F	307.0	373.0	342.0	18,300	3,550	FCA 1938.
Norfolk Lk	AR Baxter, Fulton	North Fork R	F	731.8	580.0	552.0	30,700	21,990	PL 75-761.
	MO Ozark		FP	707.0	552.0	510.0	21,990	12,320	FCA 1941
North Fork Lk	TX Williamson	N.F. San Gabriel R	F	87.6	834.0	791.0	3,220	1,310	PL 87-874.
			MC	29.2	791.0	699.0	1,310	0	HD 591-82-2.
O. C. Fisher Lk	TX Tom Green	N. Concho R	F	277.2	1,938.5	1,908.0	12,700	5,440	PL 77-228.
			M	80.4	1,908.0	1,836.0	5,440	3	
Oologah Lk	OK Rogers	Verdigris R	F	965.6	661.0	638.0	56,800	29,460	PL 75-761.
			FMN	544.1	638.0	592.0	29,460	1,120	

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Optima Lk	OK Texas	N. Candian R	F	100.5	2,779.0	2,763.5	7,640	5,340	PL 74-738.
Pat Mayse Lk	TX Lamar	Sanders Cr	FMRC	117.7	2,763.5	2,726.0	5,340	1,335	PL 87-874.
			FMCR	64.6	460.5	451.0	7,680	5,993	
Pine Cr	OK McCurtain	Little R	F	119.9	451.0	415.0	5,993	996	HD 88-71.
			FMAC	388.1	480.0	443.5	17,230	4,980	PL 85-500.
Proctor Lk	TX Comanche	Leon R	F	77.6	443.5	414.0	4,980	700	HD 170-85-1.
Sam Rayburn Res	TX Jasper, San Augustine, Angelina	Angelina R	F	310.1	1,197.0	1,162.0	14,010	4,610	PL 83-780, HD 535-81-2.
			PMC	1,099.4	173.0	164.4	142,700	114,500	HD 981-76-1.
Santa Rosa	NM Guadalupe	Pecos R	F	1,446.2	164.4	149.0	114,500	74,040	PL 83-780.
			FI	340.0	4,746.2	4,776.5	10,740	3,823	
Sardis	OK Pushmatah	Jackfork Cr	F	160.0	4,776.5	4,746.2	7,115	3,823	HD 602-79-2.
			FMR	122.6	607.0	599.0	16,960	13,610	
Somerville Lk	TX Washington, Lee, Burleson	Yegua Cr	F	274.2	599.0	542.0	13,610	40	PL 83-780.
			M	337.7	258.0	238.0	24,400	11,460	
Stiatook	OK Osage	Hominy Cr	F	143.9	238.0	200.0	11,460	0	HD 563-87.
			FMARC	178.0	729.0	714.0	13,690	10,190	
Stillhouse H. Lk	TX Bell	Lampasas R	F	311.6	714.0	657.0	10,190	1,430	PL 83-780.
			M	390.6	666.0	622.0	11,830	6,430	
Table Rock Lk	MO Taney, Stone, Barry	White R	F	204.9	622.0	498.0	6,430	0	PL 77-228.
	AR Carroll, Boone		FP	760.0	931.0	915.0	52,250	43,070	FCA 1938.
Tenkiller Ferry Lk	OK Cherokee, Sequoyah	Illinois R	F	1,181.50	915.0	881.0	43,070	27,300	RHA 1946.
			FP	576.7	667.0	632.0	20,800	12,900	PL 75-761.
Texoma Lk, Denison Dam	TX Marshall	Red R	F	371.0	632.0	594.5	12,900	7,370	
	OK Bryan, Cook, Grayson		FPM	2,669.0	640.0	617.0	144,000	88,000	HD 440-76-1.
Toronto Lk	KS Woodson	Verdigris R	F	1,612.0	617.0	590.0	88,000	41,000	
			FMA	179.8	931.0	901.5	11,740	2,660	PL 85-500.
Trinidad Lk	CO Las Animas	Purgatorie R	F	10.7	901.5	896.7	2,660	1,720	
			F	58.0	6,260.0	6,230.0	2,107	1,453	

¹ Res—Reservoir; Lk—Lake; Div—Diversion; R—River; Cr—Creek; Fk—Fork; L&D—Lock & Dam; GIWW—Gulf Intercoastal Waterway; FG—Floodgate; CS—Control Structure; DS—Drainage Structure; PS—Pump Station.

² F—Flood Control; N—Navigation; P—Hydropower; I—Irrigation; M—Municipal and/or Industrial Water/Supply; C—Fish and Wildlife Conservation; R—Recreation; A—Low Flow Augmentation or Pollution Abatement; Q—Quality or Silt Control.

³ PL—Public Law; HD—House Document; RHA—River & Harbor Act; PW—Public Works; FCA—Flood Control Act; WSA—Water Supply Act.

Project name ¹	State/county	Stream ¹	Project purpose ²	Storage 1,000 AF	Elev limits feet M.S.L.		Area in acres		Auth legis ³
					Upper	Lower	Upper	Lower	
Two Rivers Dam	NM Chaves	Rio Hondo R	FI	20.0	6,230.0	0.0	1,453	0	PL 83-780.
Waco Lk	TX McLennan	Bosque R	F	150.0	4,032.0	3,945.0	4,806	0	PL 83-780.
			M	3.3	500.0	455.0	19,440	7,270	PL 83-780.
Waurika Lk	OK Jefferson	Beaver Cr	F	100.8	455.0	370.0	7,240	0	HD 535-81-2.
			FMCAR	140.4	962.5	951.4	15,000	10,100	PL 88-253.
Whitney Lk	TX Hill, Bosquel	Brazos R	F	199.7	951.4	910.0	10,100	830	
			PM	1,372.0	571.0	533.0	49,820	23,560	PL 77-228.
Wister Lk	OK LeFlore	Pouteau R	F	381.9	533.0	425.0	23,560	475	HD 390-76-1.
Wright Patman Lk	TX Bowie, Cass	Sulphur R	F	387.0	502.5	474.6	23,070	5,000	PL 75-761.
			FM	2,363.7	259.5	220.0	119,700	20,300	PL 79-526.
				142.7	220.0	180.0	20,300	0	

¹ Res—Reservoir; Lk—Lake; Div—Diversion; R—River; Cr—Creek; Fk—Fork; L&D—Lock & Dam; GIWW—Gulf Intercoastal Waterway; FG—Floodgate; CS—Control Structure; DS—Drainage Structure; PS—Pump Station.

² F—Flood Control; N—Navigation; P—Hydropower; I—Irrigation; M—Municipal and/or Industrial Water/Supply; C—Fish and Wildlife Conservation; R—Recreation; A—Low Flow Augmentation or Pollution Abatement; Q—Quality or Silt Control.

³ PL—Public Law; HD—House Document; RHA—River & Harbor Act; PW—Public Works: FCA—Flood Control Act; WSA—Water Supply Act.

[47 FR 44544, Oct. 8, 1982, as amended at 52 FR 15804, Apr. 30, 1987; 52 FR 23816, June 25, 1987; 57 FR 35757, Aug. 11, 1992.
Redesignated at 60 FR 19851, Apr. 21, 1995]

§ 222.6 National Program for Inspection of Non-Federal Dams.

- (a) **Purpose.** This regulation states objectives, assigns responsibilities and prescribes procedures for implementation of a National Program for Inspection of Non-Federal Dams.
- (b) **Applicability.** This regulation is applicable to all Divisions and Districts having Civil Works functions.
- (c) **References.**
 - (1) The National Dam Inspection Act, Pub. L. 92-367, 8 August 1972.
 - (2) Freedom of Information Act, Pub. L. 87-487, 4 July 1967.
 - (3) ER 500-1-1.
- (d) **Authority.** The National Dam Inspection Act, Public Law 92-367, 8 August 1972 authorizes the Secretary of the Army, acting through the Chief of Engineers, to carry out a national program of inspection of non-Federal dams for the purpose of protecting human life and property.
- (e) **Scope.** The program provides for:
 - (1) An update of the National Inventory of Dams.
 - (2) Inspection of the following non-Federal dams (the indicated hazard potential categories are based upon the location of the dams relative to developed areas):
 - (i) Dams which are in the high hazard potential category (located on Federal and non-Federal lands).
 - (ii) Dams in the significant hazard potential category believed by the State to represent an immediate danger to the public safety due to the actual condition of the dam.
 - (iii) Dams in the significant hazard potential category located on Federal lands.
 - (iv) Specifically excluded from the national inspection program are:
 - (A) Dams under the jurisdiction of the Bureau of Reclamation, the Tennessee Valley Authority, the International Boundary and Water Commission and the Corps of Engineers and
 - (B) Dams which have been constructed pursuant to licenses issued under the authority of the Federal Power Act, and
 - (C) Dams which have been inspected within the 12-month period immediately prior to the enactment of this act by a State agency and which the Governor of such State requests be excluded from inspection.
- (f) **Objectives.** The objectives of the program are:
 - (1) To update the National Inventory of Dams by 30 September 1980.

- (2) To perform the initial technical inspection and evaluation of the non-Federal dams described in paragraph 222.8(e) of this section to identify conditions which constitute a danger to human life or property as a means of expediting the correction of hazardous conditions by non-Federal interests. The inspection and evaluation is to be completed by 30 September 1981.
- (3) To obtain additional information and experience that may be useful in determining if further Federal actions are necessary to assure national dam safety.
- (4) Encourage the States to establish effective dam safety programs for non-Federal dams by 30 September 1981 and assist the States in the development of the technical capability to carry out such a program.

(g) *Program execution —*

(1) *Responsibilities.*

- (i) The owner has the basic legal responsibility for potential hazards created by their dam(s). Phase II studies, as described in Chapter 4, Appendix D, and remedial actions are the owner's responsibility.
- (ii) The State has the basic responsibility for the protection of the life and property of its citizens. Once a dam has been determined to be unsafe, it is the State's responsibility to see that timely remedial actions are taken.
- (iii) The Corps of Engineers has the responsibility for executing the national program. The Federal program for inspection of dams does not modify the basic responsibilities of the States or dam owners. The Engineering Division of the Civil Works Directorate is responsible for overall program goals, guidance, technical criteria for inspections and inventory and headquarters level coordination with other agencies. The Water Resources Support Center (WRSC) located at Kingman Building, Fort Belvoir, Virginia 22060 is responsible for:
 - (A) Program Coordination of both the inventory and inspection programs.
 - (B) Developing and defining functional tasks to achieve program objectives.
 - (C) Determining resource requirements. (Budget)
 - (D) Compiling and disseminating progress reports.
 - (E) Monitoring and evaluating program progress and recommending corrective measures as needed.
 - (F) Collecting and evaluating data pertaining to inspection reports, dam owners' responses to inspection report recommendations, attitudes and capabilities of State officials, State dam safety legislation, Architect-Engineer performance, etc., for defining a comprehensive national dam safety program.
 - (G) Responding to Congressional, media, scientific and engineering organization and general public inquiries.

Division and District offices are responsible for executing the program at the State level. Assignment of Division responsibilities for States is shown in appendix A.

(2) **State participation.** Where State capability exists, every effort should be made to encourage the State to execute the inspection program either with State personnel or with Architect-Engineer (A-E) contracts under State supervision. If the State does not have the capability to carry out the inspection program, the program will be managed by the Corps of Engineers utilizing Corps employees or contracts with A-E firm.

(h) **Update of National Inventory of Dams.** (RCS-DAEN-CWE-17/OMB No. 49-R0421)

(1) The National Inventory of Dams should be updated and verified to include all Federal and non-Federal dams covered by the Act. Those dams are defined as all artificial barriers together with appurtenant works which impound or divert water and which: (1) Are twenty-five feet or more in height or (2) have an impounding capacity of fifty acre-feet or more. Barriers which are six feet or less in height, regardless of storage capacity or barriers which have a storage capacity at maximum water storage elevation of fifteen acre-feet or less regardless of height are not included.

(2) Inventory data for all dams shall be provided in accordance with appendix B.

(3) The hazard potential classification shall be in accordance with paragraph 2.1.2 *Hazard Potential* of the Recommended Guideline for Safety Inspection of Dams (Appendix D to this section).

TABLE 2—HAZARD POTENTIAL CLASSIFICATION

Category	Urban development	Economic loss
Low	No permanent structure for human habitation	Minimal (Undeveloped to occasional structures or agriculture).
Significant	No urban development and no more than a small number of habitable structures	Appreciable (Notable agriculture, industry or structures).
High	Urban development with more than a small number of habitable structures	Excessive (Extensive community, industry or agriculture).

(4) As in the original development of the inventory, the States should be encouraged to participate in the work of completing, verifying and updating the inventory. Also, when available, personnel of other appropriate Federal agencies should be utilized for the inventory work on a reimbursable basis. Work in any State may be accomplished:

(i) Under State supervision utilizing State personnel or Architect-Engineers contracts.

(ii) Under Corps supervision utilizing Corps employees, employees of other Federal agencies or Architect-Engineer contracts.

(5) A minimum staff should be assigned in Districts and Divisions to administer and monitor the inventory activities. Generally, the work should be accomplished by architect-engineers or other Federal agency personnel under State or Corps supervision. Corps personnel should participate in the inventory only to the extent needed to assure that accurate data are collected.

- (6) The National Inventory of Dams computerized data base is stored on the Boeing Computer Services (BCS) EKS computer system in Seattle, Washington. The data base uses Data Base Management System 2000 and is accessible for query by all Corps offices.
- (7) Appendix B indicates details on accessing and updating inventory data.
- (8) Appendix I describes the procedure for using NASA Land Satellite (LANDSAT) Multispectral Scanner data along with NASA's Surface Water Detection and Mapping (DAM) computer program to assist in updating and verifying and National Inventory of Dams.
- (9) All inventory data for dams will be completed and verified utilizing all available sources of information (including LANDSAT overlay maps) and will include site visitation if required. It is the responsibility of the District Engineer to insure that the inventory of each State within his area of responsibility is accurate and contains the information required by the General Instructions for completing the forms for each Federal and non-Federal dam.

(i) [Reserved]

(j) **Inspection Program.** (RCS-DAEN-CWE-17 and OMB No. 49-RO421)

- (1) **Scheduling of inspections.** The Governor of each State or his designee will continue to be involved in the selection and scheduling of the dams to be inspected. Priority will be given to inspection of those dams considered to offer the greatest potential threat to public safety.
 - (i) No inspection of a dam should be initiated until the hazard potential classification of the dam has been verified to the satisfaction of the Corps. Dams in the significant hazard category should be inspected only if requested by the State and only then if the State can provide information to show that the dam has deficiencies that pose an immediate danger to the public safety. Guidance for the selection of significant category non-Federal dams on Federal lands will be given in the near future.
 - (ii) Selection for inspection of non-Federal dams located on Federal lands or non-Federal dams designed and constructed under the jurisdiction of some Federal agency, should be coordinated with the responsible Federal agency. The appropriate State or regional representative of the Federal agency also should be contacted to obtain all available data on the dam. Representatives of the agency may participate in the inspection if they desire and should be given the opportunity to review and comment on the findings and recommendations in the inspection report prior to submission to the Governor and the dam owner. Examples of such dams are: non-Federal dams built on lands managed by National Forest Service, Bureau of Land Management, Fish and Wildlife Service, etc.; non-Federal dams designed and constructed by the Soil Conservation Service of the U.S. Department of Agriculture; high hazard mine tailings and coal mine waste dams under the jurisdiction of the Mine Safety and Health Administration, Department of Labor.
 - (iii) Indian-owned dams on trust lands are considered to be non-Federal dams. All dams in the high hazard potential category will be inspected. Privately-owned dams located on Indian lands are to be included in the program, however BIA-owned dams on Indian lands are Federal dams and are exempt.
- (2) **Procedures.** The Division Engineer is responsible for the quality of inspections and reports prepared by the District Engineer. Close liaison between the District Engineer and the State agency or A-E firm responsible for the inspections will be required in order to obtain a dependable result. To avoid

undesirable delays in the evaluation of safety of individual dams, contracts with A-E's or agreements with States which are managing the program will provide that reports be completed and furnished to the District Engineer within a specified time after completion of the on-site inspection of the dam.

- (i) **Inspection guidelines.** The inspection should be conducted in accordance with the Recommended Guidelines for Safety Inspection of Dams (Appendix D to this section). Expanded Guidance for Hydrologic and Hydraulic Assessment of Dams is provided in appendix C. The criteria in the recommended guidelines are screening criteria to be used only for initial determinations of the adequacy of the dam. Conditions found during the investigation which do not meet the guideline recommendations should be assessed as to their importance from the standpoint of the degree of risk involved.
- (ii) **Coordinators.** Experience has shown that coordination and communications among technical disciplines, Public Affairs Office, emergency officials, training officers, operations personnel, State representatives and A-E firms has been best in those districts where one person was delegated the responsibility for coordinating the actions of all involved elements. Each district should evaluate its overall coordination procedures to insure that all involved elements have the best possible access to necessary data.
- (iii) Field investigations should be carried out in a systematic manner. A detailed checklist or inspection form should be developed and used for each dam inspection and appended to the inspection report. The size of the field inspection team should be as small as practicable, generally consisting of only one representative of each required discipline in order to control the costs of the inspection without sacrificing the quality of the inspection. The inspection team for the smaller less complex dams should be limited to two or three representatives from appropriate technical areas with additional specialists used only as special conditions warrant. The larger more complex projects may require inspection teams of three or four specialists. Performance of overly detailed and precise surveys and mapping should be avoided. Necessary measurement of spillway, dam slopes, etc. can generally be made with measuring tapes and hand levels.
- (iv) **Additional engineering studies.** Dam inspections should be limited to Phase I investigations as outlined in Chapter 3 of appendix D. However, if recommended by the investigating engineer and approved by the District Engineer, some additional inexpensive investigations may be performed when a reasonable judgment on the safety of the dam cannot be made without additional investigation. Any further Phase II investigation needed to prove or disprove the findings of the District Engineer or to devise remedial measures to correct deficiencies are the responsibility of the owner and will not be undertaken by the Corps of Engineers.
- (v) **Assessment of the investigation.**
 - (A) The findings of the visual inspection and review of existing engineering data for a dam shall be assessed to determine its general condition. Dams assessed to be in generally good condition should be so described in the inspection report. Deficiencies found in a dam should be described and assessed as to the degree of risk they present. The degree of risk should consider only loss of life and/or property damage resulting from flooding due to dam failure. Loss of project benefits *i.e.*, municipal water supply, etc., should not be considered. If deficiencies are assessed to be of such a nature that, if not corrected, they could result in the failure of the dam with subsequent loss of life and/or substantial property damage, the dam should be assessed as "Unsafe." If the probable failure of an "Unsafe" dam is judged to be imminent and immediate action is required to reduce or

eliminate the hazard, the “unsafe” condition of the dam should be considered an “emergency.” If the probable failure is judged not to be imminent, the “unsafe” condition should be considered a “non-emergency.”

(B) Adequacy of spillway. The “Recommended Guidelines for Safety Inspection of Dams,” appendix D, provide current, acceptable inspection standards for spillway capacity. Any spillway capacity that does not meet the criteria in the “Guidelines” is considered inadequate. When a spillway's capacity is so deficient that it is seriously inadequate, the project must be considered *unsafe*. If all of the following conditions prevail, the Governor of the State shall be informed that such project is unsafe:

- (1) There is high hazard to loss of life from large flows downstream of the dam.
- (2) Dam failure resulting from overtopping would significantly increase the hazard to loss of life downstream from the dam over that which would exist just before overtopping failure.
- (3) The spillway is not capable of passing one-half of the probable maximum flood without overtopping the dam and causing failure.

Classification of dams with seriously inadequate spillways as “unsafe, non-emergency” is generally a proper designation of the urgency of the unsafe condition. However, there may be cases where the spillway capacity is unusually small and the consequences of dam overtopping and failure would be catastrophic. In such cases, the unsafe dam should be classified as an emergency situation.

(vi) All inspection reports will receive one level of independent review by the Corps. If the reports are prepared by the Corps, the independent review may be performed internally within the district office. However, in cases which involve significant economic, social or political impacts and technical uncertainties in evaluating the dams, advice may be obtained from the staffs of the Division Engineer and the Office, Chief of Engineers.

(3) **Reports —**

- (i) **Preparation.** A written report on the condition of each dam should be prepared as soon as possible after the completion of the field inspection and assessment. A suggested report format is attached as appendix E. It is important that the inspection report be completed in a timely manner. For inspections being done by Corps employees, it is suggested that once an inspection team has been assigned to a dam inspection it be allowed to complete the inspection and report without interruption by other work.
- (ii) **Review and approval.** The coordinating engineer should determine which disciplines should review the report and establish a procedure to accomplish the review in a timely manner. A review panel, made up of the appropriate Division and Branch Chiefs has worked well in some districts. Use of a review panel should be seriously considered by all districts. All inspection reports shall be approved by the District Engineer who will maintain a complete file of final approved reports. Any State or Federal agency having jurisdiction over the dam or the land on which the dam is built should be given the opportunity to review and comment on the report prior to submission to the Governor or dam owner. The District Engineer will transmit final approved reports to the Governor of the State and the dam owner (or the Governor only, when requested in writing by State officials). If the report is initially furnished to the Governor only, a period of up to ten days may be allowed before the report is furnished to the dam owner. If the

Governor or the owner indicates additional technical information is available that might affect the assessment of the dam's condition, the District Engineer will furnish the proposed final report to the Governor and the owner and establish a definite time period for comments to be furnished to the District Engineer prior to report approval.

- (iii) In general the Governor will be responsible for public release of an inspection report and for initiating any public Statements. However, an approved report must be treated as any other document subject to release upon request under the Freedom of Information Act. The letters of transmittal to the Governor and owner should indicate that under the provisions of the Freedom of Information Act, the documents will be subject to release upon request after receipt by the Governor. Proposed final reports will be considered as internal working papers not subject to release under the Freedom of Information Act. Corps personnel, A-E contractor personnel and others working under supervision of the Corps will be cautioned to avoid public statements about the condition of the dam until after the District Engineer has approved the report. The Corps will respond fully to inquiries after the Governor has received the approved report or been notified of an unsafe dam. An information copy of the report should be sent to the District office normally having jurisdiction if other than the District responsible for the inspection.
- (iv) **Follow-up action.** A Federal investment of the magnitude anticipated for this inspection program makes it desirable that a reporting system be established to keep the District Engineer abreast of the implementation of the recommendations in the inspection reports. The letters of transmittal to the Governor and owner will request that the District Engineer be informed of the actions taken on the recommendations in the inspection reports. However, the National Dam Inspection Act only authorizes the initial inspection of certain dams; therefore, once a report is completed no reinspection will be undertaken.
- (4) **Unsafe dams.** The investigating engineer will be required to immediately notify the District Engineer when a dam is assessed as being unsafe. He will also indicate if probable failure of the unsafe dam is judged to be imminent and immediate action is required to reduce or eliminate the threat. The District Engineer will evaluate the findings of the investigating team and will immediately notify the Governor and the owner if the findings are Unsafe Non-Emergency or Unsafe-Emergency. The appropriate State agency and the Corps of Engineers officials having emergency operation responsibility for the area in which the dam is located will also be notified. The information provided in the unsafe dam notice shall be as indicated in Appendix F. Any emergency procedures or remedial actions deemed necessary by the District Engineer will be recommended to the Governor who has the responsibility for any corrective actions. As provided in ER 500-1-1, Corps assistance under Pub. L. 84-99 "Advance Measures," may be made available to complement the owner's and Governor's action under certain conditions and subject to the approval of the Director of Civil Works. The District Engineer's Emergency Operation Officer will coordinate the advance measures request in accordance with existing procedures. Coordination will be maintained between the District responsible for emergency action under Pub. L. 84-90 and the District responsible for the inspection.
- (5) **Emergency action plans.** An emergency action plan should be available for every dam in the high and significant hazard category. Such plans should outline actions to be taken by the operator to minimize downstream effects of an emergency and should include an effective warning system. If an emergency action plan has not been developed, the inspection report should recommend that the owner develop such an action plan. However, the Corps has no authority to require an emergency action plan.

- (k) **Progress reports.** Progress reports should be submitted monthly by the Division Engineer to WRSC. The reports shall include progress through the last Saturday of the month and should be mailed by the following Monday. The reports shall contain the information and be typewritten in the format shown in appendix G. Copies of Unsafe Dam Data Sheets will be submitted with the progress report. Copies of the completed inspection report for Dams in the Unsafe-Emergency category will be submitted also. (RCS-DAEN-CWE-19)
- (l) **Contracts –**
- (1) **Corps of Engineers supervision.** Contracts for performing inventory and inspection activities under supervision of the Corps of Engineers shall be Fixed-Price Architect Engineer Contracts for Services. A sample scope of work setting forth requirements is provided in appendix H. Experience has shown that costs for individual dam inspection have been lower when multiple inspections are included in one contract. Therefore, each A-E contract should include multiple dam inspections where practicable. Corps participation in A-E inspections should be held to a minimum. Corps representatives should participate in only enough A-E inspections to assure the equality of the inspections.
- (2) **State supervision.** Contracts with States for performing inventory and inspection activities under State supervision may be either a Cost-Reimbursement type A-E Contract for Services or a Fixed-Price type contract. The selection of Architect-Engineers by the State should require approval of the Corps of Engineers Contracting Officer. The negotiated price for A-E services under cost-reimbursement type contracts with States will also require approval by the Contracting Officer. Contracts with States should require *timely* submission of the inspection reports to the District Engineer for review and approval. The contract provisions should also prevent public release of or public comment on the inspection report until the District Engineer has reviewed and approved the report. Corps of Engineers participation in State inspections should be limited to occasional selected inspections to assure the quality of the State program.
- (m) **Training.** As indicated in paragraph (f) of this section, one objective of the inspection program for non-Federal Dams is to prepare the States to provide effective dam safety programs. In many States this will require training of personnel of State agencies in the technical aspects of dam inspections. The Office, Chief of Engineers is studying the need for and content of a comprehensive Corps-sponsored training program in dam inspection technology. Pending the possible adoption of such a comprehensive plan, division and district Engineers are encouraged to take advantage of suitable opportunities to provide needed training in dam safety activities to qualified employees of State agencies and, when appropriate, to employees of architect-engineer firms engaged in the program. The following general considerations should be observed in providing such training:
- (1) Priority must be placed on inspection of dams and updating the national dam inventory; hence, diversion of resources to training activities should not deter or delay these principle program functions.
- (2) Salaries, per diem and travel expenses relating to training activities of State employees will be a State expense. There will be no tuition charge for State employees.
- (3) Architect-Engineer firms will be required to pay expenses and tuition costs for their employees participating in Corps-sponsored training activities.
- (4) Corps-sponsored training will require that each trainee is a qualified engineer or geologist and will concentrate on engineering technology related directly to dam safety. (This may require screening of proposed candidates for training.)

- (5) Under this program, the Corps will not sponsor training that is intended primarily to satisfy requirements for a degree.
- (6) Training by participation in actual dam inspections and/or management of the inspection program should be encouraged.

Appendix A to § 222.6—Division Assignments

To facilitate better coordination with the States, the Division Engineers are responsible for the dam inspection program by States as follows:

New England Division: Maine, Rhode Island, Connecticut, Vermont, New Hampshire, Massachusetts

North Atlantic Division: New York, New Jersey, Pennsylvania, Delaware, Maryland, Virginia, District of Columbia

Ohio River Division: West Virginia, Ohio, Kentucky, Tennessee, Indiana

South Atlantic Division: North Carolina, South Carolina, Georgia, Florida, Alabama, Puerto Rico, Virgin Islands

Lower Mississippi Valley Division: Mississippi, Louisiana, Missouri

North Central Division: Michigan, Wisconsin, Illinois, Minnesota, Iowa

Southwestern Division: Arkansas, Oklahoma, Texas, New Mexico

Missouri River Division: Kansas, Nebraska, South Dakota, North Dakota, Wyoming, Colorado

North Pacific Division: Oregon, Idaho, Montana, Washington, Alaska

South Pacific Division: Utah, California, Arizona, Nevada

Pacific Ocean Division: Hawaii, Trust Territories, American Samoa

Appendix B to § 222.6—Inventory of Dams

(RCS-DAEN-CWE-17 AND OMB No. 49-RO421)

1. The updating of the inventory will include the completion of all items of data for all dams now included in the inventory, verification of the data now included in the inventory, and inclusion of complete data for all appropriate existing dams not previously listed. Data completion, verification and updating will be scheduled over a three year period.

2. The inventory data will be recorded on Engineering Form 4474 and 4474A (Exhibit 2). The general instructions for completing the forms are printed on the back of the forms. Parts I and II of the forms are to be fully completed. The instruction for completing Item 29, Line 5, Para. II (Engr Form 4474A) is revised to conform identically with the hazard potential classification contained in the recommended guidelines for safety inspection of dams. Additional data has been added to designate Corps districts in which the dam is located,

Federal agency owned dams, Corps owned dams, Federal agency regulated dams, dams constructed with technical or financial assistance of the U.S. Soil Conservation Service, and privately owned dams located on Federal property.

3. All inventory data will be verified utilizing all available sources of information and will include site visitation if required.

4. The Inventory Data Base is stored on the Boeing Computer Services (BCS) EKS System in Seattle, Washington. The data is available to all Corps offices for queries using Data Base Management System 2000 (S2K).

a. To access the National Data Base log on BCS and type the following:

GET,DAMS/UN = CECELB

CALL,DAMS

b. For current information and changes to the National Inventory Data Base, type:

OLD,HOTDAM/UN = CEC1AT

LIST

5. The inventory update data will be furnished and the National Data Base will be updated on a monthly basis. The monthly submission will cover all dams whose inventory data were completed since the last report. The update data will be loaded directly onto the Boeing Computer by the field office.

a. The procedure for loading the data on the Boeing Computer can be printed by accessing the Boeing Computer and listing the information file "HOTDAM." (See paragraph 4b. above.)

b. It is the responsibility of the submitting office to edit the data prior to furnishing it for the update. Editing will be accomplished by processing the data using the Inventory Edit Computer program developed by the Kansas City District. This procedure is described in the "HOTDAM" file.

6. Federal agencies will be uniformly designated by major and minor abbreviations according to the following list whenever applicable to Items 46 through 53. Abbreviations are to be left justified within the field with one blank separating major and minor abbreviations.

	Major	Minor
a. International Boundary and Water Commission	IBWC	
b. U.S. Department of Agriculture:		
(1) Soil Conservation Service	USDA	SCS
(2) Forest Service	USDA	FS

	Major	Minor
c. U.S. Department of Energy Federal Energy Regulatory Commission	DOE	FERC
d. Tennessee Valley Authority	TVA	
e. U.S. Department of Interior:		
(1) Bureau of Sport Fisheries and Wildlife	DOI	BSFW
(2) Geological Survey	DOI	GS
(3) Bureau of Land Management	DOI	BLM
(4) Bureau of Reclamation	DOI	USBR
(5) Bureau of Indian Affairs	DOI	BIA
f. U.S. Department of Labor: (1) Mine Safety and Health Administration	DOL	MSHA
g. Corps of Engineers:		
(1) Lower Mississippi Valley Division:		
(a) Memphis District	DAEN	LMM
(b) New Orleans District	DAEN	LMN
(c) St. Louis District	DAEN	LMS
(d) Vicksburg District	DAEN	LMK
(2) Missouri River Division:		
(a) Kansas City District	DAEN	MRK
(b) Omaha District	DAEN	MRO
(3) New England Division	DAEN	NED
(4) North Atlantic Division:		
(a) Baltimore District	DAEN	NAB
(b) New York District	DAEN	NAN
(c) Norfolk District	DAEN	NAO
(d) Philadelphia District	DAEN	NAP
(5) North Central Division:		
(a) Buffalo District	DAEN	NCB
(b) Chicago District	DAEN	NCC
(c) Detroit District	DAEN	NCE
(d) Rock Island District	DAEN	NCR
(e) St. Paul District	DAEN	NCS
(6) North Pacific Division:		
(a) Alaska District	DAEN	NPA
(b) Portland District	DAEN	NPP
(c) Seattle District	DAEN	NPS
(d) Walla Walla District	DAEN	NPW
(7) Ohio River Division:		
(a) Huntington District	DAEN	ORH
(b) Louisville District	DAEN	ORL
(c) Nashville District	DAEN	ORN

	Major	Minor
(d) Pittsburgh District	DAEN	ORP
(8) Pacific Ocean Division	DAEN	POD
(9) South Atlantic Division:		
(a) Charleston District	DAEN	SAC
(b) Jacksonville District	DAEN	SAJ
(c) Mobile District	DAEN	SAM
(d) Savannah District	DAEN	SAS
(e) Wilmington District	DAEN	SAW
(10) South Pacific Division:		
(a) Los Angeles District	DAEN	SPL
(b) Sacramento District	DAEN	SPK
(c) San Francisco District	DAEN	SPN
(11) Southwestern Division:		
(a) Albuquerque District	DAEN	SWA
(b) Fort Worth District	DAEN	SWF
(c) Galveston District	DAEN	SWG
(d) Little Rock District	DAEN	SWL
(e) Tulsa District	DAEN	SWT

7. Procedures for Revising and Updating the Inventory of Dams Master File.

a. To Change Correct or Add an Item. Submit a change card that contains the identification assigned to the dams (Columns 1 thru 7), the proper card code (Column 80) and only the item or items changed, corrected or added. Data on the master file is added or replaced on an item for item basis.

b. To Delete an Item. Submit a change card that contains the identification assigned to the dam, (Columns 1 thru 7), the proper card code (Column 80), and an asterisk (*) in the left most column of the item or items to be deleted. More than one item can be changed, corrected, added on or deleted from the same card.

c. To Delete the Entire Data for a Dam from the Master File. Submit a zero (0) card punched as follows:

Columns 1 thru 7—Item 1 identification assigned to the dam

Columns 8 thru 10—Item 2, Division Code

Columns 11 thru 16—The word DELETE

Columns 17 thru 79—Blank Spaces

Column 80—A zero

8. *Keypunch Instructions and Punched Card Formats.*

- a. Table 1 describes the character set to be used for keypunch cards of Engr. Forms 4474 and 4474A.
- b. Exhibit 1 is the EDPC keypunch instructions and punch card formats defining the data fields (Items) and card columns to be used in preparing punched cards in compliance with the requirements of this regulation.
- c. Exhibit 2 are prints of Engr. Forms 4474 and 4474A which are laid out in punch card format to facilitate punching cards directly from the completed forms.

Table 1

STANDARD CHARACTER SET AND CARD CODES

A	12-1		0	0
B	12-2		1	1
C	12-3		2	2
D	12-4		3	3
E	12-5		4	4
F	12-6		5	5
G	12-7		6	6
H	12-8		7	7
I	12-9		8	8
J	11-1		9	9
K	11-2	space	blank	
L	11-3		,	0-3-8
M	11-4		.	12-3-8
N	11-5		-	11
O	11-6		*	11-4-8
P	11-7		/	0-1
Q	11-8		\$	11-3-8
R	11-9			
S	0-2			
T	0-3			
U	0-4			
V	0-5			
W	0-6			
X	0-7			
Y	0-8			
Z	0-9			

NON-STANDARD CHARACTER SET

(	12-5-8	0-8-4
)	11-5-8	12-8-4
"	8-4	
'	11-8-5	
+	12	
%	8-6	
;	12-8-7	11-8-6
:	8-2	
#	0-8-6	
@	8-5	
=	8-3	8-6

EDPC KEYPUNCH INSTRUCTIONS (Continued)							ER 1110-2-106 26 Sept 79
JOB TITLE INVENTORY OF UNITED STATES DAMS						JOB NO.	
CARD IDENTIFICATION CARDS 0&1			SOURCE ENG FORM 4474				
SOURCE BLOCK	NAME OF FIELD	COLUMNS		NO. COLS.	TYPE DATA *	JUST- IFY **	REMARKS-INSTRUCTIONS
		FROM	TO				
1	<u>Card Number 0</u> Identity (State)	1	2	2	A	L	
1	Identity (Number)	3	7	5	N	R	
2	Div	8	10	3	A	L	
3	State	11	12	2	A	L	
4	County	13	15	3	N	R	
5	Congr Dist	16	17	2	N	R	
6	State	18	19	2	A	L	
7	County	20	22	3	N	R	
8	Congr. Dist	23	24	2	N	R	
9	Name	25	61	39	A	L	
10	Latitude	62	66	5	N	R	No decimal point is punched.
11	Longitude	67	72	6	N	R	No decimal point is punched.
12	Report Data (Day)	73	74	2	N	R	
12	Report Data (Mo)	75	77	3	A	L	
12	Report Data (Yr)	78	79	2	N	R	
	Card Number	80	80	1	N		Punch a 0
1	<u>CARD NUMBER 1</u> Identity	1	7				Repeat Item 1 card 0
13	Popular Name	8	43	36	A	L	
14	Name of Impoundment	44	79	36	A	L	
	Card Number	80	80	1	N		Punch a 1

26 Sept 79

JOB TITLE	JOB NO.
INVENTORY OF UNITED STATES DAMS	

CARD IDENTIFICATION CARD 2	SOURCE ENG FORM 4474
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page 73 of 142

EDPC KEYPUNCH INSTRUCTIONS (Continued)							ER 1110-2-106 26 Sept 79
JOB TITLE INVENTORY OF UNITED STATES DAMS						JOB NO.	
CARD IDENTIFICATION CARD 3 & 4			SOURCE ENG FORM 4474A				
SOURCE BLOCK	NAME OF FIELD	COLUMNS		NO. COLS.	TYPE DATA	JUST- IFY	REMARKS-INSTRUCTIONS
		FROM	TO				
1	<u>CARD NUMBER 3</u> Identity	1	7				Repeat Item 1 card 0
21	Type of Dam	8	19	14	A	L	
22	Year Completed	20	23	4	N	R	
23	Purposes	24	33	10	A	L	
24	Structural Height	34	37	4	N	R	
25	Hydraulic Height	38	41	4	N	R	
26	Impounding Maximum	42	49	8	N	R	
27	Impounding Normal	50	57	8	N	R	
27A	C. E. District	58	60	3	A	L	
27B	Ownership	61	61	1	A		
27C	Fed. Regulated	62	62	1	A		
27D	Prv't Dams/Fed Land	63	63	1	A		
27E	Soil Con. Ser. Ass't.	64	64	1	A		
27F	Verif. Date (Day)	65	66	2	N	R	
27F	Verif. Date (Mo)	67	69	3	A	L	
27F	Verif Date (Yr)	70	71	2	N	R	
	Card Number	80	80	1	N		Punch a 3
1	<u>CARD NUMBER 4</u> Identity	1	7				Repeat Item 1 card 0
28	Remarks	8	79	72	A	L	
	Card Number	80	80	1	N		Punch a 4

ER 1110-2-106
26 Sept 79

EDPC KEYPUNCH INSTRUCTIONS (Continued)							
JOB TITLE INVENTORY OF UNITED STATES DAMS						JOB NO.	
CARD IDENTIFICATION CARD 5			SOURCE ENG FORM 4474A				
SOURCE BLOCK	NAME OF FIELD	COLUMNS		NO. COLS.	TYPE DATA	JUS- TIFY	REMARKS-INSTRUCTIONS
		FROM	TO				
1	CARD NUMBER 5 Identity	1	7				Punch from Part II Item 1
29	D/S Has	8	8	1	N		
30	Crest Length	9	13	5	N	R	
31	Spillway Type	14	14	1	A		
32	Spillway Width	15	18	4	N	R	
33	Maximum Discharge	19	25	7	N	R	
34	Volume of Dam	26	34	9	N	R	
35	Power-Installed	35	40	6	N	R	
36	Power-Proposed	41	46	6	N	R	
37	N.L. Number	47	47	1	N		
38	N.L. Length	48	51	4	N	R	
39	N.L. Width	52	54	3	N	R	
40	N.L. Length	55	58	4	N	R	
41	N.L. Width	59	61	3	N	R	
42	N.L. Length	62	65	4	N	R	
43	N.L. Width	66	68	3	N	R	
44	N.L. Length	69	72	4	N	R	
45	N.L. Width	73	75	3	N	R	
	Card Number	80	80	1	N		Punch a 5

EDPC KEYPUNCH INSTRUCTIONS (Continued)							ER 1110-2-106 26 Sept 79
JOB TITLE INVENTORY OF UNITED STATES DAMS						JOB NO.	
CARD IDENTIFICATION CARD 6,7,8, & 9			SOURCE ENG FORM 4474A				
SOURCE BLOCK	NAME OF FIELD	COLUMNS		NO. COLS	TYPE DATA	SUB- TYPE	REMARKS-INSTRUCTIONS
		FROM	TO				
1	<u>CARD NUMBER 6</u> Identity	1	7				Repeat Item 1 card 5
46	Owner	8	31	24	A	L	
47	Engineering By	32	55	24	A	L	
48	Construction By	56	79	24	A	L	
	Card Number	80	80	1	N		Punch a 6
1	<u>CARD NUMBER 7</u> Identity	1	7				Repeat Item 1 card 5
49	Design	8	25	18	A	L	
50	Construction	26	43	18	A	L	
51	Operation	44	61	18	A	L	
52	Maintenance	62	79	18	A	L	
	Card Number	80	80	1	N		Punch a 7
1	<u>CARD NUMBER 8</u> Identity	1	7				Repeat Item 1 card 5
53	Inspection By	8	40	33	A	L	
54	Inspection (Day)	41	42	2	N	R	
54	Inspection (Mo)	43	45	3	N	R	
54	Inspection (Yr)	46	47	2	N	R	
55	Authority	48	79	32	A	L	
	Card Number	80	80	1	N		Punch a 8
1	<u>CARD NUMBER 9</u> Identity	1	7				Repeat Item 1 card 5
56	Remarks	8	79	72	A	L	
	CARD NUMBER	80	80	1	N		Punch a 9

PART I - INVENTORY OF DAMS IN THE UNITED STATES (PURSUANT TO PUBLIC LAW 92-467)										FIRM APPROVED OMB NO. 49-R0421 REQUIREMENTS CONTROL SYMBOL DAEN-CWE-17		FIRM NUMBER	
										STATE		FIRM NUMBER	
										11		12	
										13		14	
										15		16	
										17		18	
										19		20	
										21		22	
										23		24	
										25		26	
										27		28	
										29		30	
										31		32	
										33		34	
										35		36	
										37		38	
										39		40	
										41		42	
										43		44	
										45		46	
										47		48	
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										67		68	
										69		70	
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										75		76	
										77		78	
										79		80	
										81		82	
										83		84	
										85		86	
										87		88	
										89		90	
										91		92	
										93		94	
										95		96	
										97		98	
										99		100	

GENERAL INSTRUCTIONS

This form is for use in preparing the inventory of dams in the United States under the requirements of the National Program for the Inspection of Dams, P.L. 92-367. All items of Part I and Part II (Lines 0-9) must be completed as instructed below. Print entries distinctly in ink or pencil. For letters o, z, and i, write Ø, Z, and I.

Write only one letter or numeral in each space; do not use more letters than blocks allowed for an item. Do not abbreviate on Part I. Leave one space between words and no space between code letters.

For all letter codes or word entries place first letters in left block of field. In word fields any alphabetic, numeric or special character may be entered. For all numerical entries, use only numerals placing the last digit of number in the right block of field, including trailing zeros. Do not include a decimal point! In fields where decimals are required values are to be placed around the decimal point printed on the form.

Leave blank those spaces where item does not apply, e.g., do not write "N/A", "--", "None", etc., unless instructed to do so by specific instructions. Use the remarks line when additional space is needed for an item, or to clarify an entry. Preface each remark with the item number. (See Item #28 or #56 instructions)

PART I

Item # 1 IDENTITY: The Division Engineer will assign and control the identity for dams in the states for which he is responsible. The first two characters of the identity will be the two-letter state abbreviation in accordance with Federal Information Processing Standards Publication, June 15, 1970 (FIPS PUB 6-1). In cases where a dam is physically located in two or more states, one state will be designated as the principal state for the identity. The last five (5) characters of the identity will be a sequential number assigned to identify dams within a state.

LINE 0:

Item # 2 DIVISION: Enter the three (3) letter office symbol for the division making the report in accordance with ABBR Report Code, Appendix B, ER 18-2-1, Civil Works Information System; e.g., NAD, ORD, SWD, etc.

Location:

Item # 3 STATE: Enter two (2) letter principal state abbreviation in accordance with FIPS PUB 6-1.

Item # 4 COUNTY: Enter three (3) digit county identification in accordance with FIPS PUB 6-1.

Item # 5 CONG DIST: Enter one (1) or two (2) digit number for congressional districts in which dam is located.

Item # 6, 7, and 8 (Use second location for structures situated in more than one state.)

Item # 9 DAM NAME: Enter official name of dam. Do not abbreviate unless the abbreviation is a part of the official name. For dams that do not have a name, create a name by combining the two (2) letter state abbreviation plus "NO NAME" plus a sequential number. Example: if two dams in the State of Alabama do not have names, they would be named as ALNONAME1 and ALNONAME2.

Item #10 & #11 LATITUDE AND LONGITUDE: Enter the latitude and longitude in degrees, minutes and tenths of a minute. All geographical location items pertain to dam as its maximum section.

Item #12 REPORT DATE: Enter the one (1) or two (2) digits for day, the first three (3) letters of the month and a two (2) digit year (e.g., 12 JAN74) in which the data has been revised, updated or otherwise changed.

LINE 1:

Item #13 POPULAR NAME OF DAM: If (other than the official name of the dam) in common use, enter the name in this space. Leave blank if not applicable.

Item #14 NAME OF IMPOUNDMENT: Enter official name of lake or reservoir. Leave blank if reservoir does not have a name.

LINE 2

Item #15 & #16 **REGION AND BASIN**: Enter two (2) digit numbers for Region and Basin in accordance with Appendix C, ER 18-2-1, Civil Works Information System.

Item #17 **RIVER OR STREAM**: Enter official name of river or stream on which the dam is built. If stream is without name indicate as tributary to river named, e.g., TR-COLORADO. If off stream, enter name of river plus "OFF-STREAM".

Item #18 **NEAREST DOWNSTREAM CITY-TOWN-VILLAGE**: Enter the nearest downstream city-town-village of such size which can be located on a general map.

Item #19 **DISTANCE FROM DAM**: Enter distance from dam to nearest downstream city-town-village to the nearest mile.

Item #20 **POPULATION**: Enter population of city-town-village given in Item #18.

LINE 3

Item #21 **TYPE OF DAM**: Enter two (2) letter codes, in any order, to describe type of dam.

EARTH	RE	BUTTRESS	CB	OTHER - OT
ROCKILL	ER	ARCH	VA	(Describe "other" in remarks)
GRAVITY	PG	MULTI-ARCH	MV	

Item #22 **YEAR COMPLETED**: Enter year when the main dam structure was completed and ready for use. If only approximate year can be determined, note this in remarks.

Item #23 **PURPOSES**: Enter one (1) letter codes that describe the purposes for which the reservoir is used. The order entered should indicate the relative decreasing importance of the project purposes.

IRRIGATION	- I	WATER SUPPLY	- S	DEBRIS CONTROL	- D
HYDROELECTRIC	- H	RECREATION	- R	OTHER	- O
FLOOD CONTROL	- C	STOCK OR SMALL		(Describe "other" in remarks)	
NAVIGATION	- N	FARM POND	- P		

Item #24 **STRUCTURAL HEIGHT**: Enter, to the nearest foot, the structural height of the dam which is defined as: the overall vertical distance from the lowest point of foundation surface to the top of the dam.

Item #25 **HYDRAULIC HEIGHT**: Enter, to the nearest foot, the hydraulic height of the dam which is defined as: the effective height of the dam with respect to the maximum storage capacity, measured from the natural bed of the stream or watercourse at the downstream toe of the barrier, or if it is not across a stream or watercourse, the height from the lowest elevation of the outside limit of the barrier to the maximum storage elevation.

Impounding Capabilities:

Item #26 **MAXIMUM**: Enter the acre feet for maximum storage which is defined as: the total storage space in a reservoir below the maximum attainable water surface elevation, including any surcharge storage.

Item #27 **NORMAL**: Enter the acre feet for normal storage which is defined as: the total storage space in a reservoir below the normal retention level, including dead and inactive storage and excluding any flood control or surcharge storage.

Item #27A **CORPS OF ENGINEERS DISTRICT**: Enter the three character Corps of Engineers ABBR report code in which the dam is geographically located, in accordance with Appendix B, ER 19-2-1, Civil Works Information System, e.g., NAN, ORH, SWF, etc.

Item #27B **OWNERSHIP**: Enter N, for Non-Federal; G, for Federal Gov't. Agencies other than the Corps of Engineers; C for Corps of Engineers.

Item #27C **FEDERALLY REGULATED**: Enter N for No; Enter Y for Yes.

Item #27D **PRIVATE DAMS ON FEDERAL LAND**: Enter N for No; Enter Y for Yes.

Item #27E **ASSISTANCE BY SOIL CONSERVATION SERVICE**: Enter N for None; T for Technical Assistance; F for Financial Assistance; B for Both Technical and Financial Assistance.

Item #27F **VERIFICATION**: Date the data was verified as being complete and correct. Enter date as described in Item #12.

LINE 4

Item #28 **REMARKS**: Preface remarks with the item number to which it pertains, e.g., 22-ORIGINALLY CONSTRUCTED IN 1928. 23-SETTLING BASIN. Only one remark line should be used for PART I remarks.

page 80 of 142

PART II:

Item 1 **IDENTITY:** Enter Identity per **GENERAL INSTRUCTIONS** on PART I.

LINE 5:

Item 129 **D/S HAZ:** Enter the digit that most closely represents the hazard potential that could occur to the downstream (D/S) area resulting from failure or mis-operation of the dam or facilities.

HAZARD POTENTIAL

CATEGORY	LOSS OF LIFE (Extent of Development)	ECONOMIC LOSS (Extent of Development)
3 = Low	None expected (No permanent structures for human habitation)	Minimal (Undeveloped to occasional structures or agriculture)
2 = Significant	Few (No urban developments and no more than a small number of inhabitable structures)	Appreciable (Notable agriculture, industry or structures)
1 = High	More than few	Excessive (Extensive community, industry or agriculture)

Item 130 **CREST LENGTH:** Enter, to the nearest foot, the crest length of the dam which is defined as: the total horizontal distance measured along the axis at the elevation of the top of dam between abutments or ends of dam. Note that this includes spillway width, powerhouse sections, and navigation locks where they form a continuous part of the dam water retaining structure. Detached spillways, locks, and powerhouses shall not be included.

Spillway:

Item 131 **TYPE:** Enter the one letter code that applies.

CONTROLLED = C

UNCONTROLLED = U

NONE = N

Item 132 **WIDTH:** Enter to the nearest foot, the width of the spillway available for discharge when the reservoir is at its maximum designed water surface elevation.

Item 133 **MAXIMUM DISCHARGE:** Enter the number of cubic feet per second which the spillway is capable of discharging when the reservoir is at its maximum designed water surface elevation.

Volume of Dam:

Item 134 **VOLUME OF DAM:** Enter the total number of cubic yards occupied by the materials used in the dam structure. If volume of separate materials is known, enter in remarks. Include portions of powerhouses, locks and spillways only if integral with the dam and required for structural stability.

Power Capacity:

Item 135 **INSTALLED:** Enter installed capacity to one tenth (1/10) Megawatt as of the report date.

Item 136 **PROPOSED:** Enter the future additional capacity proposed to one tenth (1/10) Megawatt.

Navigation Locks:

Item 137 NUMBER: Enter the number of existing navigation locks for the project.

Item 138 LENGTH: Enter to the nearest foot the length of the navigation lock.

Item 139 WIDTH: Enter to the nearest foot the width of the navigation lock.

Item 140 thru 145: Enter the lengths and widths of additional locks.

LINE 6:

Item 146 OWNER: Enter name of owner. Abbreviate as necessary.

Item 147 ENGINEERING BY: Enter name of organization that engineered the main dam structure. Abbreviate as required.

Item 148 CONSTRUCTION BY: Enter name of construction agency responsible for construction of main structure. Abbreviate as required.

LINE 7:

Regulatory Agency:

Item 149 DESIGN: Enter the name of the organization other than the owner having regulatory or approval authority over the design of the dam. If no organization other than the owner has regulatory or approval authority over the design of the dam indicate NONE.

Item 150 CONSTRUCTION: Enter the name of the organization other than the owner having regulatory authority or inspection responsibilities over the construction of the dam. If no organization other than the owner has regulatory authority or inspection responsibilities over the construction of the dam indicate NONE.

Item 151 OPERATION: Enter the name of the organization other than the owner having regulatory authority, operational control, or surveillance responsibilities over the operation of the dam. If no organization other than the owner has regulatory authority, operational control or surveillance responsibilities over the operation of the dam indicate NONE.

Item 152 MAINTENANCE: Enter the name of the organization other than the owner having regulatory authority or inspection or surveillance responsibilities over the maintenance of the dam. If no organization other than the owner has regulatory authority or inspection or surveillance responsibilities over the maintenance of the dam indicate NONE.

LINE 8:

Inspection:

Item 153 BY: Enter the name of the organization that performed the last safety inspection. Abbreviate as required. If no inspection has been performed enter NONE.

Item 154 DATE: Enter the one (1) or two (2) digits for day, the first three (3) letters of the month and a two (2) digit year when the inspection was performed. If not applicable, leave blank.

Item 155 AUTHORITY FOR INSPECTION: Enter the legislative or regulatory authority for performing the inspection indicated in item 53, e.g., P.L. 92-367; Div 3, Water Code, State of Calif; F.R. 1110-2-100; etc.

LINE 9:

Item 156 REMARKS: Preface remarks with the item number to which it pertains. e.g., 34.2, 500,000 c.y. conc. 475,000 c.y. earthfill. Only one Remarks line should be used for PART II remarks.

EXHIBIT 2

Appendix C to § 222.6—Hydrologic and Hydraulic Assessment of Dams

1. Phase I inspections are not intended to provide detailed hydrologic and hydraulic analyses of dam and reservoir capabilities. However, when such analyses are available, they should be evaluated for reliability and completeness. If a project's ability to pass the appropriate flood (see Table 3, page D-12 of Recommended Guidelines) can be determined from available information of a brief study, such an assessment should be made. It should be noted that hydrologic and hydraulic analyses connected with the Phase I inspections should be based on approximate methods or systematized computer programs that take minimal effort. The Hydrologic Engineering Center (HEC) has developed a special computer program for hydrologic and hydraulic analyses to be used with the Phase I inspection program. Other Field Operating Agencies have developed similar computer programs or generalized procedures which are acceptable for use. All such efforts should be completed with minimum resources.

2. A finding that a dam will not safely pass the flood indicated in the Recommended Guidelines does not necessarily indicate that the dam should be classified as unsafe. The degree of inadequacy of the spillway to pass the appropriate flood and the probable adverse impacts of dam failure because of overtopping must be considered in making such classification. The following criteria have been selected which indicate when spillway capacity is so seriously inadequate that a project must be classified as unsafe. All of the following conditions must prevail before designating a dam unsafe:

- a. There is high hazard to loss of life from large flows downstream of the dam.
- b. Dam failure resulting from overtopping would significantly increase the hazard to loss of life downstream from the dam from that which would exist just before overtopping failure.
- c. The spillway is not capable of passing one-half of the probable maximum flood without overtopping the dam and causing failure.

3. The above criteria are generally adequate for evaluating most non-Federal dams. However, in a few cases the increased hazard potential from overtopping and failure is so great as to result in catastrophic consequences. In such cases, the evaluation of condition 2c should utilize a flood more closely approximating the full probable maximum flood rather than one-half the flood. An example of such a situation would be a large dam immediately above a highly populated flood plain, with little likelihood of time for evacuation in the event of an emergency.

4. Conditions 2a and 2b require an approximation of housing location in relation to flooded areas. Resources available in Phase I inspections do not permit detailed surveys or time-consuming studies to develop such relationships. Therefore, rough estimates will generally be made from data obtained during the inspection and from readily available maps and drawings. Brief computer routings such as the HEC-1 dam break analysis, using available data, are recommended in marginal cases. The HEC-1, dam break version, is available on the Boeing Computer Services or may be obtained from the Hydrologic Engineering Center, Davis, California. Available resources do not permit detailed studies or investigations to establish the amount of overtopping that would cause a dam to fail, as designated in condition 2c. Professional judgment and available information will have to be used in these determinations. When detailed investigations and studies are required to make a reasonable judgment of the conditions which designate an unsafe dam, the inspection report should recommend that such studies be the responsibility of the dam owner.

5. During the inspection of a dam, consideration should be given to impacts on other dams located downstream from the project being inspected. When failure of a dam would be likely to cause failure of another dam(s) downstream, its designation as an unsafe dam could result in multiple impacts. Therefore, the information should be explicitly described in the inspection report. Such information may be vital to the priorities established by State Governors for dam improvements. Similarly, when the failure of an upstream dam (classified as unsafe) could cause failure of the dam being inspected, this information should be prominently displayed in the inspection report.

6. The criteria established in paragraph 2 for designating unsafe dams because of seriously inadequate spillways are considered reasonable and prudent. They provide a consistent bases for declaring unsafe dams and also serve as an effective compromise between the Recommended Guidelines and unduly low standards suggested by special interests and individuals unfamiliar with flood hazard potential.

7. The Hydrometeorological Branch (HMB) of the National Weather Service has reviewed some 500 experienced large storms in the United States. The purpose of the review was to ascertain the relative magnitude of experienced large storms to probable maximum precipitation (PMP) and their distribution throughout the country. Their review reveals that about 25 percent of the major storms have exceeded 50 percent of the probable maximum precipitation for one or more combinations of area and duration. In fact some storms have very closely approximated the PMP values. Exhibits C-1 thru C-5 indicate locations where experienced storms have exceeded 50 percent of the PMP.

8. There are several options to consider when selecting mitigation measures to avoid severe consequences of a dam failure from overtopping. The following measures may be required by a Governor when sufficient legal authority is available under State laws and a dam presents a serious threat to loss of life.

- a. Remove the dam.
- b. Increase the height of dam and/or spillway size to pass the probable maximum flood without overtopping the dam.
- c. Purchase downstream land that would be adversely impacted by dam failure and restrict human occupancy.
- d. Enhance the stability of the dam to permit overtopping by the probable maximum flood without failure.

- e. Provide a highly reliable flood warning system (generally does not prevent damage but avoids loss of life).

TABLE 1—STORMS WITH RAINFALL $\geq 150\%$ OF PMP, U.S. EAST OF THE 105TH MERIDIAN (FOR 10 MI², 6 HOURS; 200 MI², 24 HOURS AND/OR 1,000 MI², 48 HOURS)

Storm date	Index No.	Corps assignment No. (if available)	Storm center		Latitude	Longitude
			Town	State		
July 26, 1819	1		Catskill	NY	42°12'	73°53'
Aug. 5, 1843	2		Concordville	PA	39°53'	75°32'
Sept. 10-13, 1878	3	OR 9-19	Jefferson	OH	41°45'	80°46'
Sept. 20-24, 1882	4	NA 1-3	Paterson	NJ	40°55'	74°10'
June 13-17, 1886	5	LMV 4-27	Alexandria	LA	31°19'	92°33'
June 27-July 11, 1899	6	GM 3-4	Turnersville	TX	30°52'	96°32'
Aug. 24-28, 1903	7	MR 1-10	Woodburn	IA	40°57'	93°35'
Oct. 7-11, 1903	8	GL 4-9	Paterson	NJ	40°55'	74°10'
July 18-23, 1909	9	UMV 1-11B	Ironwood	MI	46°27'	90°11'
July 18-23, 1909	10	UMV 1-11A	Beaulieu	MN	47°21'	95°48'
July 22-23, 1911	11		Swede Home	NB	40°22'	96°54'
July 19-24, 1912	12	GL 2-29	Merrill	WI	45°11'	89°41'
July 13-17, 1916	13	SA 2-9	Altapass	NC	35°33'	82°01'
Sept. 8-10, 1921	14	GM 4-12	Taylor	TX	30°35'	97°18'
Oct. 4-11, 1924	15	SA 4-20	New Smyrna	FL	29°07'	80°55'
Sept. 17-19, 1926	16	MR 4-24	Boyden	IA	43°12'	96°00'
Mar. 11-16, 1929	17	UMV 2-20	Elba	AL	31°25'	86°04'
June 30-July 2, 1932	18	GM 5-1	State Fish Hatchery	TX	30°01'	99°07'
Sept. 16-17,	19		Ripogenus	ME	45°53'	69°09'

Storm date	Index No.	Corps assignment No. (if available)	Storm center		Latitude	Longitude
			Town	State		
1932			Dam			
July 22-27, 193	20	LMV 2-26	Logansport	LA	31°58'	94°00'
Apr. 3-4 1934	21	SW 2-11	Cheyenne	OK	35°37'	99°40'
May 30-31, 1935	22	MR 3-28A	Cherry Creek	CO	39°13'	104°32'
May 31, 1935	23	GM 5-20	Woodward	TX	29°20'	99°28'
July 6-10, 1935	24	NA 1-27	Hector	NY	42°30'	76°53'
Sept. 2-6, 1935	25	SA 1-26	Easton	MD	38°46'	76°01'
Sept. 14-18, 1936	26	GM 5-7	Broome	TX	31°47'	100°50'
June 19-20, 1939	27		Snyder	TX	32°44'	100°55'
July 4-5, 1939	28		Simpson	KY	38°13'	83°22'
Aug. 19, 1939	29	NA 2-3	Manahawkin	NJ	39°42'	74°16'
June 3-4, 1940	30	MR 4-5	Grant Township	NB	42°01'	96°53'
Aug. 6-9, 1940	31	LMV 4-24	Miller Isl	LA	29°45'	92°10'
Aug. 10-17, 1940	32	SA 5-19A	Keysville	VA	37°03'	78°30'
Sept. 1, 1940	33	NA 2-4	Ewan	NJ	39°42'	75°12'
Sept. 2-6, 1940	34	SW 2-18	Hallet	OK	36°15'	96°36'
Aug. 28-31, 1941	35	UMV 1-22	Haywood	WI	46°00'	91°28'
Oct. 17-22, 1941	36	SA 5-6	Trenton	FL	29°48'	82°57'
July 17-18, 1942	37	OR 9-23	Smethport	PA	41°50'	78°25'
Oct. 11-17, 1942	38	SA 1-28A	Big Meadows	VA	38°31'	78°26'
May 6-12, 1943	39	SW 2-20	Warner	OK	35°29'	95°18'
May 12-20, 1943	40	SW 2-21	Nr. Mounds	OK	35°52'	96°04'
July 27-29, 1943	41	GM 5-21	Devers	TX	30°02'	94°35'
Aug. 4-5, 1943	42	OR 3-30	Nr. Glenville	WV	38°56'	80°50'
June 10-13, 1944	43	MR 6-15	Nr. Stanton	NB	41°52'	97°03'
Aug. 12-15, 1946	44	MR 7-2A	Cole Camp	MO	38°40'	93°13'
Aug. 12-16,	45	MR 7-2B	Nr. Collinsville	IL	38°40'	89°59'

Storm date	Index No.	Corps assignment No. (if available)	Storm center		Latitude	Longitude
			Town	State		
1946						
Sept. 26-27, 1946	46	GM 5-24	Nr. San Antonio	TX	29°20'	98°29'
June 23-24, 1948	47		Nr. Del Rio	TX	29°22'	100°37'
Sept. 3-7, 1950	48	SA 5-8	Yankeetown	FL	29°03'	82°42'
June 23-28, 1954	49	SW 3-22	Vic Pierce	TX	30°22'	101°23'
Aug. 17-20, 1955	50	NA 2-22A	Westfield	MA	42°07'	72°45'
May 15-16, 1957	51		Hennessey	OK	36°02'	97°56'
June 14-15, 1957	52		Nr. E. St. Louis	IL	38°37'	90°24'
June 23-24, 1963	53		David City	NB	41°14'	97°05'
June 13-20, 1965	54		Holly	CO	37°43'	102°23'
June 24, 1966	55		Glenullin	ND	47°21'	101°19'
Aug. 12-13, 1966	56		Nr. Greely	NB	41°33'	98°32'
Sept. 19-24, 1967	57	SW 3-24	Falfurrias	TX	27°16'	98°12'
July 16-17, 1968	58		Waterloo	IA	42°30'	92°19'
July 4-5, 1969	59		Nr. Wooster	OH	40°50'	82°00'
Aug. 19-20, 1969	60	NA 2-3	Nr. Tyro	VA	37°49'	79°00'
June 9, 1972	61		Rapid City	SD	44°12'	103°31'
June 19-23, 1972	62		Zerbe	PA	40°37'	76°31'
July 21-22, 1972	63		Nr. Cushing	MN	46°10'	94°30'
Sept. 10-12, 1972	64		Harlan	IA	41°43'	95°15'
Oct. 10-11,	65		Enid	OK	36°25'	97°52'

Storm date	Index No.	Corps assignment No. (if available)	Storm center		Latitude	Longitude
			Town	State		
1973						

TABLE 2—STORMS WITH RAINFALL $\geq 50\%$ OF PMP, U.S. WEST OF CONTINENTAL DIVIDE (FOR 10 MI² 6 HOURS OR 1,000 MI² FOR ONE DURATION BETWEEN 6 AND 72 HOURS)

Storm date	Index No.	Storm center		Latitude	Longitude	Duration for 1,000 mi ²
		Town	State			
Aug. 11, 1890	1	Palmetto	NV	37°27'	117°42'	
Aug. 12, 1891	2	Campo	CA	32°36'	116°28'	
Aug. 28, 1898	3	Ft. Mohave	AZ	35°03'	114°36'	
Oct. 4-6, 1911	4	Gladstone	CO	37°53'	107°39'	
Dec. 29, 1913-Jan. 3, 1914	5		CA	39°55'	121°25'	
Feb. 17-22, 1914	6	Colby Ranch	CA	34°18'	118°07'	
Feb. 20-25, 1917	7		CA	37°35'	119°36'	
Sept. 13, 1918	8	Red Bluff	CA	40°10'	122°14'	
Feb. 26-Mar 4, 1938	9		CA	34°14'	117°11'	
Mar. 30-Apr. 2, 1931	10		ID	46°30'	114°50'	24
Feb. 26, 1932	11	Big Four	WA	48°05'	121°30'	
Nov. 21, 1933	12	Tatoosh Is	WA	48°23'	124°44'	
Jan. 20-25, 1935	13		WA	47°30'	123°30'	6
Jan. 20-25, 1935	14		WA	47°00'	122°00'	72
Feb. 4-8, 1937	15	Cyamaca Dam	CA	33°00'	116°35'	
Dec. 9-12, 1937	16		CA	38°51'	122°43'	
Feb. 27-Mar. 4, 1938	17		AZ	34°57'	111°44'	12
Jan. 19-24, 1943	18		CA	37°35'	119°25'	18
Jan. 19-24, 1943	19	Hoegge's Camp	CA	34°13'	118°02'	
Jan. 30-Feb. 3, 1945	20		CA	37°35'	119°30'	
Dec. 27, 1945	21	Mt. Tamalpias	CA	37°54'	122°34'	
Nov. 13-21, 1950	22		CA	36°30'	118°30'	24
Aug. 25-30, 1951	23		AZ	34°07'	112°21'	72

Storm date	Index No.	Storm center		Latitude	Longitude	Duration for 1,000 mi ²
		Town	State			
July 19, 1955	24	Chiatovich Flat	CA	37°44'	118°15'	
Aug. 16, 1958	25	Morgan	UT	41°03'	111°38'	
Sept. 18, 1959	26	Newton	CA	40°22'	122°12'	
June 7-8, 1964	27	Nyack Ck	MT	48°30'	113°38'	12
Sept. 3-7, 1970	28		UT	37°38'	109°04'	6
Sept. 3-7, 1970	29		AZ	33°49'	110°56'	6
June 7, 1972	30	Bakersfield	CA	35°25'	119°03'	

Storm date	Index No.	Storm center		Latitude	Longitude	Duration for 1,000 mi ²
		Town	State			
Dec. 9-12, 1937	31		CA	39°45'	121°30'	48

ER 1110-2-106
26 Sept 79

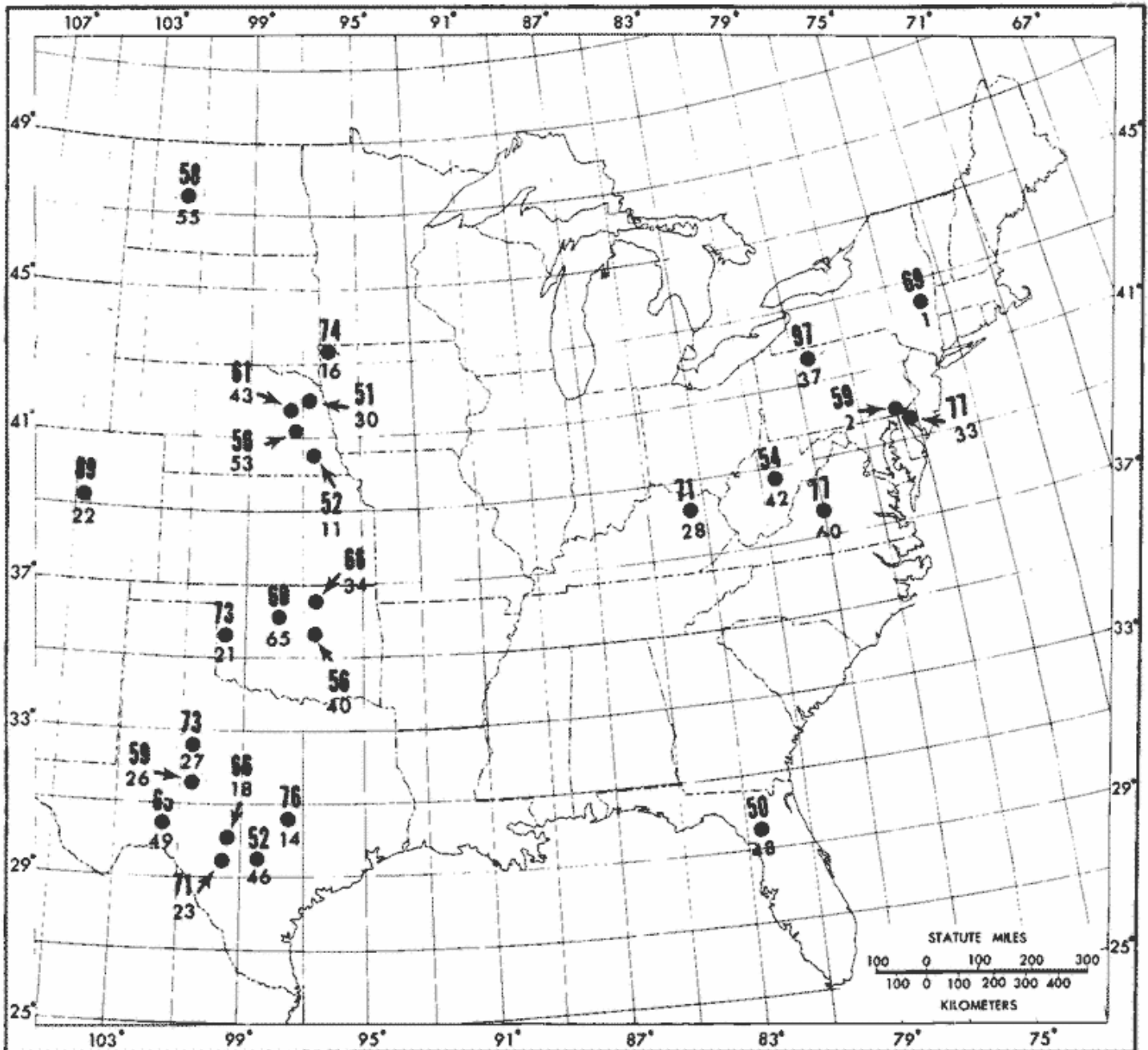


Plate 1: Observed point rainfalls $\geq 50\%$ of all-season PMP, U.S. east of 105th meridian for 10 mi² 6 hours. (Large number is % of PMP, small number is storm index, see table 1.)

Exhibit C-1

ER 1110-2-106'
26 Sept 79

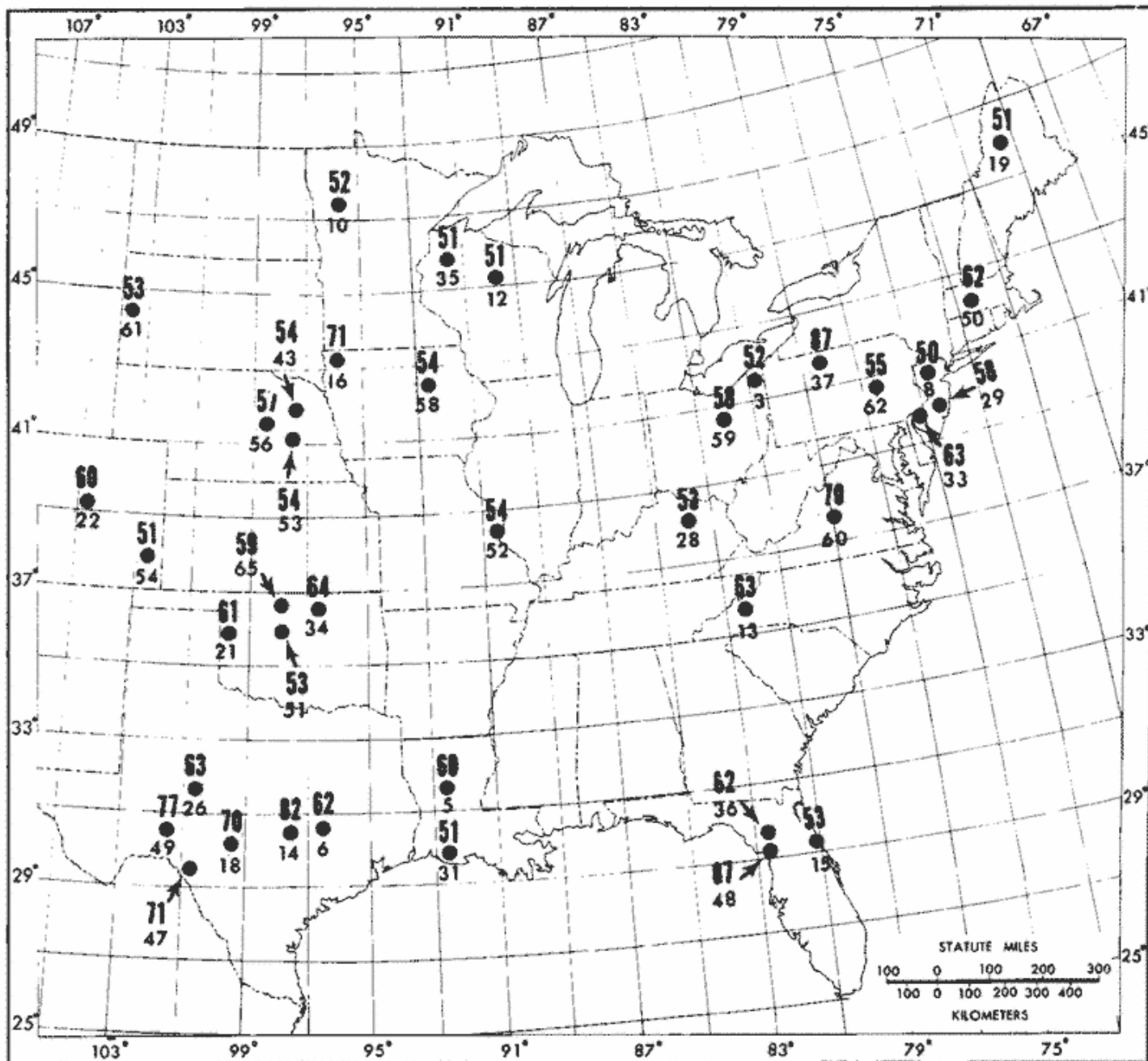


Plate 2: Observed rainfalls $\geq 50\%$ of all-season PMP, U.S. east of 105th meridian for 200 mi² 24 hours. (Large number is % of PMP, small number is storm index, see table 1.)

Exhibit C-2

ER 1110-2-106
26 Sept 79

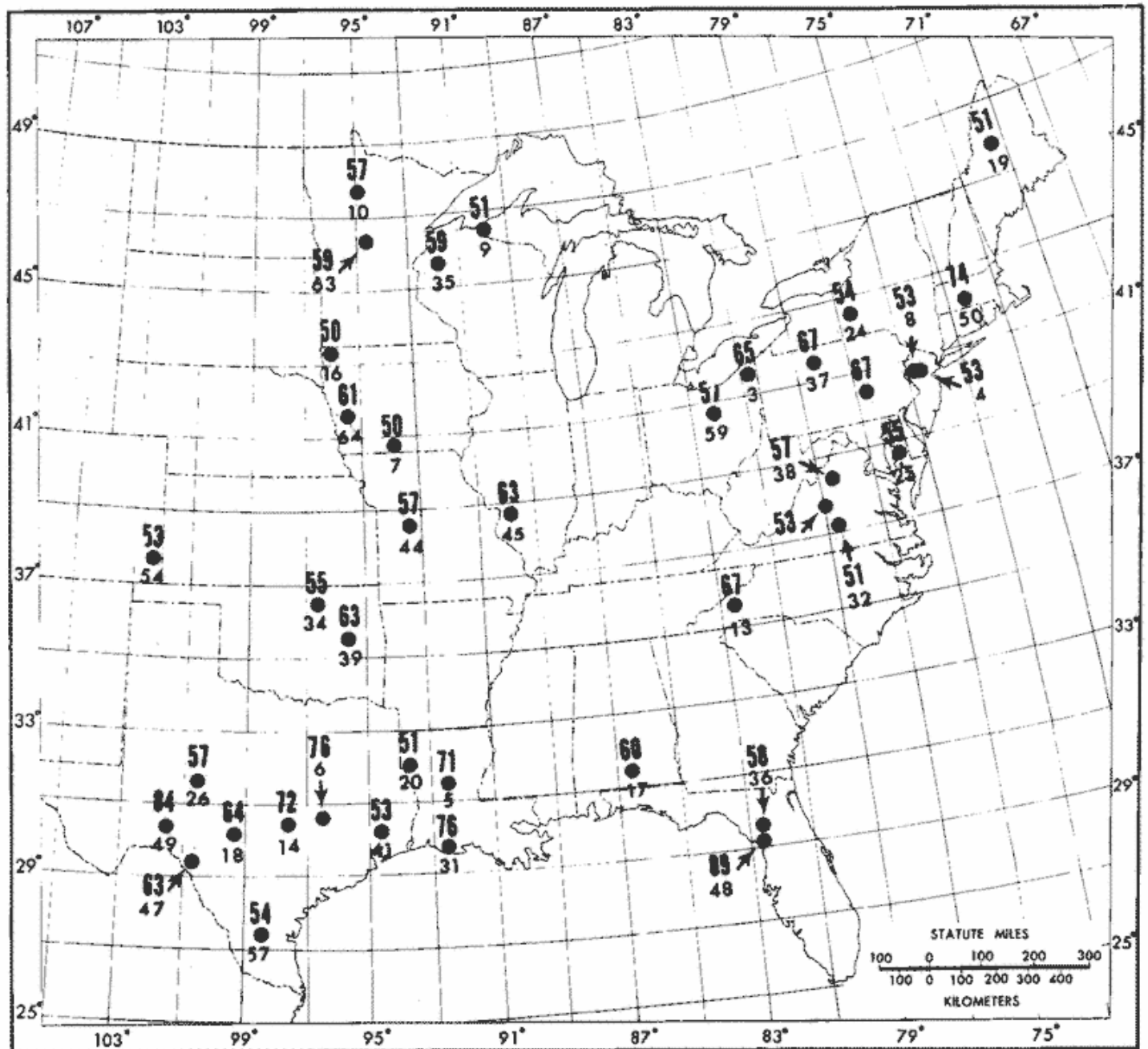


Plate 3: Observed rainfalls $\geq 50\%$ of all-season PMP, U.S. east of the 105th meridian for 1000 mi^2 48 hours. (Large number is % of PMP, small number is storm index, see table 1.)

Exhibit C-3

ER 1110-2-106
26 Sept 79

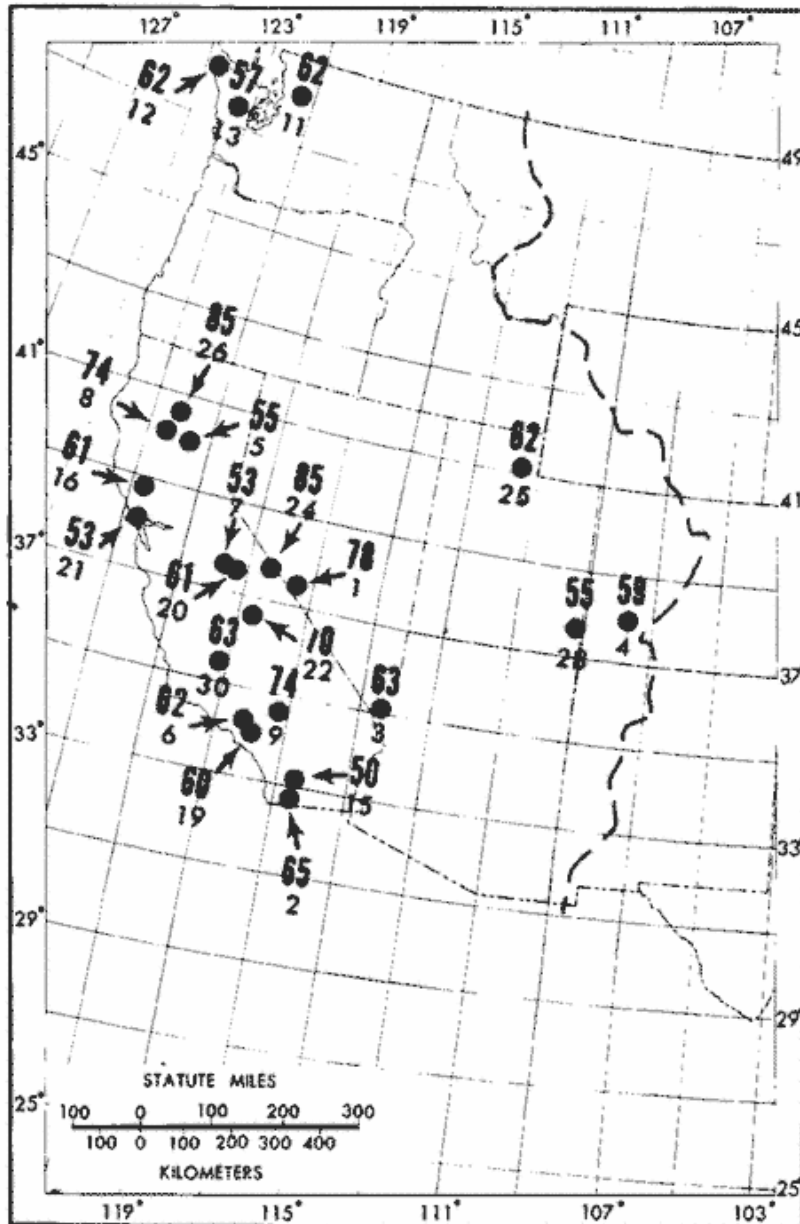


Plate 4: Observed point rainfalls $\geq 50\%$ of all-season PMP, U.S. west of the Continental Divide for 10 mi² for 6 hours. (Large number is % of PMP. Small number is storm index, see table 2.)

ER 1110-2-106
20 Sept 79

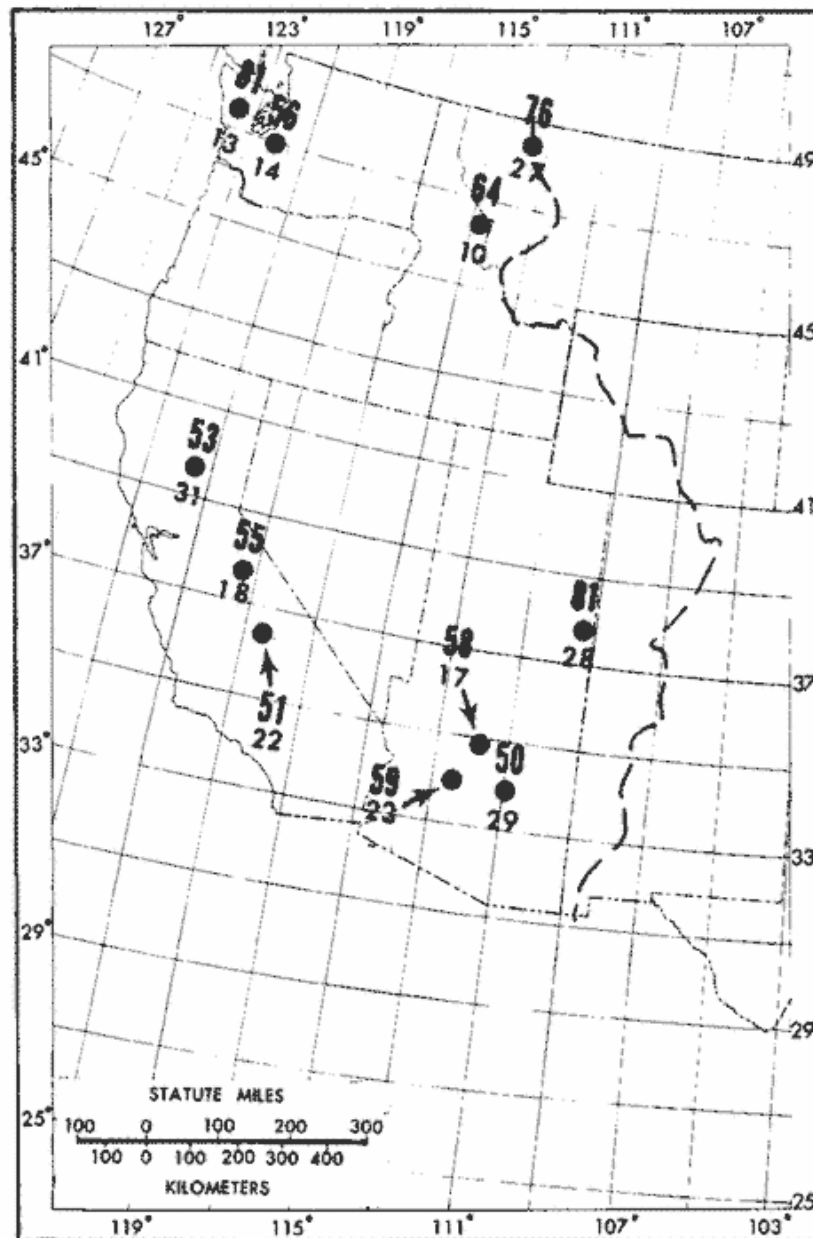


Plate 5: Observed rainfalls $\geq 50\%$ of all-season PMP, U.S. west of the Continental Divide for 1000 mi^2 for one duration between 6 and 72 hours. (Large number is % of PMP. Small number is storm index, see table 2.)

Appendix D to § 222.6—Recommended Guidelines for Safety Inspection of Dams

DEPARTMENT OF THE ARMY—OFFICE OF THE CHIEF OF ENGINEERS

Preface

The recommended guidelines for the safety inspection of dams were prepared to outline principal factors to be weighed in the determination of existing or potential hazards and to define the scope of activities to be undertaken in the safety inspection of dams. The establishment of rigid criteria or standards is not intended. Safety must be evaluated in the light of peculiarities and local conditions at a particular dam and in recognition of the many factors involved, some of which may not be precisely known. This can only be done by competent, experienced engineering judgment, which the guidelines are intended to supplement and not supplant. The guidelines are intended to be flexible, and the proper flexibility must be achieved through the employment of experienced engineering personnel.

Conditions found during the investigation which do not meet guideline recommendations should be assessed by the investigator as to their import from the standpoint of the involved degree of risk. Many deviations will not compromise project safety and the investigator is expected to identify them in this manner if that is the case. Others will involve various degrees of risk, the proper evaluation of which will afford a basis for priority of subsequent attention and possible remedial action.

The guidelines present procedures for investigating and evaluating existing conditions for the purpose of identifying deficiencies and hazardous conditions. The two phases of investigation outlined in the guidelines are expected to accomplish only this and do not encompass in scope the engineering which will be required to perform the design studies for corrective modification work.

It is recognized that some States may have established or will adopt inspection criteria incongruous in some respects with these guidelines. In such instances assessments of project safety should recognize the State's requirements as well as guideline recommendations.

The guidelines were developed with the help of several Federal agencies and many State agencies, professional engineering organizations, and private engineers. In reviewing two drafts of the guidelines they have contributed many helpful suggestions. Their contributions are deeply appreciated and have made it possible to evolve a document representing a consensus of the engineering fraternity. As experience is gained with use of the guidelines, suggestions for future revisions will be generated. All such suggestions should be directed to the Chief of Engineers, U.S. Army, DAEN-CWE-D, Washington, D.C. 20314.

Recommended Guidelines for Safety Inspection of Dams

Table of Contents

PREFACE

Chapter 1—Introduction

Para.

1.1 Purpose.

1.2 Applicability.

1.3 Authority.

Chapter 2—General Requirements

2.1 Classification of dams.

2.1.1 Size.

2.1.2 Hazard potential.

2.2 Selection of dams to be investigated.

2.3 Technical investigations.

2.4 Qualifications of investigators.

2.5 Reports.

Chapter 3—Phase I Investigation

3.1 Purpose.

3.2 Scope.

3.3 Engineering data.

3.4 Field inspections.

3.5 Evaluation of hydraulic and hydrologic features.

3.5.1 Design data.

3.5.2 Experience data.

3.6 Evaluation of structural stability.

3.6.1 Design and construction data.

3.6.2 Operating records.

3.6.3 Post construction changes.

3.6.4 Seismic stability.

Chapter 4—Phase II Investigation

4.1 Purpose.

4.2 Scope.

4.3 Hydraulic and hydrologic analysis.

4.3.1 Maximum water surface based on SDF peak inflow.

4.3.1.1 Peak for 100-year flood.

4.3.1.2 Peak for PMF or fraction thereof.

4.3.2 Maximum water surface based on SDF hydrograph.

4.3.3 Acceptable procedures.

4.3.4 Freeboard allowances.

4.4 Stability investigations.

4.4.1 Foundation and material investigations.

4.4.2 Stability assessment.

4.4.2.1 Seismic stability.

4.4.2.2 Clay shale foundation.

4.4.3 Embankment dams.

4.4.3.1 Liquefaction.

4.4.3.2 Shear failure.

4.4.3.3 Loading conditions.

4.4.3.4 Safety factors.

4.4.3.5 Seepage failure.

4.4.3.6 Seepage analyses.

4.4.4 Concrete dams and appurtenant structures.

4.4.4.1 Requirements for stability.

4.4.4.2 Loads.

4.4.4.3 Stresses.

4.4.4.4 Overturning.

4.4.4.5 Sliding.

4.4.4.5.1 Sliding resistance.

4.4.4.5.2 Downstream resistance.

4.4.4.5.3 Safety factor.

Chapter 5—Reports

5.1 General.

5.2 Preparation of report.

5.2.1 Phase I reports.

5.2.2 Phase II reports.

Tables

Table

1 Size classification.

2 Hazard potential classification.

3 Hydrologic evaluation guidelines.

4 Factors of safety (embankment dams).

Figures

Fig.

1 Seismic zone map of contiguous States.

2 Seismic zone map of California, Nevada and Arizona.

3 Seismic zone map of Alaska.

4 Seismic zone map of Hawaii.

5 Design envelope for Case I (Table 4).

6 Design envelope for Cases II and III (Table 4).

Appendixes

Appendix I to App. D—Engineering data

Appendix II to App. D—Inspection items

Appendix III to App. D—Pub. L. 92-367

Chapter 1—Introduction

1.1. *Purpose.* This document provides recommended guidelines for the inspection and evaluation of dams to determine if they constitute hazards to human life or property.

1.2. *Applicability.* The procedures and guidelines outlined in this document apply to the inspection and evaluation of all dams as defined in the National Dam Inspection Act, Public Law 92-367. Included in this program are all artificial barriers together with appurtenant works which impound or divert water and which (1) are twenty-five feet or more in height or (2) have an impounding capacity of fifty acre-feet or more. Not included are barriers which are six feet or less in height, regardless of storage capacity, or barriers which have a storage capacity at maximum water storage elevation of fifteen acre-feet or less regardless of height.

1.3. *Authority.* The Dam Inspection Act, Public Law 92-367 (Appendix III), authorized the Secretary of the Army, through the Corps of Engineers, to initiate a program of safety inspection of dams throughout the United States. The Chief of Engineers issues these guidelines pursuant to that authority.

Chapter 2—General Requirements

2.1. *Classification of dams.* Dams should be classified in accordance with size and hazard potential in order to formulate a priority basis for selecting dams to be included in the inspection program and also to provide compatibility between guideline requirements and involved risks. When possible the initial classifications

should be based upon information listed in the National Inventory of Dams with respect to size, impoundment capacity and hazard potential. It may be necessary to reclassify dams when additional information becomes available.

2.1.1. *Size.* The classification for size based on the height of the dam and storage capacity should be in accordance with Table 1. The height of the dam is established with respect to the maximum storage potential measured from the natural bed of the stream or watercourse at the downstream toe of the barrier, or if it is not across a stream or watercourse, the height from the lowest elevation of the outside limit of the barrier, to the maximum water storage elevation. For the purpose of determining project size, the maximum storage elevation may be considered equal to the top of dam elevation. Size classification may be determined by either storage or height, whichever gives the larger size category.

TABLE 1—SIZE CLASSIFICATION

Category	Impoundment	
	Storage (ac-ft)	Height (ft)
Small	<1,000 and ≥ 50	<40 and ≥ 25 .
Intermediate	$\geq 1,000$ and <50,000	≥ 40 and <100.
Large	$\geq 50,000$	≥ 100 .

2.1.2. *Hazard Potential.* The classification for potential hazards should be in accordance with Table 2. The hazards pertain to potential loss of human life or property damage in the area downstream of the dam in event of failure or misoperation of the dam or appurtenant facilities. Dams conforming to criteria for the low hazard potential category generally will be located in rural or agricultural areas where failure may damage farm buildings, limited agricultural land, or township and country roads. Significant hazard potential category structures will be those located in predominantly rural or agricultural areas where failure may damage isolated homes, secondary highways or minor railroads or cause interruption of use or service of relatively important public utilities. Dams in the high hazard potential category will be those located where failure may cause serious damage to homes, extensive agricultural, industrial and commercial facilities, important public utilities, main highways, or railroads.

TABLE 2—HAZARD POTENTIAL CLASSIFICATION

Category	Loss of life (extent of development)	Economic loss (extent of development)
Low	None expected (No permanent structures for human habitation)	Minimal (Undeveloped to occasional structures or agriculture).
Significant	Few (No urban developments and no more than a	Appreciable (Notable agriculture,

Category	Loss of life (extent of development)	Economic loss (extent of development)
High	small number of inhabitable structures) More than few	industry or structures). Excessive (Extensive community, industry or agriculture).

2.2. Selection of dams to be investigated. The selection of dams to be investigated should be based upon an assessment of existing developments in flood hazard areas. Those dams possessing a hazard potential classified high or significant as indicated in Table 2 should be given first and second priorities, respectively, in the inspection program. Inspection priorities within each category may be developed from a consideration of factors such as size classification and age of the dam, the population size in the downstream flood area, and potential developments anticipated in flood hazard areas.

2.3. Technical Investigations. A detailed, systematic, technical inspection and evaluation should be made of each dam selected for investigation in which the hydraulic and hydrologic capabilities, structural stability and operational adequacy of project features are analyzed and evaluated to determine if the dam constitutes a danger to human life or property. The investigation should vary in scope and completeness depending upon the availability and suitability of engineering data, the validity of design assumptions and analyses and the condition of the dam. The minimum investigation will be designated Phase I, and an in-depth investigation designated Phase II should be made where deemed necessary. Phase I investigations should consist of a visual inspection of the dam, abutments and critical appurtenant structures, and a review of readily available engineering data. It is not intended to perform costly explorations or analyses during Phase I. Phase II investigations should consist of all additional engineering investigations and analyses found necessary by results of the Phase I investigation.

2.4. Qualifications of investigators. The technical investigations should be conducted under the direction of licensed professional engineers experienced in the investigation, design, construction and operation of dams, applying the disciplines of hydrologic, hydraulic, soils and structural engineering and engineering geology. All field inspections should be conducted by qualified engineers, engineering geologists and other specialists, including experts on mechanical and electrical operation of gates and controls, knowledgeable in the investigation, design, construction and operation of dams.

Chapter 3—Phase I Investigation

3.1. Purpose. The primary purpose of the Phase I investigation program is to identify expeditiously those dams which may pose hazards to human life or property.

3.2. Scope. The Phase I investigation will develop an assessment of the general condition with respect to safety of the project based upon available data and a visual inspection, determine any need for emergency measures and conclude if additional studies, investigation and analyses are necessary and warranted. A review will be made of pertinent existing and available engineering data relative to the design, construction and operation of the dam and appurtenant structures, including electrical and mechanical operating equipment and measurements from inspection and performance instruments and devices; and a detailed systematic visual inspection will be performed of those features relating to the stability and operational adequacy of the project.

Based upon findings of the review of engineering data and the visual inspection, an evaluation will be made of the general condition of the dam, including where possible the assessment of the hydraulic and hydrologic capabilities and the structural stability.

3.3. *Engineering data.* To the extent feasible the engineering data listed in Appendix I relating to the design, construction and operation of the dam and appurtenant structures, should be collected from existing records and reviewed to aid in evaluating the adequacy of hydraulic and hydrologic capabilities and stability of the dam. Where the necessary engineering data are unavailable, inadequate or invalid, a listing should be made of those specific additional data deemed necessary by the engineer in charge of the investigation and included in the Phase I report.

3.4. *Field inspections.* The field inspection of the dam, appurtenant structures, reservoir area, and downstream channel in the vicinity of the dam should be conducted in a systematic manner to minimize the possibility of any significant feature being overlooked. A detailed checklist should be developed and followed for each dam inspected to document the examination of each significant structural and hydraulic feature including electrical and mechanical equipment for operation of the control facilities that affect the safety of the dam.

3.4.1. Particular attention should be given to detecting evidence of leakage, erosion, seepage, slope instability, undue settlement, displacement, tilting, cracking, deterioration, and improper functioning of drains and relief wells. The adequacy and quality of maintenance and operating procedures as they pertain to the safety of the dam and operation of the control facilities should also be assessed.

3.4.2. Photographs and drawings should be used freely to record conditions in order to minimize descriptions.

3.4.3. The field inspection should include appropriate features and items, including but not limited to those listed in Appendix II, which may influence the safety of the dam or indicate potential hazards to human life or property.

3.5. *Evaluation of hydraulic and hydrologic Features.*

3.5.1. *Design data.* Original hydraulic and hydrologic design assumptions obtained from the project records should be assessed to determine their acceptability in evaluating the safety of the dam. All constraints on water control such as blocked entrances, restrictions on operation of spillway and outlet gates, inadequate energy dissipators or restrictive channel conditions, significant reduction in reservoir capacity by sediment deposits and other factors should be considered in evaluating the validity of discharge ratings, storage capacity, hydrographs, routings and regulation plans. The discharge capacity and/or storage capacity should be capable of safely handling the recommended spillway design flood for the size and hazard potential classification of the dam as indicated in Table 3. The hydraulic and hydrologic determinations for design as obtained from project records will be acceptable if conventional techniques similar to the procedures outlined

in paragraph 4.3. were used in obtaining the data. When the project design flood actually used exceeds the recommended spillway design flood, from Table 3, the project design flood will be acceptable in evaluating the safety of the dam.

**TABLE 3—HYDROLOGIC EVALUATION GUIDELINES
[RECOMMENDED SPILLWAY DESIGN FLOODS]**

Hazard	Size	Spillway design flood (SDF) ¹
Low	Small	50 to 100-yr frequency.
	Intermediate	100-yr to $1\frac{1}{2}$ PMF.
	Large	$1\frac{1}{2}$ PMF to PMF.
Significant	Small	100-yr to $1\frac{1}{2}$ PMF.
	Intermediate	$1\frac{1}{2}$ PMF to PMF.
	Large	PMF.
High	Small	$1\frac{1}{2}$ PMF to PMF.
	Intermediate	PMF.
	Large	PMF.

¹ The recommended design floods in this column represent the magnitude of the spillway design flood (SDF), which is intended to represent the largest flood that need be considered in the evaluation of a given project, regardless of whether a spillway is provided; i.e., a given project should be capable of safely passing the appropriate SDF. Where a range of SDF is indicated, the magnitude that most closely relates to the involved risk should be selected.

1000-yr = 100-Year Exceedence Interval. The flood magnitude expected to be exceeded, on the average, of once in 100 years. It may also be expressed as an exceedence frequency with a one-percent chance of being exceeded in any given year.

PMF = Probable Maximum Flood. The flood that may be expected from the most severe combination of critical meteorologic and hydrologic conditions that are reasonably possible in the region. The PMF is derived from probable maximum precipitation (PMP), which information is generally available from the National Weather Service, NOAA. Most Federal agencies apply reduction factors to the PMP when appropriate. Reductions may be applied because rainfall isohyets are unlikely to conform to the exact shape of the drainage basin and/or the storm is not likely to center exactly over the drainage basin. In some cases local topography will cause changes from the generalized PMP values, therefore it may be advisable to contact Federal construction agencies to obtain the prevailing practice in specific areas.

3.5.2. *Experience data.* In some cases where design data are lacking, an evaluation of overtopping potential may be based on watershed characteristics and rainfall and reservoir records. An estimate of the probable maximum flood may also be developed from a conservative, generalized comparison of the drainage area size and the magnitude of recently adopted probable maximum floods for damsites in comparable hydrologic regions. Where the review of such experience data indicates that the recommended spillway design flood would not cause overtopping additional hydraulic and hydrologic determinations will be unnecessary.

3.6. *Evaluation of structural stability.* The Phase I evaluations of structural adequacy of project features are expected to be based principally on existing conditions as revealed by the visual inspection, together with available design and construction information and records of performance. The objectives are to determine the existence of conditions which are hazardous, or which with time might develop into safety hazards, and to formulate recommendations pertaining to the need for any additional studies, investigations, or analyses. The results of this phase of the inspection must rely very substantially upon the experience and judgment of the inspecting engineer.

3.6.1. *Design and construction data.* The principal design assumptions and analyses obtained from the project records should be assessed. Original design and construction records should be used judiciously, recognizing the restricted applicability of such data as material strengths and permeabilities, geological factors and construction descriptions. Original stability studies and analyses should be acceptable if conventional techniques and procedures similar to those outlined in paragraph 4.4 were employed, provided that review of operational and performance data confirm that the original design assumptions were adequately conservative. The need for such analyses where either none exist or the originals are incomplete or unsatisfactory will be determined by the inspecting engineer based upon other factors such as condition of structures, prior maximum loadings and the hazard degree of the project. Design assumptions and analyses should include all applicable loads including earthquake and indicate the structure's capability to resist overturning, sliding and overstressing with adequate factors of safety. In general seepage and stability analyses comparable to the requirements of paragraph 4.4 should be on record for all dams in the high hazard category and large dams in the significant hazard category. This requirement for other dams will be subject to the opinion of the inspecting engineer.

3.6.2. *Operating records.* The performance of structures under prior maximum loading conditions should in some instances provide partial basis for stability evaluation. Satisfactory experience under loading conditions not expected to be exceeded in the future should generally be indicative of satisfactory stability, provided adverse changes in physical conditions have not occurred. Instrumentation observations of forces, pressures, loads, stresses, strains, displacements, deflections or other related conditions should also be utilized in the safety evaluation. Where such data indicate abnormal behavior, unsafe movement or deflections, or loadings which adversely affect the stability or functioning of the structure, prompt reporting of such circumstances is required without the delay for preparation of the official inspection report.

3.6.3. *Post construction changes.* Data should be collected on changes which have occurred since project construction that might influence the safety of the dam such as road cuts, quarries, mining and groundwater changes.

3.6.4. *Seismic stability.* An assessment should be made of the potential vulnerability of the dam to seismic events and a recommendation developed with regard to the need for additional seismic investigation. In general, projects located in Seismic Zones 0, 1 and 2 may be assumed to present no hazard from earthquake provided static stability conditions are satisfactory and conventional safety margins exist. Dams in Zones 3 and 4 should, as a minimum, have on record suitable analyses made by conventional equivalent static load methods. The seismic zones together with appropriate coefficients for use in such analyses are shown in Figures 1 through 4. Boundary lines are approximate and in the event of doubt about the proper zone, the higher zone should be used. All high hazard category dams in Zone 4 and high hazard dams of the hydraulic fill type in Zone 3 should have a stability assessment based upon knowledge of regional and local geology, engineering seismology, in situ properties of materials and appropriate dynamic analytical and testing procedures. The assessment should include the possibility of physical displacement of the structures due to

movements along active faults. Departure from this general guidance should be made whenever in the judgment of the investigating engineer different seismic stability requirements are warranted because of local geological conditions or other reasons.

Chapter 4—Phase II Investigation

4.1. *Purpose.* The Phase II investigation will be supplementary to Phase I and should be conducted when the results of the Phase I investigation indicate the need for additional in-depth studies, investigations or analyses.

4.2. *Scope.* The Phase II investigation should include all additional studies, investigations and analyses necessary to evaluate the safety of the dam. Included, as required, will be additional visual inspections, measurements, foundation exploration and testing, materials testing, hydraulic and hydrologic analysis and structural stability analyses.

4.3. *Hydraulic and hydrologic analysis.* Hydraulic and hydrologic capabilities should be determined using the following criteria and procedures. Depending on the project characteristics, either the spillway design flood peak inflow or the spillway design flood hydrograph should be the basis for determining the maximum water surface elevation and maximum outflow. If the operation or failure of upstream water control projects would have significant impact on peak flow or hydrograph analyses, the impact should be assessed.

4.3.1. *Maximum water surface based on SDF peak inflow.* When the total project discharge capability at maximum pool exceeds the peak inflow of the recommended SDF, and operational constraints would not prevent such a release at controlled projects, a reservoir routing is not required. The maximum discharge should be assumed equal to the peak inflow of the spillway design flood. Flood volume is not controlling in this situation and surcharge storage is either absent or is significant only to the extent that it provides the head necessary to develop the release capability required.

4.3.1.1. *Peak for 100-year flood.* When the 100-year flood is applicable under the provisions of Table 3 and data are available, the spillway design flood peak inflow may be determined by use of “A Uniform Technique for Determining Flood Frequencies,” Water Resources Council (WRC), Hydrology Committee, Bulletin 15, December 1967. Flow frequency information from regional analysis is generally preferred over single station results when available and appropriate. Rainfall-runoff techniques may be necessary when there are inadequate runoff data available to make a reasonable estimate of flow frequency.

4.3.1.2. *Peak for PMF or fraction thereof.* When either the Probable Maximum Flood peak or a fraction thereof is applicable under the provisions of Table 3, the unit hydrograph—infiltration loss technique is generally the most expeditious method of computing the spillway design flood peak for most projects. This technique is discussed in the following paragraph.

4.3.2. *Maximum water surface based on SDF hydrograph.* Both peak and volume are required in this analysis. Where surcharge storage is significant, or where there is insufficient discharge capability at maximum pool to pass the peak inflow of the SDF, considering all possible operational constraints, a flood hydrograph is required. When there are upstream hazard areas that would be imperiled by fast rising reservoirs levels, SDF hydrographs should be routed to ascertain available time for warning and escape. Determination of probable maximum precipitation or 100-year precipitation, which ever is applicable, and unit hydrographs or runoff models will be required, followed by the determination of the PMF or 100-year flood. Conservative loss rates (significantly reduced by antecedent rainfall conditions where appropriate) should be estimated for computing

the rainfall excess to be utilized with unit hydrographs. Rainfall values are usually arranged with gradually ascending and descending rates with the maximum rate late in the storm. When applicable, conservatively high snowmelt runoff rates and appropriate releases from upstream projects should be assumed. The PMP may be obtained from National Weather Service (NWS) publications such as Hydrometeorological Report (HMR) 33. Special NWS publications for particular areas should be used when available. Rainfall for the 100-year frequency flood can be obtained from the NWS publication "Rainfall Frequency Atlas of the United States," Technical Paper No. 40; Atlas 2, "Precipitation Frequency Atlas of Western United States;" or other NWS publications. The maximum water surface elevation and spillway design flood outflow are then determined by routing the inflow hydrograph through the reservoir surcharge storage, assuming a starting water surface at the bottom of surcharge storage, or lower when appropriate. For projects where the bottom of surcharge space is not distinct, or the flood control storage space (exclusive of surcharge) is appreciable, it may be appropriate to select starting water surface elevations below the top of the flood control storage for routings. Conservatively high starting levels should be estimated on the basis of hydrometeorological conditions reasonably characteristic for the region and flood release capability of the project. Necessary adjustment of reservoir storage capacity due to existing or future sediment or other encroachment may be approximated when accurate determination of deposition is not practicable.

4.3.3. *Acceptable procedures.* Techniques for performing hydraulic and hydrologic analyses are generally available from publications prepared by Federal agencies involved in water resources development or textbooks written by the academic community. Some of these procedures are rather sophisticated and require expensive computational equipment and large data banks. While results of such procedures are generally more reliable than simplified methods, their use is generally not warranted in studies connected with this program unless they can be performed quickly and inexpensively. There may be situations where the more complex techniques have to be employed to obtain reliable results; however, these cases will be exceptions rather than the rule. Whenever the acceptability of procedures is in question, the advice of competent experts should be sought. Such expertise is generally available in the Corps of Engineers, Bureau of Reclamation and Soil Conservation Service. Many other agencies, educational facilities and private consultants can also provide expert advice. Regardless of where such expertise is based, the qualification of those individuals offering to provide it should be carefully examined and evaluated.

4.3.4. *Freeboard allowances.* Guidelines on specific minimum freeboard allowances are not considered appropriate because of the many factors involved in such determinations. The investigator will have to assess the critical parameters for each project and develop its minimum requirement. Many projects are reasonably safe without freeboard allowance because they are designed for overtopping, or other factors minimize possible overtopping. Conversely, freeboard allowances of several feet may be necessary to provide a safe condition. Parameters that should be considered include the duration of high water levels in the reservoir during the design flood; the effective wind fetch and reservoir depth available to support wave generation; the probability of high wind speed occurring from a critical direction; the potential wave runup on the dam based on roughness and slope; and the ability of the dam to resist erosion from overtopping waves.

4.4 *Stability investigations.* The Phase II stability investigations should be compatible with the guidelines of this paragraph.

4.4.1 *Foundation and material investigations.* The scope of the foundation and materials investigation should be limited to obtaining the information required to analyze the structural stability and to investigate any suspected condition which would adversely affect the safety of the dam. Such investigations may include borings to obtain concrete, embankment, soil foundation, and bedrock samples; testing specimens from these samples to determine the strength and elastic parameters of the materials, including the soft seams, joints, fault gouge

and expansive clays or other critical materials in the foundation; determining the character of the bedrock including joints, bedding planes, fractures, faults, voids and caverns, and other geological irregularities; and installing instruments for determining movements, strains, suspected excessive internal seepage pressures, seepage gradients and uplift forces. Special investigations may be necessary where suspect rock types such as limestone, gypsum, salt, basalt, claystone, shales or others are involved in foundations or abutments in order to determine the extent of cavities, piping or other deficiencies in the rock foundation. A concrete core drilling program should be undertaken only when the existence of significant structural cracks is suspected or the general qualitative condition of the concrete is in doubt. The tests of materials will be necessary only where such data are lacking or are outdated.

4.4.2. *Stability assessment.* Stability assessments should utilize in situ properties of the structure and its foundation and pertinent geologic information. Geologic information that should be considered includes groundwater and seepage conditions; lithology, stratigraphy, and geologic details disclosed by borings, "as-built" records, and geologic interpretation; maximum past overburden at site as deduced from geologic evidence; bedding, folding and faulting; joints and joint systems; weathering; slickensides, and field evidence relating to slides, faults, movements and earthquake activity. Foundations may present problems where they contain adversely oriented joints, slickensides or fissured material, faults, seams of soft materials, or weak layers. Such defects and excess pore water pressures may contribute to instability. Special tests may be necessary to determine physical properties of particular materials. The results of stability analyses afford a means of evaluating the structure's existing resistance to failure and also the effects of any proposed modifications. Results of stability analyses should be reviewed for compatibility with performance experience when possible.

4.4.2.1. *Seismic stability.* The inertial forces for use in the conventional equivalent static force method of analysis should be obtained by multiplying the weight by the seismic coefficient and should be applied as a horizontal force at the center of gravity of the section or element. The seismic coefficients suggested for use with such analyses are listed in Figures 1 through 4. Seismic stability investigations for all high hazard category dams located in Seismic Zone 4 and high hazard dams of the hydraulic fill type in Zone 3 should include suitable dynamic procedures and analyses. Dynamic analyses for other dams and higher seismic coefficients are appropriate if in the judgment of the investigating engineer they are warranted because of proximity to active faults or other reasons. Seismic stability investigations should utilize "state-of-the-art" procedures involving seismological and geological studies to establish earthquake parameters for use in dynamic stability analyses and, where appropriate, the dynamic testing of materials. Stability analyses may be based upon either time-history or response spectra techniques. The results of dynamic analyses should be assessed on the basis of whether or not the dam would have sufficient residual integrity to retain the reservoir during and after the greatest or most adverse earthquake which might occur near the project location.

4.4.2.2. *Clay shale foundation.* Clay shale is a highly overconsolidated sedimentary rock comprised predominantly of clay minerals, with little or no cementation. Foundations of clay shales require special measures in stability investigations. Clay shales, particularly those containing montmorillonite, may be highly susceptible to expansion and consequent loss of strength upon unloading. The shear strength and the resistance to deformation of clay shales may be quite low and high pore water pressures may develop under increase in load. The presence of slickensides in clay shales is usually an indication of low shear strength. Prediction of field behavior of clay shales should not be based solely on results of conventional laboratory tests since they may be misleading. The use of peak shear strengths for clay shales in stability analyses may be unconservative because of nonuniform stress distribution and possible progressive failures. Thus the available shear resistance may be less than if the peak shear strength were mobilized simultaneously along the entire failure surface. In such cases, either greater safety factors or residual shear strength should be used.

4.4.3. *Embankment dams.*

4.4.3.1. *Liquefaction.* The phenomenon of liquefaction of loose, saturated sands and silts may occur when such materials are subjected to shear deformation or earthquake shocks. The possibility of liquefaction must presently be evaluated on the basis of empirical knowledge supplemented by special laboratory tests and engineering judgment. The possibility of liquefaction in sands diminishes as the relative density increases above approximately 70 percent. Hydraulic fill dams in Seismic Zones 3 and 4 should receive particular attention since such dams are susceptible to liquefaction under earthquake shocks.

4.4.3.2. *Shear failure.* Shear failure is one in which a portion of an embankment or of an embankment and foundation moves by sliding or rotating relative to the remainder of the mass. It is conventionally represented as occurring along a surface and is so assumed in stability analyses, although shearing may occur in a zone of substantial thickness. The circular arc or the sliding wedge method of analyzing stability, as pertinent, should be used. The circular arc method is generally applicable to essentially homogeneous embankments and to soil foundations consisting of thick deposits of fine-grained soil containing no layers significantly weaker than other strata in the foundation. The wedge method is generally applicable to rockfill dams and to earth dams on foundations containing weak layers. Other methods of analysis such as those employing complex shear surfaces may be appropriate depending on the soil and rock in the dam and foundation. Such methods should be in reputable usage in the engineering profession.

4.4.3.3. *Loading conditions.* The loading conditions for which the embankment structures should be investigated are (I) Sudden drawdown from spillway crest elevation or top of gates, (II) Partial pool, (III) Steady state seepage from spillway crest elevation or top of gate elevation, and (IV) Earthquake. Cases I and II apply to upstream slopes only; slopes; and Case IV applies to both upstream and downstream Case III applies to downstream slopes. A summary of suggested strengths and safety factors are shown in Table 4.

TABLE 4—FACTORS OF SAFETY ¹

Case and loading condition	Factor of safety	Shear ² strength	Remarks
I Sudden drawdown from spillway crest or top of gates to minimum drawdown elevation	³ 1.2	Minimum composite of R and S shear strengths. See Figure 5	Within the drawdown zone submerged unit weights of materials are used for computing forces resisting sliding and saturated unit weights are used for computing forces contributing to sliding.
II Partial pool with assumed horizontal steady seepage saturation	1.5	R + S/2 for R<S S for R>S	Composite intermediate envelope of R and S shear strengths. See Figure 6.
III Steady seepage from spillway crest or top of gates with $K_h/K_v = 9$ assumed ⁴	1.5	Same as Case II	
IV Earthquake (Cases II and III with seismic	1.0	(⁵)	See Figures 1 through 4 for Seismic Coefficients.

¹ Not applicable to embankments on clay shale foundation. Experience has indicated special problems in determination of design shear strengths for clay shale foundations and acceptable safety factors should be compatible with the confidence level in shear strength assumptions.

² Other strength assumptions may be used if in common usage in the engineering profession.

³ The safety factor should not be less than 1.5 when drawdown rate and pore water pressure developed from flow nets are used in stability analyses.

⁴ K_h/K_v is the ratio of horizontal to vertical permeability. A minimum of 9 is suggested for use in compacted embankments and alluvial sediments.

⁵ Use shear strength for case analyzed without earthquake. It is not necessary to analyze sudden drawdown for earthquake loading. Shear strength tests are classified according to the controlled drainage conditions maintained during the test. R tests are those in which specimen drainage is allowed during consolidation (or swelling) under initial stress conditions, but specimen drainage is not allowed during application of shearing stresses. S tests allow full drainage during initial stress application and shearing is at a slow rate so that complete specimen drainage is permitted during the complete test.

Case and loading condition	Factor of safety	Shear ² strength	Remarks
loading)			

¹ Not applicable to embankments on clay shale foundation. Experience has indicated special problems in determination of design shear strengths for clay shale foundations and acceptable safety factors should be compatible with the confidence level in shear strength assumptions.

² Other strength assumptions may be used if in common usage in the engineering profession.

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4.4.3.4. *Safety factors.* Safety factors for embankment dam stability studies should be based on the ratio of available shear strength to developed shear strength, S_D :

$$S_D = \frac{C}{F.S.} + \frac{\sigma \tan \phi}{F.S.} \quad (1)$$

Where:

C = Cohesion

ϕ = Angle of internal friction

σ = Normal stress

The factors of safety listed in Table 4 are recommended as minimum acceptable. Final accepted factors of safety should depend upon the degree of confidence the investigating engineer has in the engineering data available to him. The consequences of a failure with respect to human life and property damage are important considerations in establishing factors of safety for specific investigations.

4.4.3.5. *Seepage failure.* A critical uncontrolled underseepage or through seepage condition that develops during a rising pool can quickly reduce a structure which was stable under previous conditions, to a total structural failure. The visually confirmed seepage conditions to be avoided are (1) the exit of the phreatic surface on the downstream slope of the dam and (2) development of hydrostatic heads sufficient to create in the area downstream of the dam sand boils that erode materials by the phenomenon known as "piping" and (3)

localized concentrations of seepage along conduits or through pervious zones. The dams most susceptible to seepage problems are those built of or on pervious materials of uniform fine particle size, with no provisions for an internal drainage zone and/or no underseepage controls.

4.4.3.6. *Seepage analyses.* Review and modifications to original seepage design analyses should consider conditions observed in the field inspection and piezometer instrumentation. A seepage analysis should consider the permeability ratios resulting from natural deposition and from compaction placement of materials with appropriate variation between horizontal and vertical permeability. An underseepage analysis of the embankment should provide a critical gradient factor of safety for the maximum head condition of not less than 1.5 in the area downstream of the embankment.

$$F.S. = \frac{i_c}{i} = \frac{H_c/D_b}{H/D_b} = D_b \frac{(\gamma_m - \gamma_w)}{H\gamma_w} \quad (2)$$

Where:

i_c = Critical gradient

i = Design gradient

H = Uplift head at downstream toe of dam measured above tailwater

H_c = The critical uplift

D_b = The thickness of the top impervious blanket at the downstream toe of the dam

γ_m = The estimated saturated unit weight of the material in the top impervious blanket

γ_w = The unit weight of water

Where a factor of safety less than 1.5 is obtained the provision of an underseepage control system is indicated. The factor of safety of 1.5 is a recommended minimum and may be adjusted by the responsible engineer based on the competence of the engineering data.

4.4.4. *Concrete dams and appurtenant structures.*

4.4.4.1. *Requirements for stability.* Concrete dams and structures appurtenant to embankment dams should be capable of resisting overturning, sliding and overstressing with adequate factors of safety for normal and maximum loading conditions.

4.4.4.2. *Loads.* Loadings to be considered in stability analyses include the water load on the upstream face of the dam; the weight of the structure; internal hydrostatic pressures (uplift) within the body of the dam, at the base of the dam and within the foundation; earth and silt loads; ice pressure, seismic and thermal loads, and other loads as applicable. Where tailwater or backwater exists on the downstream side of the structure it should be considered, and assumed uplift pressures should be compatible with drainage provisions and uplift measurements if available. Where applicable, ice pressure should be applied to the contact surface of the structure of normal pool elevation. A unit pressure of not more than 5,000 pounds per square foot should be

used. Normally, ice thickness should not be assumed greater than two feet. Earthquake forces should consist of the inertial forces due to the horizontal acceleration of the dam itself and hydrodynamic forces resulting from the reaction of the reservoir water against the structure. Dynamic water pressures for use in a conventional methods of analysis may be computed by means of the "Westergaard Formula" using the parabolic approximation (H.M. Westergaard, "Water Pressures on Dams During Earthquakes," Trans., ASCE, Vol 98, 1933, pages 418-433), or similar method.

4.4.4.3. *Stresses.* The analysis of concrete stresses should be based on in situ properties of the concrete and foundation. Computed maximum compressive stresses for normal operating conditions in the order of $\frac{1}{3}$ or less of in situ strengths should be satisfactory. Tensile stresses in unreinforced concrete should be acceptable only in locations where cracks will not adversely affect the overall performance and stability of the structure. Foundation stresses should be such as to provide adequate safety against failure of the foundation material under all loading conditions.

4.4.4.4. *Overturning.* A gravity structure should be capable of resisting all overturning forces. It can be considered safe against overturning if the resultant of all combinations of horizontal and vertical forces, excluding earthquake forces, acting above any horizontal plane through the structure or at its base is located within the middle third of the section. When earthquake is included the resultant should fall within the limits of the plane or base, and foundation pressures must be acceptable. When these requirements for location of the resultant are not satisfied the investigating engineer should assess the importance to stability of the deviations.

4.4.4.5. *Sliding.* Sliding of concrete gravity structures and of abutment and foundation rock masses for all types of concrete dams should be evaluated by the shear-friction resistance concept. The available sliding resistance is compared with the driving force which tends to induce sliding to arrive at a sliding stability safety factor. The investigation should be made along all potential sliding paths. The critical path is that plane or combination of planes which offers the least resistance.

4.4.4.5.1. *Sliding resistance.* Sliding resistance is a function of the unit shearing strength at no normal load (cohesion) and the angle of friction on a potential failure surface. It is determined by computing the maximum horizontal driving force which could be resisted along the sliding path under investigation. The following general formula is obtained from the principles of statics and may be derived by resolving forces parallel and perpendicular to the sliding plane:

$$R_R = V \tan (\phi + \alpha) + \frac{cA}{\cos \alpha (1 - \tan \phi \tan \alpha)} \quad (3)$$

Where:

R_R = Sliding Resistance (maximum horizontal driving force which can be resisted by the critical path)

ϕ = Angle of internal friction of foundation material or, where applicable, angle of sliding friction

V = Summation of vertical forces (including uplift)

c = Unit shearing strength at zero normal loading along potential failure plane

A = Area of potential failure plane developing unit shear strength "c"

α = Angle between inclined plane and horizontal (positive for uphill sliding)

For sliding downhill the angle α is negative and Equation (1) becomes:

$$R_R = V \tan (\phi - \alpha) + \frac{cA}{\cos \alpha (1 + \tan \phi \tan \alpha)} \quad (4)$$

When the plane of investigation is horizontal, and the angle α is zero and Equation (1) reduced to the following:

$$R_R = V \tan \phi + cA \quad (5)$$

4.4.4.5.2. *Downstream resistance.* When the base of a concrete structure is embedded in rock or the potential failure plane lies below the base, the passive resistance of the downstream layer of rock may sometimes be utilized for sliding resistance. Rock that may be subjected to high velocity water scouring should not be used. The magnitude of the downstream resistance is the lesser of (a) the shearing resistance along the continuation of the potential sliding plane until it daylights or (b) the resistance available from the downstream rock wedge along an inclined plane. The theoretical resistance offered by the passive wedge can be computed by a formula equivalent to formula (3):

$$P_p = W \tan (\phi + \alpha) + \frac{cA}{\cos \alpha (1 - \tan \phi \tan \alpha)} \quad (6)$$

Where:

P_p = Passive resistance of rock wedge

W = Weight (buoyant weight if applicable) of downstream rock wedge above inclined plane of resistance, plus any superimposed loads

ϕ = Angle of internal friction or, if applicable, angle of sliding friction

α = Angle between inclined failure plane and horizontal

c = Unit shearing strength at zero normal load along failure plane

A = Area of inclined plane of resistance

When considering cross-bed shear through a relatively shallow, competent rock strut, without adverse jointing or faulting, W and α may be taken at zero and 45°, respectively, and an estimate of passive wedge resistance per unit width obtained by the following equation:

$$P_p = 2 cD \quad (7)$$

Where:

D = Thickness of the rock strut

4.4.4.5.3. *Safety factor.* The shear-friction safety factor is obtained by dividing the resistance R_R by H, the summation of horizontal service loads to be applied to the structure:

$$S_{S-f} = \frac{R_R}{H} \quad (8)$$

When the downstream passive wedge contributes to the sliding resistance, the shear friction safety factor formula becomes:

$$S_{S-f} = \frac{R_R + P_P}{H} \quad (9)$$

The above direct superimposition of passive wedge resistance is valid only if shearing rigidities of the foundation components are similar. Also, the compressive strength and buckling resistance of the downstream rock layer must be sufficient to develop the wedge resistance. For example, a foundation with closely spaced, near horizontal, relatively weak seams might not contain sufficient buckling strength to develop the magnitude of wedge resistance computed from the cross-bed shear strength. In this case wedge resistance should not be assumed without resorting to special treatment (such as installing foundation anchors). Computed sliding safety factors approximating 3 or more for all loading conditions without earthquake, and 1.5 including earthquake, should indicate satisfactory stability, depending upon the reliability of the strength parameters used in the analyses. In some cases when the results of comprehensive foundation studies are available, smaller safety factors may be acceptable. The selection of shear strength parameters should be fully substantiated. The bases for any assumptions; the results of applicable testing, studies and investigations; and all pre-existing, pertinent data should be reported and evaluated.

Chapter 5—Reports

5.1. *General.* This chapter outlines the procedures for reporting the results of the technical investigations. Hazardous conditions should be reported immediately upon detection to the owner of the dam, the Governor of the State in which the dam is located and the appropriate regulatory agency without delay for preparation of the formal report.

5.2. *Preparation of report.* A formal report should be prepared for each dam investigated for submission to the regulatory agency and the owner of the dam. Each report should contain the information indicated in the following paragraphs. The signature and registration identification of the professional engineer who directed the investigation and who was responsible for evaluation of the dam should be included in the report.

5.2.1. *Phase I reports.* Phase I reports should contain the following information:

5.2.1.1. Description of dam including regional vicinity map showing location and plans, elevations and sections showing the essential project features and the size and hazard potential classifications.

5.2.1.2. Summary of existing engineering data, including geologic maps and information.

5.2.1.3. Results of the visual inspection of each project feature including photographs and drawings to minimize descriptions.

5.2.1.4. Evaluation of operational adequacy of the reservoir regulation plan and maintenance of the dam and operating facilities and features that pertain to the safety of the dam.

5.2.1.5. Description of any warning system in effect.

5.2.1.6. Evaluation of the hydraulic and hydrologic assumptions and structural stability.

5.2.1.7. An assessment of the general condition of the dam with respect to safety based upon the findings of the visual inspection and review of engineering data. Where data on the original design indicate significant departure from or non-conformance with guidelines contained herein, the engineer-in-charge of the investigation will give his opinion of the significance, with regard to safety, of such factors. Any additional studies, investigations and analyses considered essential to assessment of the safety of the dam should be listed, together with an opinion about the urgency of such additional work.

5.2.1.8. Indicate alternative possible remedial measures or revisions in operating and maintenance procedures which may (subject to further evaluation) correct deficiencies and hazardous conditions found during the investigation.

5.2.2. *Phase II reports.* Phase II reports should describe the detailed investigations and should supplement Phase I reports. They should contain the following information:

5.2.2.1. Summary of additional engineering data obtained to determine the hydraulic and hydrologic capabilities and/or structural stability.

5.2.2.2. Results of all additional studies, investigations, and analyses performed.

5.2.2.3. Technical assessment of dam safety including deficiencies and hazardous conditions found to exist.

5.2.2.4. Indicate alternative possible remedial measures or revision in maintenance and operating procedures which may (subject to further evaluation) correct deficiencies and hazardous conditions found during the investigation.

From TM 5-809-10/NAVFAC P-355/AFM 88-3, Chapter 13, April 1973

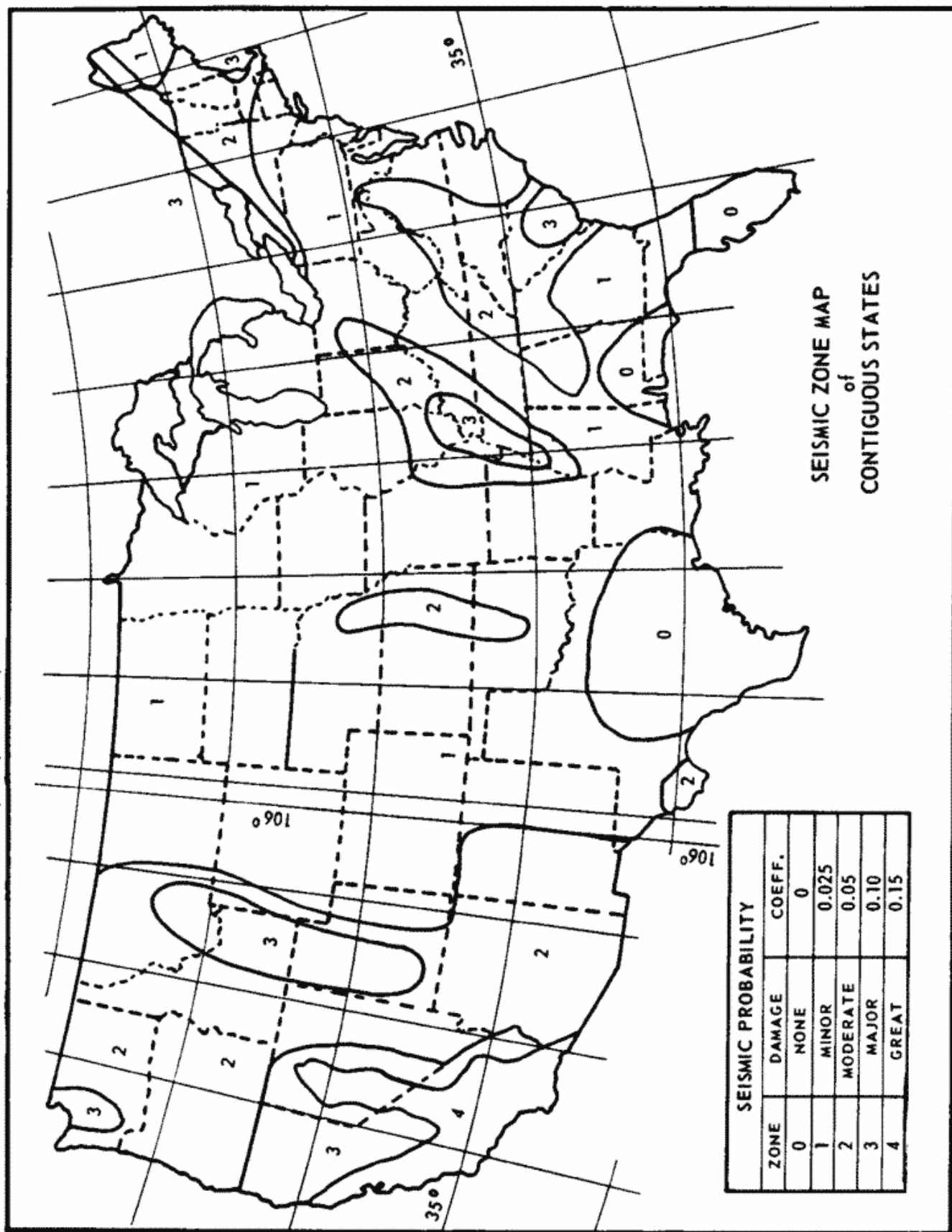
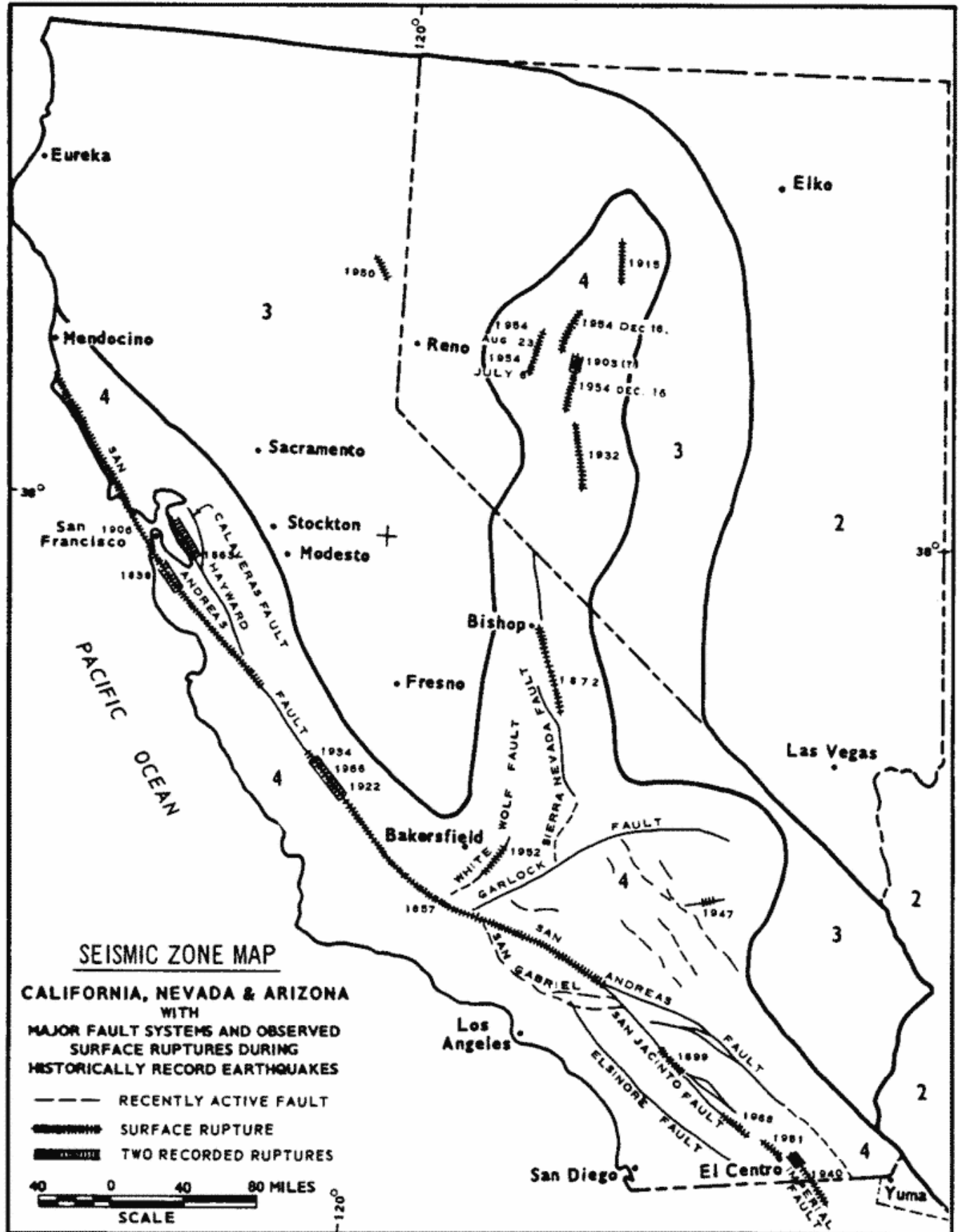


Figure 1

From TM 5-809-10/ NAVFAC P-355/AFM 88-3, Chapter 13; April 1973



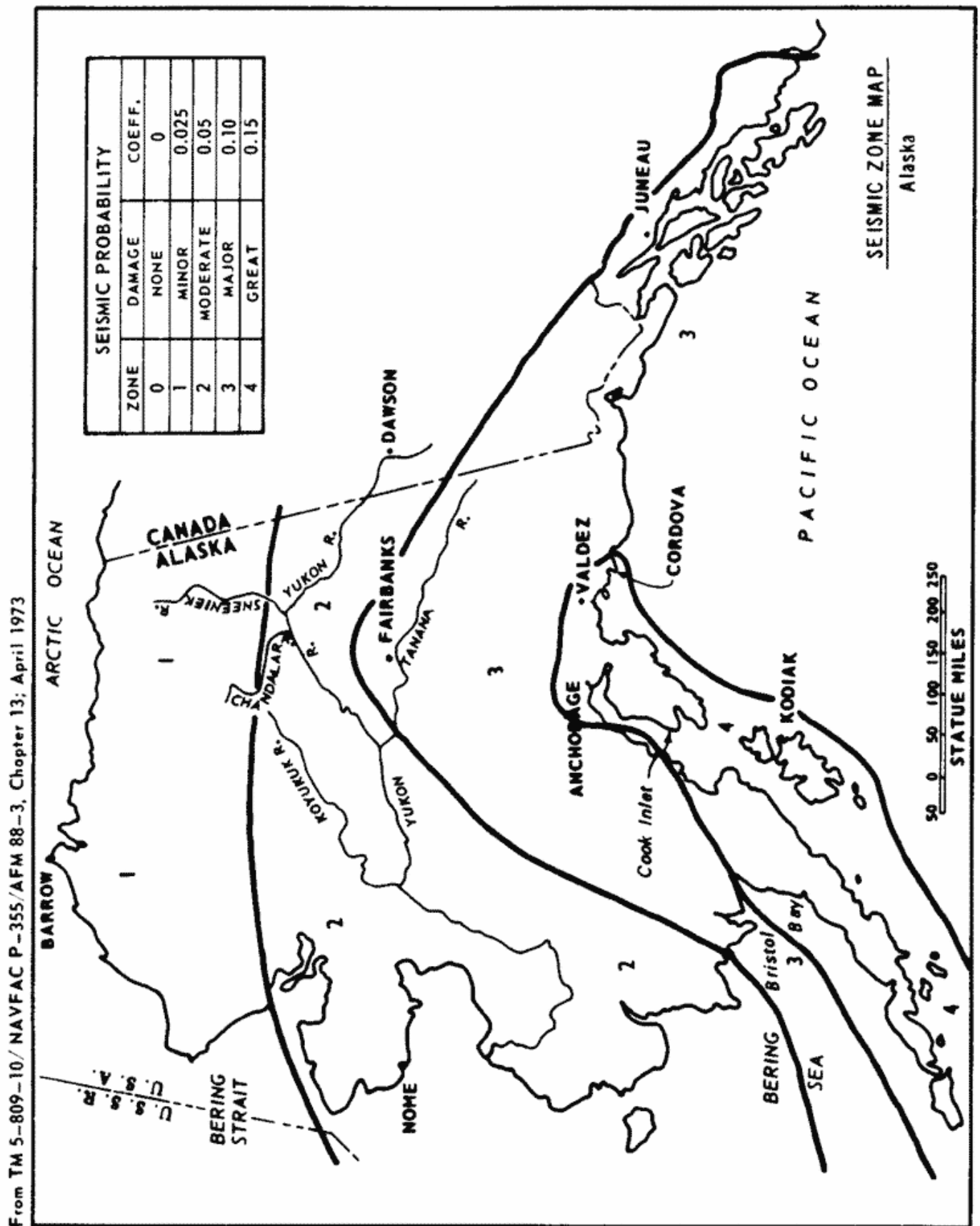


Figure 3

From TM 5-809-10, NAVFAC P-355, AFM 88-3, Chapter 13, April 1973

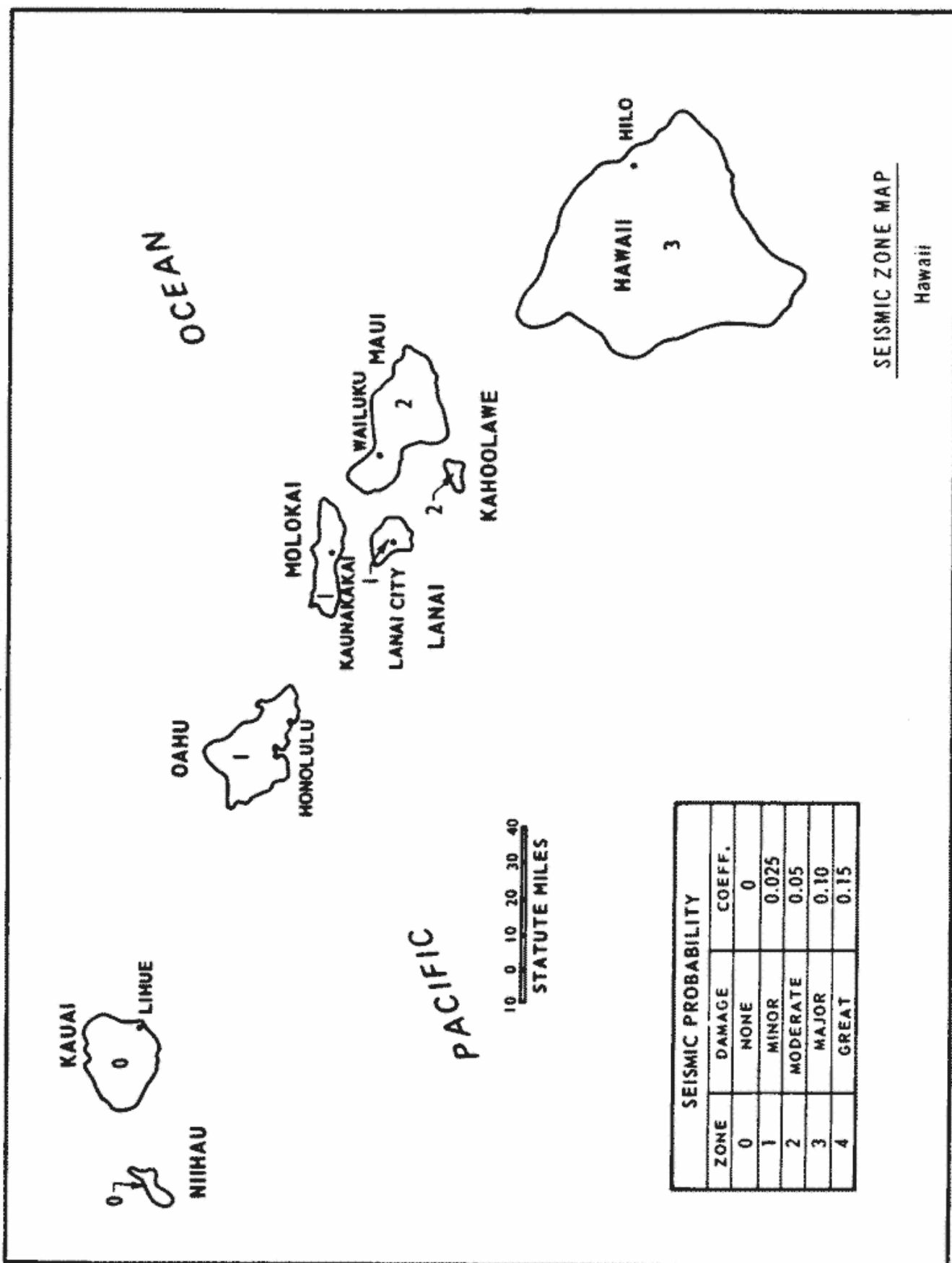


Figure 4

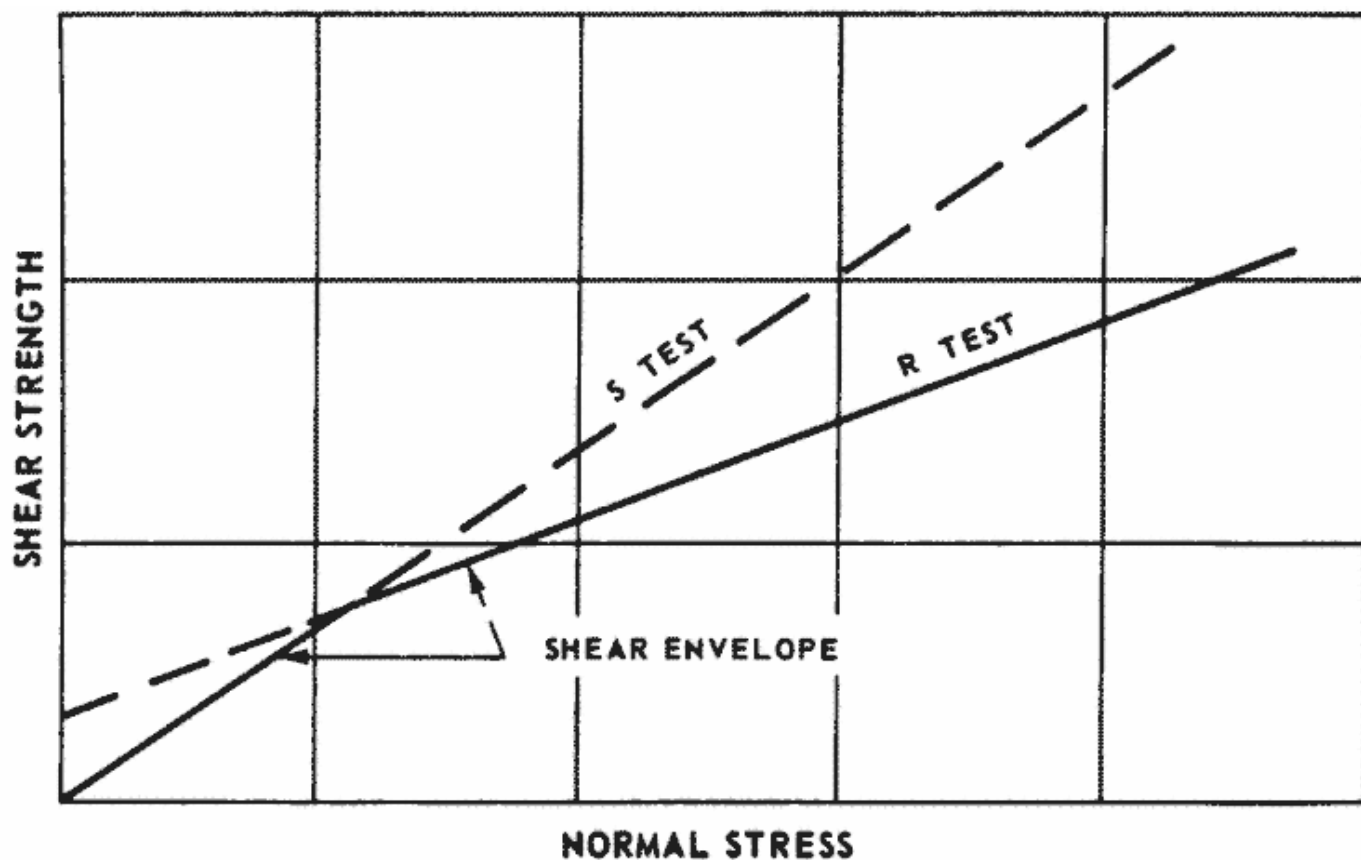
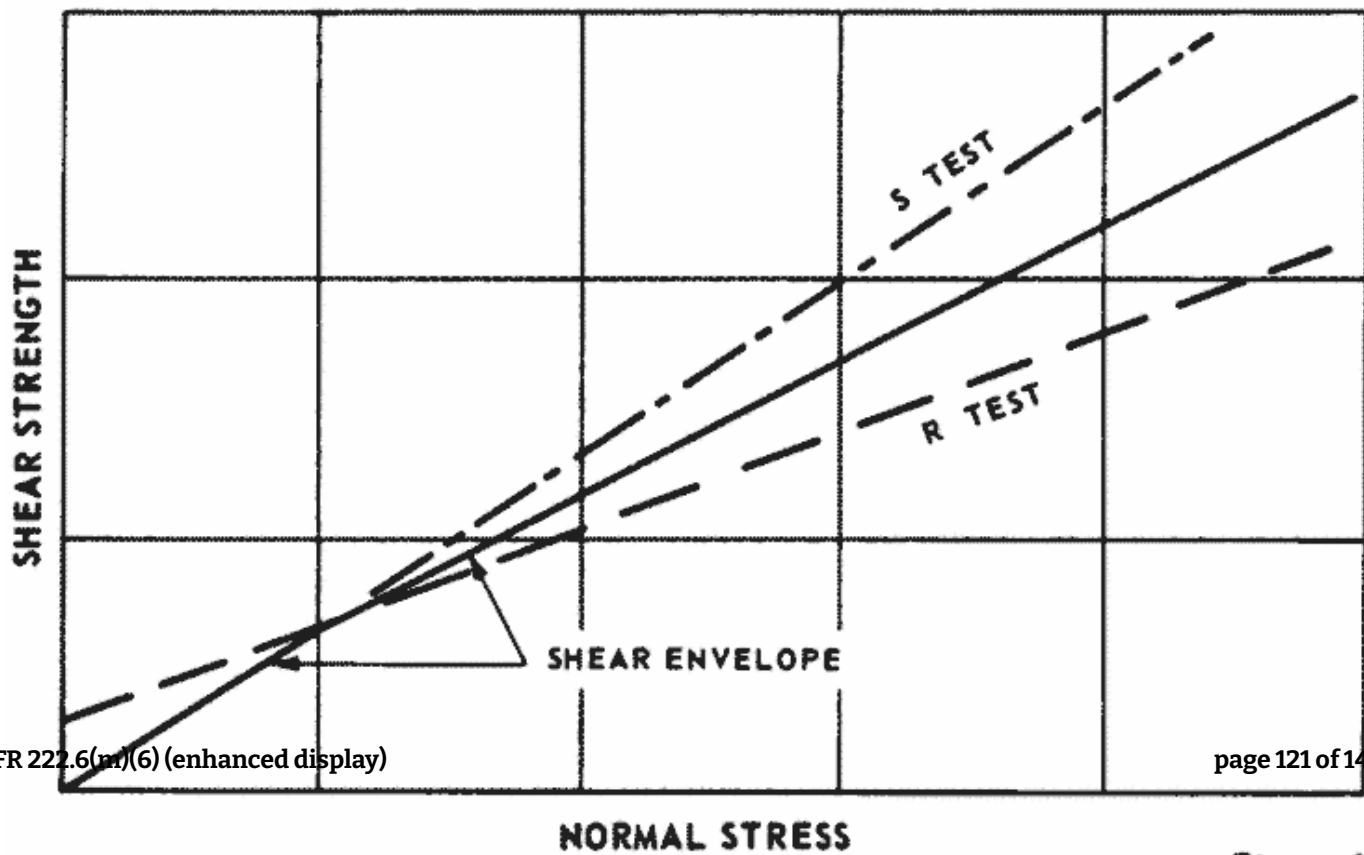


Figure 5

SHEAR ENVELOPE FOR CASE I



Appendix I to App. D to § 222.6—Engineering Data

This appendix lists engineering data which should be collected from project records and, to the extent available, included in the Phase I investigation report. The list is intended to serve as a checklist and not to establish rigid data requirements. Such a compilation should also facilitate future inspections and investigations. Only data readily available will be included in Phase I reports, but data lacking and deemed necessary for an adequate safety evaluation should be identified.

1. *General Project Data.*

- a. *Regional Vicinity Map* showing the location of the dam, the upstream drainage area and the downstream area subject to potential damage due to failure of the dam and misoperation or failure of the operating equipment.
- b. *As-Built Drawings* indicating plans, elevations and sections of the dam and appurtenant structures including the details of the discharge facilities such as outlet works, limited service and emergency spillways, flashboards, fuse plugs and operating equipment.

2. *Hydrologic and Hydraulic Data* including the following:

- a. Drainage area and basin runoff characteristics (indicating pending changes).
- b. Elevation of top of conservation pool or normal upper retention water surface elevation, as applicable (base level of any flood impoundment).
- c. Storage capacity including dead or inactive storage, corresponding to top of conservation or normal upper retention level (cumulative, excluding flood control and surcharge storage).
- d. Elevation of the top of flood control pool.
- e. Storage capacity of flood control zone (incremental).
- f. Elevation of maximum design pool (corresponding to top of surcharge storage or spillway design flood).
- g. Storage capacity of surcharge zone (incremental, above top of flood control pool or, above normal upper retention level if flood control space not provided).
- h. Height of freeboard (distance between maximum design flood water surface and top of dam).
- i. Elevation of top of dam (lowest point of embankment or non-overflow structure).
- j. Elevation of crest, type, width, crest length and location of spillways (number, size and type of gates if controlled).
- k. Type, location, entrance and exit inverts of outlet works and emergency drawdown facilities (number, size and shape of conduits and gates, including penstocks and sluices).

- l. Location, crest elevation, description of invert and abutments (concrete, rock, grass, earth) and length of limited service and emergency spillways.
- m. Location and description of flashboards and fuse plugs, including hydraulic head (pool elevation) and other conditions required for breaching, along with the assumed results of breaching.
- n. Location and top elevation of dikes and floodwalls (overflow and non-overflow) affected by reservoir. Include information on low reaches of reservoir rim.
- o. Type, location, observations and records of hydrometeorological gages appurtenant to the project.
- p. Maximum non-damaging discharge, or negligible damage rate, at potential damage locations downstream.
- 3. *Foundation Data and Geological Features* including logs of borings, geological maps, profiles and cross sections, and reports of foundation treatment.
- 4. *Properties of Embankments and Foundation Materials* including results of laboratory tests, field permeability tests, construction control tests, and assumed design properties for materials.
- 5. *Concrete Properties* including the source and type of aggregate, cement used, mix design data and the results of testing during construction.
- 6. *Electrical and Mechanical Equipment* type and rating of normal and emergency power supplies, hoists, cranes, valves and valve operator, control and alarm systems and other electrical and mechanical equipment and systems that could affect the safe operation of the dam.
- 7. *Construction History* including diversion scheme, construction sequence, pertinent construction problems, alterations, modifications and maintenance repairs.
- 8. *Water Control Plan* including regulation plan under normal conditions and during flood events or other emergency conditions. The availability of dam tenders, means of communication between dam tenders and authority supervising water control, and method of gate operation (manual, automatic, or remote control) should be included. Flood warning systems should be described in sufficient detail to enable assessment of their reduction in the flood hazard potential.
- 9. *Operation Record*.
 - a. Summary of past major flood events including any experiences that presented a serious threat to the safety of the project or to human life or property. The critical project feature, date and duration of event, causative factor, peak inflow and outflow, maximum elevation of water surface, wind and wave factors if significant, issuance of alert or evacuation warnings and adequacy of project feature involved should be included in the summary of past experience of serious threat to the safety of the project.
 - b. Records of performance observations including instrumentation records.
 - c. List of any known deficiencies that pose a threat to the safety of the dam or to human life or property.

d. History of previous failures or deficiencies and pending remedial measures for correcting known deficiencies and the schedule for accomplishing remedial measures should be indicated.

10. *Earthquake History* including a summary of the seismic data of significant recorded earthquakes in the vicinity of the dam and information on major damage in the vicinity of the dam from both recorded and unrecorded earthquakes. Regional geologic maps and other documents showing fault locations should be collected.

11. *Inspection History* including the results of the last safety inspection, the organization that performed the inspection, the date inspection performed and the authority for conducting the inspection.

12. *Principal Design Assumptions and Analyses.*

a. *Hydrologic and Hydraulic Determinations.*

(1) Quantity, time and area distribution, and reference source of depth-area-duration data of spillway design storm precipitation (point precipitation if applicable).

(2) Maximum design flood inflow hydrograph including loss rates (initial and average for design flood conditions) and time of runoff concentration of reservoir watershed (peak inflow only when applicable).

(3) Maximum design flood outflow hydrograph (maximum outflow only when applicable).

(4) Discharge-frequency relationship, preferably at damsite, including estimated frequency of spillway design flood for small dams, when appropriate.

(5) Reservoir area and storage capacity versus water surface elevation (table or curves).

(6) Rating curves (free flow and partial gate openings) for all discharge facilities contributing to the maximum design flood outflow hydrograph. Also a composite-rating of all contributing facilities, if appropriate.

(7) Tailwater rating curve immediately below damsite including elevation corresponding to maximum design flood discharge and approximate nondamaging channel capacity.

(8) Hydrologic map of watershed above damsite including reservoir area, watercourse, elevation contours, and principal stream-flow and precipitation gaging stations.

b. *Stability and Stress Analysis* of the dam, spillway and appurtenant structures and features including the assumed properties of materials and all pertinent applied loads.

c. *Seepage and Settlement Analyses.* The determination of distribution, direction and magnitude of seepage forces and the design and construction measures for their control. Settlement estimates and steps adopted to compensate for total settlement and to minimize differential settlements.

Appendix II to App. D to § 222.6—Inspection Items

This appendix provides guidance for performing field inspections and may serve as the basis for developing a detailed checklist for each dam.

1. *Concrete Structures in General.*

a. *Concrete Surfaces.* The condition of the concrete surfaces should be examined to evaluate the deterioration and continuing serviceability of the concrete. Descriptions of concrete conditions should conform with the appendix to "Guide for Making a Condition Survey of Concrete in Service," American Concrete Institute (ACI) Journal, Proceedings Vol. 65, No. 11, November 1968, page 905-918.

b. *Structural Cracking.* Concrete structures should be examined for structural cracking resulting from overstress due to applied loads, shrinkage and temperature effects or differential movements.

c. *Movement—Horizontal and Vertical Alignment.* Concrete structures should be examined for evidence of any abnormal settlements, heaving, deflections, or lateral movements.

d. *Junctions.* The conditions at the junctions of the structure with abutments or embankments should be determined.

e. *Drains—Foundation, Joint, Face.* All drains should be examined to determine that they are capable of performing their design function.

f. *Water Passages.* All water passages and other concrete surfaces subject to running water should be examined for erosion, cavitation, obstructions, leakage or significant structural cracks.

g. *Seepage or Leakage.* The faces, abutments and toes of the concrete structures should be examined for evidence of seepage or abnormal leakage, and records of flow of downstream springs reviewed for variation with reservoir pool level. The sources of seepage should be determined if possible.

h. *Monolith Joints—Construction Joints.* All monolith and construction joints should be examined to determine the condition of the joint and filler material, any movement of joints, or any indication of distress or leakage.

i. *Foundation.* Foundation should be examined for damage or possible undermining of the downstream toe.

j. *Abutments.* The abutments should be examined for sign of instability or excessive weathering.

2. *Embankment Structures.*

a. *Settlement.* The embankments and downstream toe areas should be examined for any evidence of localized or overall settlement, depressions or sink holes.

b. *Slope Stability.* Embankment slopes should be examined for irregularities in alignment and variances from smooth uniform slopes, unusual changes from original crest alignment and elevation, evidence of movement at or beyond the toe, and surface cracks which indicate movement.

c. *Seepage*. The downstream face of abutments, embankment slopes and toes, embankment—structure contacts, and the downstream valley areas should be examined for evidence of existing or past seepage. The sources of seepage should be investigated to determine cause and potential severity to dam safety under all operating conditions. The presence of animal burrows and tree growth on slopes which might cause detrimental seepage should be examined.

d. *Drainage Systems*. All drainage systems should be examined to determine whether the systems can freely pass discharge and that the discharge water is not carrying embankment or foundation material. Systems used to monitor drainage should be examined to assure they are operational and functioning properly.

e. *Slope Protection*. The slope protection should be examined for erosion-formed gullies and wave-formed notches and benches that have reduced the embankment cross-section or exposed less wave resistant materials. The adequacy of slope protection against waves, currents, and surface runoff that may occur at the site should be evaluated. The condition of vegetative cover should be evaluated where pertinent.

3. *Spillway Structures*. Examination should be made of the structures and features including bulkheads, flashboards, and fuse plugs of all service and auxiliary spillways which serve as principal or emergency spillways for any condition which may impose operational constraints on the functioning of the spillway.

a. *Control Gates and Operating Machinery*. The structural members, connections, hoists, cables and operating machinery and the adequacy of normal and emergency power supplies should be examined and tested to determine the structural integrity and verify the operational adequacy of the equipment. Where cranes are intended to be used for handling gates and bulkheads, the availability, capacity and condition of the cranes and lifting beams should be investigated. Operation of control systems and protective and alarm devices such as limit switches, sump high water alarms and drainage pumps should be investigated.

b. *Unlined Saddle Spillways*. Unlined saddle spillways should be examined for evidence of erosion and any conditions which may impose constraints on the functioning of the spillway. The ability of the spillway to resist erosion due to operation and the potential hazard to the safety of the dam from such operation should be determined.

c. *Approach and Outlet Channels*. The approach and outlet channels should be examined for any conditions which may impose constraints on the functioning of the spillway and present a potential hazard to the safety of the dam.

d. *Stilling Basin (Energy Dissipators)*. Stilling basins including baffles, flip buckets or other energy dissipators should be examined for any conditions which may pose constraints on the ability of the stilling basin to prevent downstream scour or erosion which may create or present a potential hazard to the safety of the dam. The existing condition of the channel downstream of the stilling basin should be determined.

4. *Outlet Works*. The outlet works examination should include all structures and features designed to release reservoir water below the spillway crest through or around the dam.

a. *Intake Structure*. The structure and all features should be examined for any conditions which may impose operational constraints on the outlet works. Entrances to intake structure should be examined for conditions such as silt or debris accumulation which may reduce the discharge capabilities of the outlet works.

b. *Operating and Emergency Control Gates*. The structural members, connections, guides, hoists, cables and operating machinery including the adequacy of normal and emergency power supplies should be examined and tested to determine the structural integrity and verify the operational adequacy of the operating and emergency gates, valves, bulkheads, and other equipment.

c. *Conduits, Sluices, Water Passages, Etc.* The interior surfaces of conduits should be examined for erosion, corrosion, cavitation, cracks, joint separation and leakage at cracks or joints.

d. *Stilling Basin (Energy Dissipator)*. The stilling basin or other energy dissipator should be examined for conditions which may impose any constraints on the ability of the stilling basin to prevent downstream scour or erosion which may create or present a potential hazard to the safety of the dam. The existing condition of the channel downstream of the stilling basin should be determined by soundings.

e. *Approach and Outlet Channels*. The approach and outlet channels should be examined for any conditions which may impose constraints on the functioning of the discharge facilities of the outlet works, or present a hazard to the safety of the dam.

f. *Drawdown Facilities*. Facilities provided for drawdown of the reservoir to avert impending failure of the dam or to facilitate repairs in the event of stability or foundation problems should be examined for any conditions which may impose constraints on their functioning as planned.

5. *Safety and Performance Instrumentation*. Instruments which have been installed to measure behavior of the structures should be examined for proper functioning. The available records and readings of installed instruments should be reviewed to detect any unusual performance of the instruments or evidence of unusual performance or distress of the structure. The adequacy of the installed instrumentation to measure the performance and safety of the dam should be determined.

a. *Headwater and Tailwater Gages*. The existing records of the headwater and tailwater gages should be examined to determine the relationship between other instrumentation measurements such as stream flow, uplift pressures, alignment, and drainage system discharge with the upper and lower water surface elevations.

b. *Horizontal and Vertical Alignment Instrumentation (Concrete Structures)*. The existing records of alignment and elevation surveys and measurements from inclinometers, inverted plumb bobs, gage points across cracks and joints, or other devices should be examined to determine any change from the original position of the structures.

c. *Horizontal and Vertical Movement, Consolidation, and Pore-Water Pressure Instrumentation (Embankment Structures)*. The existing records of measurements from settlement plates or gages, surface reference marks, slope indicators and other devices should be examined to determine the movement history of the embankment. Existing piezometer measurements should be examined to determine if the pore-water pressures in the embankment and foundation would under given conditions impair the safety of the dam.

d. *Uplift Instrumentation*. The existing records of uplift measurements should be examined to determine if the uplift pressures for the maximum pool would impair the safety of the dam.

e. *Drainage System Instrumentation.* The existing records of measurements of the drainage system flow should be examined to establish the normal relationship between pool elevations and discharge quantities and any changes that have occurred in this relationship during the history of the project.

f. *Seismic Instrumentation.* The existing records of seismic instrumentation should be examined to determine the seismic activity in the area and the response of the structures of past earthquakes.

6. *Reservoir.* The following features of the reservoir should be examined to determine to what extent the water impounded by the dam would constitute a danger to the safety of the dam or a hazard to human life or property.

a. *Shore line.* The land forms around the reservoir should be examined for indications of major active or inactive landslide areas and to determine susceptibility of bedrock stratigraphy to massive landslides of sufficient magnitude to significantly reduce reservoir capacity or create waves that might overtop the dam.

b. *Sedimentation.* The reservoir and drainage area should be examined for excessive sedimentation or recent developments in the drainage basin which could cause a sudden increase in sediment load thereby reducing the reservoir capacity with attendant increase in maximum outflow and maximum pool elevation.

c. *Potential Upstream Hazard Areas.* The reservoir area should be examined for features subject to potential backwater flooding resulting in loss of human life or property at reservoir levels up to the maximum water storage capacity including any surcharge storage.

d. *Watershed Runoff Potential.* The drainage basin should be examined for any extensive alterations to the surface of the drainage basin such as changed agriculture practices, timber clearing, railroad or highway construction or real estate developments that might extensively affect the runoff characteristics. Upstream projects that could have impact on the safety of the dam should be identified.

7. *Downstream Channel.* The channel immediately downstream of the dam should be examined for conditions which might impose any constraints on the operation of the dam or present any hazards to the safety of the dam. Development of the potential flooded area downstream of the dam should be assessed for compatibility with the hazard classification.

8. *Operation and Maintenance Features.*

a. *Reservoir Regulation Plan.* The actual practices in regulating the reservoir and discharges under normal and emergency conditions should be examined to determine if they comply with the designed reservoir regulation plan and to assure that they do not constitute a danger to the safety of the dam or to human life or property.

b. *Maintenance.* The maintenance of the operating facilities and features that pertain to the safety of the dam should be examined to determine the adequacy and quality of the maintenance procedures followed in maintaining the dam and facilities in safe operating condition.

Appendix III to App. D to § 222.6—Pub. L. 92-367



Public Law 92-367
92nd Congress, H. R. 15951
August 8, 1972

An Act

To authorize the Secretary of the Army to undertake a national program of inspection of dams.

Be it enacted by the Senate and House of Representatives of the United States of America in Congress assembled, That the term "dam" as used in this Act means any artificial barrier, including appurtenant works, which impounds or diverts water, and which (1) is twenty-five feet or more in height from the natural bed of the stream or watercourse measured at the downstream toe of the barrier, or from the lowest elevation of the outside limit of the barrier, if it is not across a stream channel or watercourse, to the maximum water storage elevation or (2) has an impounding capacity at maximum water storage elevation of fifty acre-feet or more. This Act does not apply to any such barrier which is not in excess of six feet in height, regardless of storage capacity or which has a storage capacity at maximum water storage elevation not in excess of fifteen acre-feet, regardless of height.

National dam inspection program.
"Dam."

SEC. 2. As soon as practicable, the Secretary of the Army, acting through the Chief of Engineers, shall carry out a national program of inspection of dams for the purpose of protecting human life and property. All dams in the United States shall be inspected by the Secretary except (1) dams under the jurisdiction of the Bureau of Reclamation, the Tennessee Valley Authority, or the International Boundary and Water Commission, (2) dams which have been constructed pursuant to licenses issued under the authority of the Federal Power Act, (3) dams which have been inspected within the twelve-month period immediately prior to the enactment of this Act by a State agency and which the Governor of such State requests be excluded from inspection, and (4) dams which the Secretary of the Army determines do not pose any threat to human life or property. The Secretary may inspect dams which have been licensed under the Federal Power Act upon request of the Federal Power Commission and dams under the jurisdiction of the International Boundary and Water Commission upon request of such Commission.

Army, authorization.

Exceptions.

41 Stat. 1063;
49 Stat. 863.
16 USC 791a.

86 STAT. 506
86 STAT. 507

SEC. 3. As soon as practicable after inspection of a dam, the Secretary shall notify the Governor of the State in which such dam is located the results of such investigation. The Secretary shall immediately notify the Governor of any hazardous conditions found during an inspection. The Secretary shall provide advice to the Governor, upon request, relating to timely remedial measures necessary to mitigate or obviate any hazardous conditions found during an inspection.

Notice to Governors.

SEC. 4. For the purpose of determining whether a dam (including the waters impounded by such dam) constitutes a danger to human life or property, the Secretary shall take into consideration the possibility that the dam might be endangered by overtopping, seepage, settlement, erosion, sediment, cracking, earth movement, earthquakes, failure of bulkheads, flashboards, gates on conduits, or other conditions which exist or which might occur in any area in the vicinity of the dam.

page 130 of 142

SEC. 5. The Secretary shall report to the Congress on or before

Report to Congress

86 STAT. 507 Pub. Law 92-367 - 2 - August 8, 1972

Liability. (3) recommendations for a comprehensive national program for the inspection, and regulation for safety purpose of dams of the Nation, and the respective responsibilities which should be assumed by Federal, State, and local governments and by public and private interests.

SEC. 6. Nothing contained in this Act and no action or failure to act under this Act shall be construed (1) to create any liability in the United States or its officers or employees for the recovery of damages caused by such action or failure to act; or (2) to relieve an owner or operator of a dam of the legal duties, obligations, or liabilities incident to the ownership or operation of the dam.

Approved August 8, 1972.

LEGISLATIVE HISTORY:

HOUSE REPORT No. 92-1232 (Comm. on Public Works).
CONGRESSIONAL RECORD, Vol. 118 (1972):

July 24, considered and passed House.

July 25, considered and passed Senate.

WEEKLY COMPILATION OF PRESIDENTIAL DOCUMENTS, Vol. 8, No. 33:

Aug. 9, Presidential statement.

Appendix E to § 222.6—Suggested Outline

INSPECTION REPORT—NATIONAL DAM INSPECTION PROGRAM (RCS-DAEN-CWE-17 AND OMB No.
49-R0421)

Title Sheet

Name of Dam

ID Number from Inventory

State, County and River or Stream where dam is located

Owner

Size and Hazard Classification

Names of Inspectors

Names of Review Board

Approval Signature of District Engineer

TABLE OF CONTENTS

GENERAL ASSESSMENT

Give brief assessment of general condition of dam with respect to safety, including a listing of deficiencies, and recommendations indicating degree of urgency.

1. INTRODUCTION

- a. Authority
- b. Purpose and Scope of Inspection

2. PROJECT INFORMATION

- a. Site Information
- b. Description of Structures—Dam, Outlet, Spillway and other principal features.
- c. Purpose of Dam
- d. Design, Construction and Operating History

3. FIELD INSPECTION

Briefly describe physical condition of the dam and appurtenant structures as they were observed during the field inspection. (If field inspection form is appended, only present summary.) Describe operational procedures, including any warning system, condition of operating equipment, and provision for emergency procedures. Describe any pertinent observations of the reservoir area and downstream channel adjacent to dam.

4. EVALUATION

- a. Structural and Geotechnical

(1) General

(2) Embankment and/or Foundation Condition

(3) Stability—Briefly discuss pertinent information such as design, construction and operating records. Assess stability under maximum loading on basis of the record data, together with observations of field inspection and results of any additional, brief calculations performed by inspectors. If additional, detailed stability analyses are considered necessary, recommend that the owner engage a qualified engineer or firm to provide the analysis.

b. Hydrologic and Hydraulic

(1) Spillway Adequacy—Briefly describe pertinent record information such as hydrologic and hydraulic design data, flood of record, and previous analyses. Describe any hydraulic and hydrologic analyses made for this inspection. Present conclusion with respect to adequacy of spillway to pass the recommended spillway design flood without overtopping dam. If overtopping would occur, and if available from the type of analysis used, give maximum depth over top of dam and duration of overtopping, assuming the dam does not fail. Also indicate the largest flood, as a percentage of the probable maximum flood which can be passed without overtopping.

(2) Effects of overtopping—If dam is overtopped by the recommended spillway design flood, provide assessment as to whether or not dam would likely fail, and if, in case of failure, the hazard to loss of life downstream of the dam would be substantially increased over that which would exist without failure. If information upon which to base a reasonable assessment is insufficient, so state and describe the needed data, and recommend that the necessary studies be performed by engineers engaged by the owner.

c. Operation and Maintenance

Assess operating equipment and procedures, emergency power for gate operation, and Emergency Action Plan. Assess quality of maintenance as it pertains to dam safety.

5. CONCLUSIONS

Provide conclusions on condition of dam and list all deficiencies. If dam is considered unsafe, so state and give reason.

6. RECOMMENDATIONS

List all recommended actions, including additional studies, installation of new surveillance procedures and devices, development of Emergency Action Plans, and remedial work. Recommend that a qualified engineering firm be retained to accomplish any recommended additional investigations and studies and also to design and supervise remedial works.

APPENDIXES

a. Inspection Checklist (if available)

b. Other Illustrations as follows:

- (1) Include a map showing location of the dam. Usually a portion of a USGS quadrangle sheet can be used which will show the topography of the area, location of the dam, extent of the lake and drainage basin, and perhaps indicate the downstream development.
- (2) If available, include a plan and section of the dam.
- (3) General photographs of the dam and downstream channel should be included.
- (4) Color photographs of deficiencies should be included. These should be held to the minimum required to illustrate the deficiencies.
- (5) Available engineering data including Hydrologic/Hydraulic calculation and physical test results that might be available.

Appendix F to § 222.6

INSTRUCTIONS FOR UNSAFE DAM DATA SHEET (RCS-DAEN-CWE-17 AND OMB No. 49-R0421)

The indicated information shall be provided in the format shown on Pg F-3 for each dam assessed to be unsafe during the reporting period. A separate data sheet should be provided for each unsafe dam. The information supplied should conform to the following.

- a. *Name*—Name of dam.
- b. *Id. No.*—Dam inventory identity number.
- c. *Location*—List state county, river or stream and nearest D/S city or town where the dam is located.
- d. *Height*—Maximum hydraulic height of dam.
- e. *Maximum Impoundment Capacity*—List the capacity of the reservoir at maximum attainable water surface elevation including any surcharge loading.
- f. *Type*—Type of dam, i.e., earth, rockfill, gravity, combination earth-gravity, etc.
- g. *Owner*—Owner of dam.
- h. *Date Governor Notified of Unsafe Condition*—The date and method of notification, such as, by telegram, letter, report, etc.
- i. *Condition of Dam Resulting in Unsafe Assessment*—Brief description of the deficiencies discovered which resulted in the unsafe assessment.
- j. *Description of Danger Involved*—Downstream (D/S) hazard potential category and a brief description of the danger involved.

k. *Recommendations Given to Governor*—Brief description of the actions recommended to Governor at time of notification of unsafe condition to eliminate or reduce the danger.

l. *Urgency Category*—State whether the unsafe condition of the dam is an emergency or non-emergency situation. An emergency situation should be considered to exist if the failure of the dam is judged to be imminent and requires immediate action to eliminate or reduce the danger.

m. *Emergency Actions Taken*—In case of an emergency situation, list the actions taken. For non-emergency situation, put NA for “not applicable.”

n. *Remedial Action Taken*—For non-emergency situations list remedial actions taken.

o. *Remarks*—For other pertinent information.

Format for Unsafe Dam Data Sheet (RCS-DAEN-CWE-17 and OMB No. 49-R0421

NATIONAL PROGRAM OF INSPECTION OF NON-FEDERAL DAMS—UNSAFE DAM DATA SHEET

a. *Name:*

b. *Type:*

c. *Height:*

d. *Id. No.*

e. *Location:*

State: County:

Nearest D/S City, Town or Village:

River or Stream:

f. *Owner:*

g. *Date Governor Notified of Unsafe Condition:*

h. *Condition of Dam Resulting in Unsafe Assessment:*

i. *Description of Danger Involved:*

j. *Recommendations Given to Governor:*

k. *Urgency Category:*

l. *Emergency Actions Taken:*

m. *Remarks:*

Appendix G to § 222.6

NATIONAL PROGRAM FOR INSPECTION OF NON-FEDERAL DAMS—MONTHLY PROGRESS REPORT (RCS-DAEN-CWE-19)

l. *Instructions for Monthly Progress Report.* The indicated information shall be provided in the format shown on page G-2.

1. Division Reporting:

2. Date:

3. Information Required for Each State Regarding Total Number of Inspections Performed (AE Inspections included) (Cumulative):

3.1. Number of Inspections Initiated by on-site inspection or the review of engineering data from project records.^[1]

3.2. Number of Inspections Completed (The number of inspection reports which have been submitted to the District Engineer for review and approval).

3.3 Number of Dams Reported to the Governor as Unsafe.^[2]

3.4. Number of Approved Inspection Reports Submitted to the Governor.

4. Information Required for Each State Regarding Inspections Performed Under AE Contracts (Cumulative):

4.1. Number of Dams Contracted for Inspection by AE's with State or Corps.

4.2. Number of Inspections Initiated by AE's by on-site inspection or the review of engineering data from project records.^[1]

^[1] Each of the initiated inspections reported should be planned for completion within a reasonable period of time (30 days.)

^[2] An unsafe dam is defined as a dam with deficiencies of such a nature that if not corrected could result in the failure of the dam with subsequent loss of lives or substantial property damage.

^[1] See footnote on previous page.

4.3. Number of Inspections Completed by AE's (The number of inspection reports which have been submitted to the District Engineer for review and approval).

4.4. Number of Approved Inspection Reports Prepared by AE's Submitted to the Governor.

II. Formation for Monthly Progress Report.

NATIONAL PROGRAM FOR INSPECTION OF NON-FEDERAL DAMS—MONTHLY PROGRESS REPORT

1. Division Reporting:

2. Date:

3. Information Required for Each State Regarding Total Number of Inspections Performed (Cumulative):

State	Inspection Initiated (3.1)	Inspection Completed (3.2)	Unsafe Dams Reported (3.3)	Approved Reports (3.4)
Total				

4. Information Required for Each State Regarding Inspections Performed Under A/E Contracts (Cumulative):

State	Dams Under A/E Contract (4.1)	A/E Inspections Initiated (4.2)	A/E Inspections Completed (4.3)	A/E Reports Approved (4.4)
Totals				

Appendix H to § 222.6

SUGGESTED SCOPE OF WORK CONTRACT FOR ARCHITECT-ENGINEER SERVICES FOR SAFETY INSPECTION OF DAMS WITHIN THE STATE OF _____

1. *General Description of Scope of Work.* The services to be rendered by the Architect-Engineer (AE) under the proposed contract shall include all engineering functions, hereinafter described, as needed to inspect the dams listed in Appendix A of this contract for the purpose of evaluating their risk of failure. A report which (a) describes the assessed condition of the dam, (b) provides conclusions as to which particular conditions could cause failure, (c) makes recommendations on remedial measures believed necessary, and (d) makes recommendations on whether and what type of future investigation should be conducted shall be provided for

each inspected dam. The work shall proceed in accordance with Phase I of the Recommended Guidelines for Safety Inspection of Dams established by the Office of the Chief of Engineers (OCE) and the supplemented requirements listed in paragraph 3 below. The OCE guidelines are listed in Appendix B of this contract.

2. Information and Services To Be Furnished by the Government. The Contracting Officer will furnish the following information and services to the AE:

- a. All information pertaining to each dam to be inspected as contained in the National Inventory of Dams.
- b. Copies of recommended format for preparation of inspection report, engineering data check list and visual inspection check list.
- c. All available pertinent information pertaining to the Dam Inspection Program and previous investigations having a bearing on inspections to be performed under this contract.
- d. Right-of-entry for access to each dam site.

3. Services To Be Rendered by the Architect-Engineer. The principal services, subject to the optional provisions of the contract, to be rendered by the AE are itemized below:

a. Technical Investigations.

(1) *Engineering Data Collection.* To the extent feasible, the engineering data listed in Appendix I of the OCE guidelines relating to the design, construction and operation of the dam and appurtenant structures, should be collected from existing records and reviewed to aid in evaluating the general condition of each dam, including an assessment of the hydraulic and hydrologic features and structural stability of the dam. Where the necessary engineering data are unavailable, inadequate or invalid, a listing shall be made of those specific additional data deemed necessary by the engineer in charge of the investigation and included in the inspection report. The engineering data checklist provided by the Contracting Officer shall be used as a guide to compile this data.

(2) *Field Inspections.* The field inspection of each dam shall include examination of the items listed in Appendix II of the OCE guidelines, electrical and mechanical equipment for operation of the control facilities, reservoir area, downstream channel in the vicinity of the dam and any other significant feature to determine how these features affect the risk of failure of the dam. The inspection shall be conducted in a systematic manner to minimize the possibility of any significant feature being overlooked. The visual inspection checklist provided by the Contracting Officer shall be used as a guide to document the examination of each significant feature.

Particular attention shall be given to detecting evidence of leakage, erosion, seepage, slope instability, undue settlement, displacement, tilting, cracking, deterioration, and improper functioning of drains and relief wells. The degree and quality of maintenance and regulating procedures for operation of the control facilities shall be assessed. The design and existing condition of such control facilities (*i.e.*, spillway, outlet works, etc.) shall be evaluated. An assessment of the degree of siltation that is evident and its effect on the dam's reservoir shall be performed. Photographs and drawings should be used to record conditions in order to minimize written descriptions.

(3) Engineering Analyses.

(a) *Evaluation of Hydraulic and Hydrologic (H&H) Features.* Evaluation of the hydraulic and hydrological features of each dam shall be based on criteria set forth in the OCE guidelines. If it is determined that the available H&H data are insufficient, the Contracting Officer must be so informed and may exercise an option of requiring the AE to perform an overtopping analysis at additional agreed-upon compensation. The methodology to be used by the AE for this analysis will be based on the OCE guidelines and subject to the approval of the Contracting Officer.

(b) *Evaluation of Structural Stability.* The evaluation of structural stability of each dam is to be based principally on existing conditions as revealed by the visual inspection, available design and construction information, and records of performance. The objectives are to determine the existence of conditions, identifiable by visual inspection or from records, which may pose a high risk of failure and to formulate recommendations pertaining to the need for any remedial improvements, additional studies, investigations, or analysis. The results of this phase of the inspection must rely substantially upon the experience and judgment of the inspecting engineer. Should it be determined that sufficient data are not available for a reasonable evaluation of the structural stability of a dam and appurtenances, the Contracting Officer should be informed which information is required prior to attempting to evaluate the risk of failure of the dam.

(c) *Evaluation of Operational Features.* Where critical mechanical/electrical operating equipment is used in controlling the reservoir of a dam, an evaluation of the operational characteristics of this equipment from the standpoint of risk of failure must be performed.

(d) *Evaluation of Reservoir Regulation Plan and Warning System.* The operational characteristics of each dam's existing reservoir regulation plan and warning system in event of a threatened failure shall be investigated.

b. *Emergency Situations.* The Contracting Officer must be immediately notified of any observed condition which is deemed to require immediate remedial action. After being notified, the Contracting Officer will contact the appropriate State personnel and will meet the AE at the site to determine the appropriate course of action. This will not relieve the AE of his responsibility to prepare a comprehensive inspection report at the earliest practicable date.

c. *Qualifications of Investigators.* The technical investigations shall be conducted by licensed professional engineers with a minimum of five years experience after licensing in the investigation, design and construction of earthfill, rockfill and concrete dams and/or in making risk of failure evaluations of completed dams. These engineers must be knowledgeable in the disciplines of hydrology, hydraulics, geotechnical, electrical, mechanical and structural engineering, as necessary. All field inspections should be conducted by engineers, engineering geologists and other specialists who are knowledgeable in the investigation, design, construction and operation of dams, including experts on mechanical and electrical operation of gates and controls, where needed.

d. *Preparation of Report.* A formal report shall be prepared for each dam inspected for submission to the Contracting Officer. Each report should contain the information specified in OCE guidelines and any other pertinent information. The recommended format provided by the Contracting Officer shall be used to document each report. The signature and registration identification of the professional engineer who directed the investigation and who was responsible for evaluation of the dam should be included in the report.

4. *Supervision and Approval of Work.* All work performed under this contract shall be subject to the review and approval of the Contracting Officer or his designee. Meetings will be held on a regular basis in the District office, during which the progress of inspections will be discussed and questions relating to inspection reports previously received by the Contracting Officer will be addressed. Reports will be revised as necessary when required by the Contracting Officer.

5. *Coordination.* During the progress of work, the AE shall maintain liaison with the * ____ and other local authorities through the Contracting Officer as required to assure the orderly progression of the inspection. Copies of all correspondence with such authorities shall be provided to the Contracting Officer.

6. *Submission of Report.*

a. Each inspection report will be submitted for review to the Contracting Officer. Reports will be revised as required by the Contracting Officer. After all revisions have been made, the original and __ copies of each inspection report shall be submitted to the Contracting Officer.

b. Text of all reports shall be typewritten and printed on both sides of 8" × 10¹/₂" paper. All notes, inspection forms, sketches or similar matter shall be legible, distinct and suitable for reproduction.

7. *Period of Services.*

a. All inspections and reports included under this contract shall be completed within __ days from date of Notice to Proceed.

b. If the option for performing an H&H analysis for any particular site is exercised, the AE shall complete such analysis within __ days from date of Notice to Proceed. However, the overall completion time stated in paragraph 7a above shall not change.

* NOTE: Write in the designated State Authority.

Appendix I to § 222.6

PROCEDURE FOR USING NASA LAND SATELLITE MULTISPECTRAL SCANNER DATA FOR VERIFICATION AND UPDATING THE NATIONAL INVENTORY OF DAMS

1. *Purpose.* This appendix states the objective, defines the scope, prescribes procedures, and assigns responsibilities for using NASA Land Satellite (LANDSAT) Multispectral Scanner data along with NASA's Surface Water Detection And Mapping (DAM) Computer program to assist in verification and updating the National Inventory of Dams.

2. *Applicability.* This appendix is applicable to all divisions and districts having Civil Works responsibilities except POD.

3. *Reference.* NASA, DETECTION AND MAPPING PACKAGE, Users Manuals, Volumes 1, 2a, 2b, and 3 dated June 1976, published by the Johnson Space Center, Houston, Texas.

4. *Objectives.* Provide a uniform method, nation-wide, to help insure that all dams subject to Public Law 92-367, 8 August 1972 are properly identified and located in the National Inventory of Dams.

5. *Scope.* The computer printer overlay maps produced by the procedure described in reference 3b will be used by district and/or state or contractor personnel as a tool to assist in verification and updating of the National Inventory of Dams.

6. *Exceptions.* a. If a Division/District attempts the use of the procedure for a given region within their area of responsibility and finds the overlay maps cannot be used to assist in verification and updating the National Inventory of Dams, they may request an exception for a selected region. A selected region may include areas where conditions can reasonably be assumed to be the same as the region where the procedure was tried.

b. Request for exceptions should be documented to include firm boundary definitions and appropriate justification to demonstrate why the procedure cannot be used. This request should be submitted to WRSC WASH DC 20314, through the normal engineering chain of command.

c. Map overlays will be produced for all areas of the Continental United States even if they are not used in a few selected regions. This processing is required for a future Computer Water Body Change Detection system.

7. *Procedures.* Acquisition of LANDSAT data, registration of satellite coordinates to earth latitude and longitude and computer processing to produce overlay maps will be accomplished by two Regional Centers. Nashville District and Seattle District have been designated as the Regional Centers, with each responsible for processing maps by state based on Divisional assignments in Appendix A. Regional Centers will support divisions as follows:

Regional Center	Division
Nashville District	New England
	North Atlantic
	South Atlantic
	Ohio River
	Lower Mississippi Valley
	North Central
Seattle District	Southwestern
	Missouri River
	North Pacific
	South Pacific

8. *Responsibilities.* a. The Water Resources Support Center at Fort Belvoir has overall responsibility for coordination and monitoring of this activity between NASA, Division Offices, and Regional Centers, and for providing Regional Center funding.

b. Regional Centers are responsible for:

(1) Acquiring proper LANDSAT data tape from EROS Data Center (Sioux Falls, South Dakota). Actual data scene selection will be coordinated with Division and/or District to insure proper consideration is given to local priorities and seasonal coverage.

(2) Arranging computer processing support using NASA's DAM package.

(3) Establishing proper control between satellite scanner-oriented coordinates and earth latitude/longitude.

(4) Producing total coverage of map overlays at a scale of 1:24,000 and/or smaller scales as required by Divisions and/or Districts.

(5) Instructing District, State, or contractor personnel in the assembly and use of map overlays.

c. Divisions/Districts are responsible for:

(1) Designating one person from each Division and District as the point of contact with the Regional Center and provide this person's name and phone number to the Regional Center.

(2) Providing the Regional Center with map coverage of their area of responsibility. This will include state indexes and 7¹/₂ minute quadrangle sheets (scale 1:24,000) where available.

(3) Coordinating with the Regional Center in selecting LANDSAT data tapes.

(4) Providing information to Regional Center on scale and priorities of desired computer produced map overlays.

(5) Assembling computer print-outs into overlay maps, and using as appropriate to assist in verification and updating the National Inventory of Dams.

9. *Points of Contact.* The points of contact in the Regional Centers for this program are as follows:

Name, Office Symbol, and Telephone

Jim Cook—DAEN-ORNED, (615) 251-7366; FTS 852-7366.

Jack Erlandson—DAEN-NPSEN, (206) 764-3535; FTS 399-3535.

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